Machine Learning Track Inference in the Dead Regions of DUNE's Near Detector Prototype:

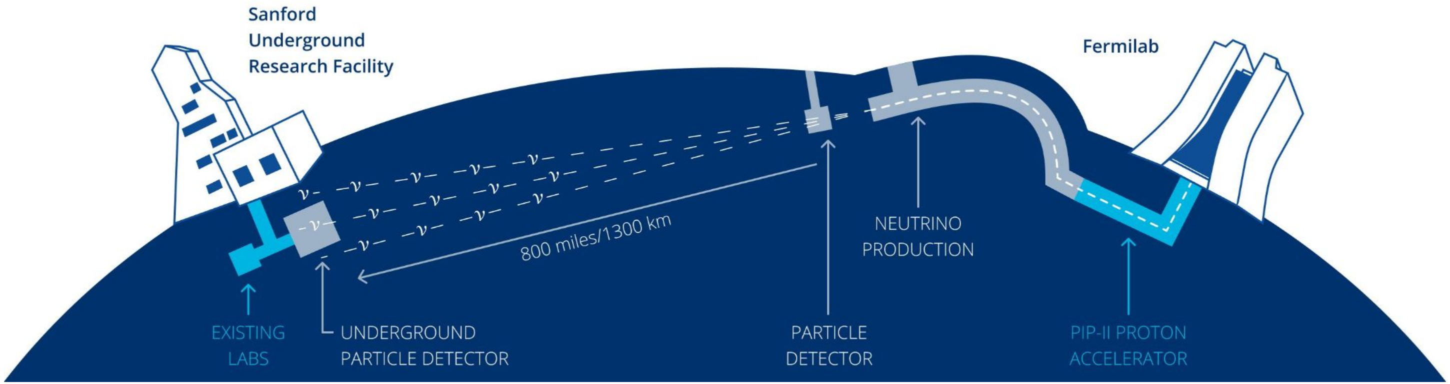
The Liquid Argon TPC Dead Region Inference Project

Hilary Utaegbulam, on behalf of the DUNE Collaboration

Neutrino Physics and Machine Learning 2026

I am a person who stutters.

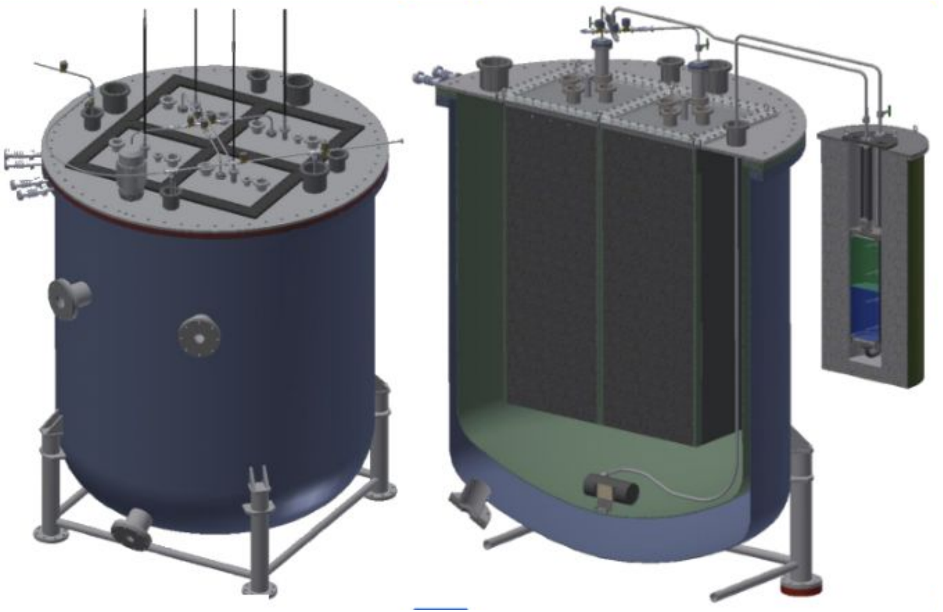
The Deep Underground Neutrino Experiment (DUNE)



- Sends a high-intensity neutrino beam 1300 km from Fermilab to South Dakota
- Uses massive liquid-argon TPC detectors to image neutrino interactions in 3D
- Main goals: measure neutrino oscillations, CP violation, mass ordering, supernova neutrinos, and rare physics
- ML is essential because DUNE produces complex, high-dimensional detector data

The 2×2 Demonstrator

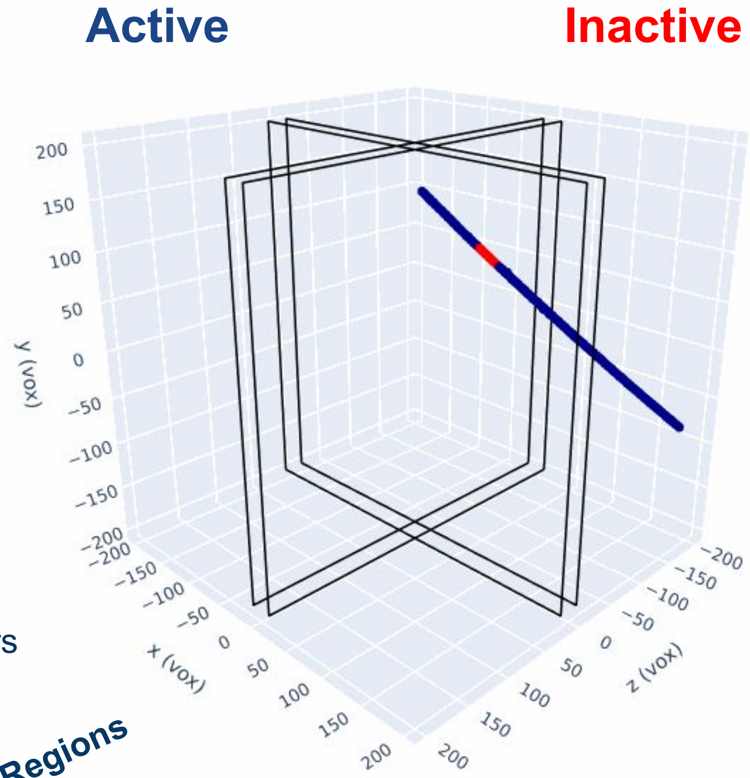
- Prototype for DUNE ND-LAr, operated in the NuMI beam at Fermilab
- Built from four modular liquid-argon TPCs in a 2×2 layout
- Uses pixelated charge readout + light readout for true 3D imaging
- Tests reconstruction using the NDLAr design

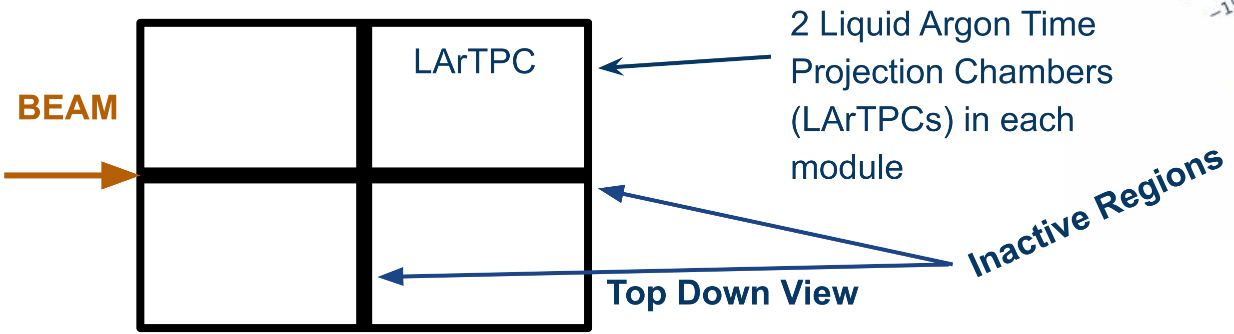


Inactive Regions in the 2×2 Demonstrator

Liquid Argon Time Projection Chamber (LArTPC)

- A ~5 cm dead gap splits the detector into four modules

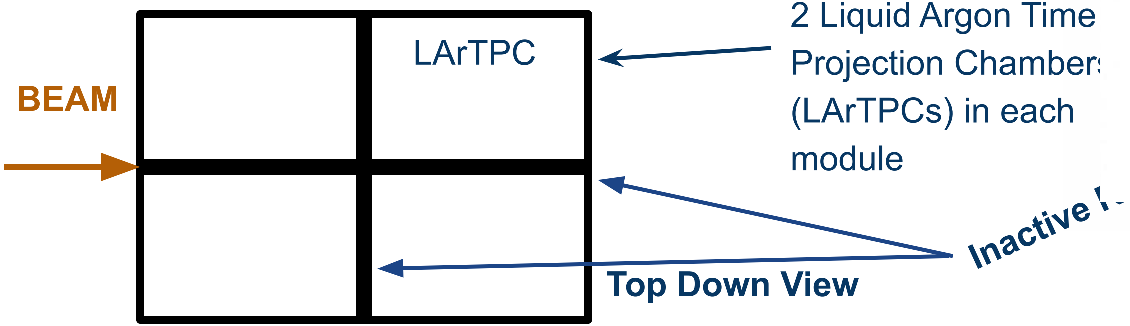


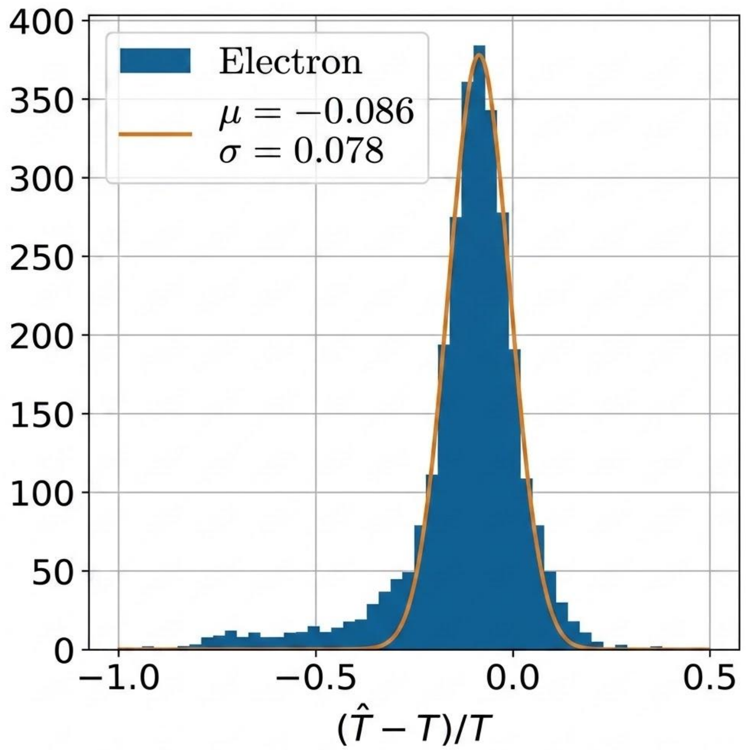


Inactive Regions in the 2×2 Demonstrator

Liquid Argon Time Projection Chamber (LArTPC)

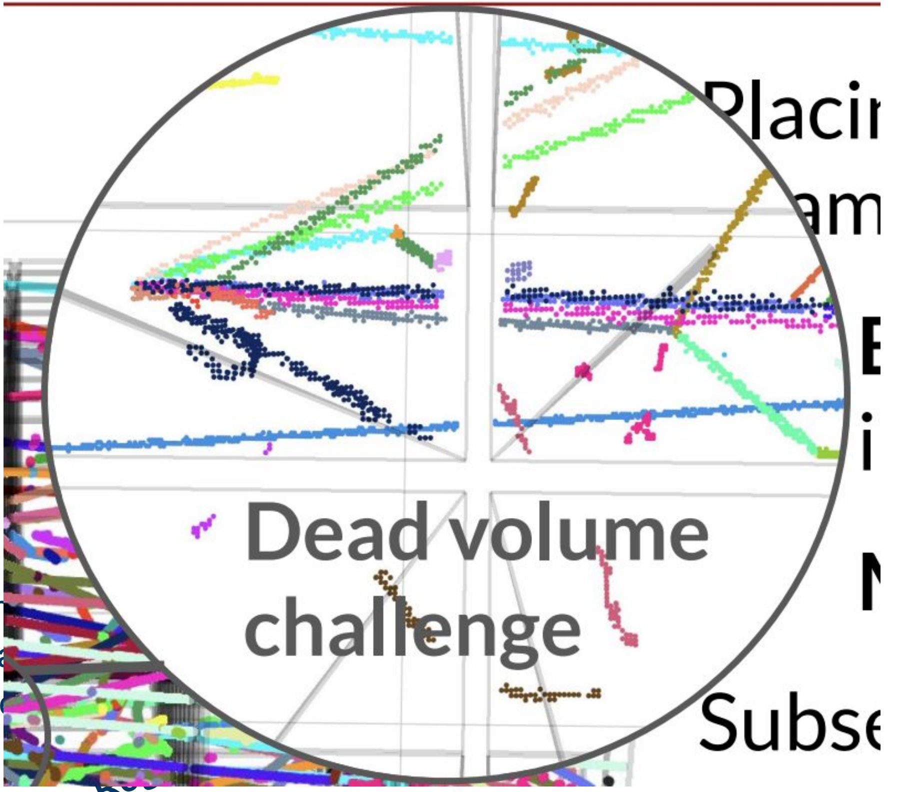
- A ~5 cm dead gap splits the detector into four modules
  - For EM showers, bias + energy resolution hampered by inactive region

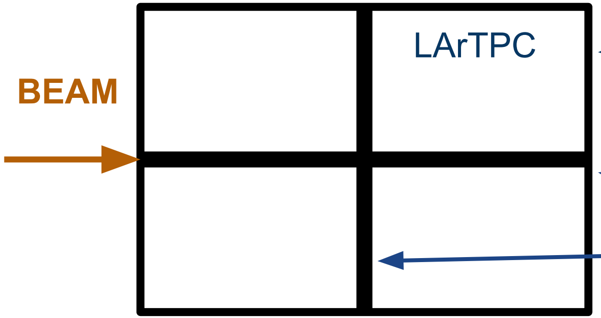




Inactive Regions in the 2×2 Demonstrator

- A ~5 cm dead gap splits the detector into four modules
    - For EM showers, bias + energy resolution hampered by inactive region
    - Makes Clustering step for SPINE harder!





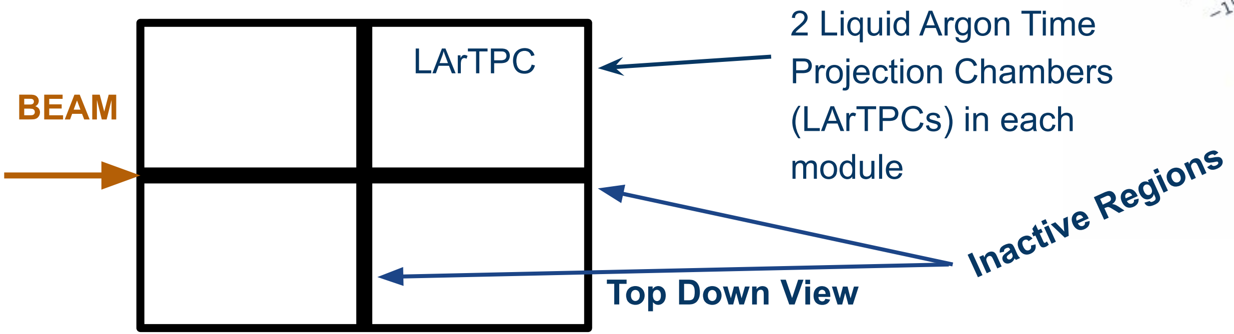
2 Liquid Argon
Projection Cha
(LArTPCs) in
module

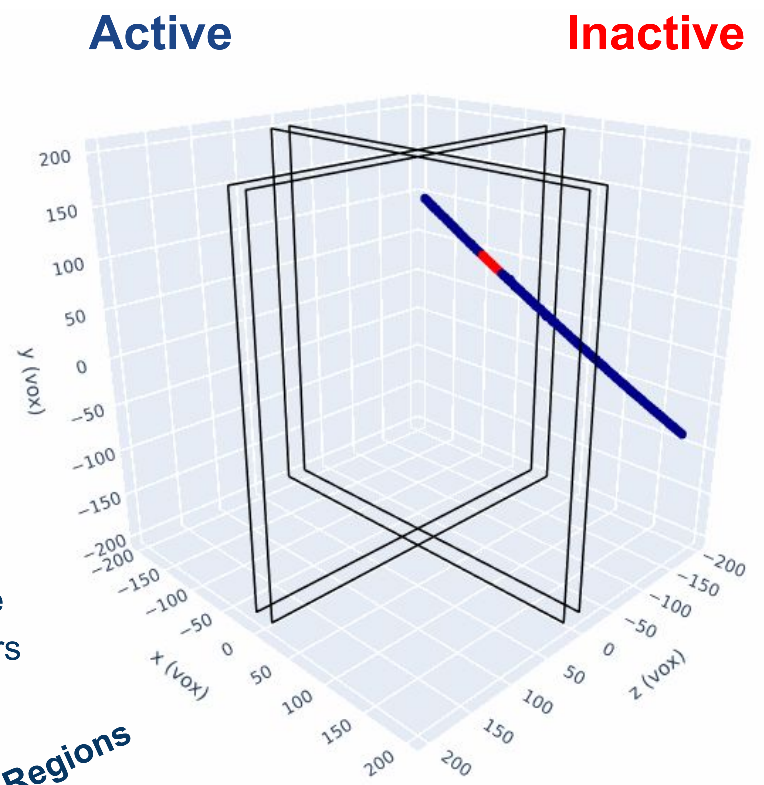
Inactive Re

Top Down View

Inactive Regions in the 2×2 Demonstrator

- A ~5 cm dead gap splits the detector into four modules
- Can a neural network infer hits across the gap?



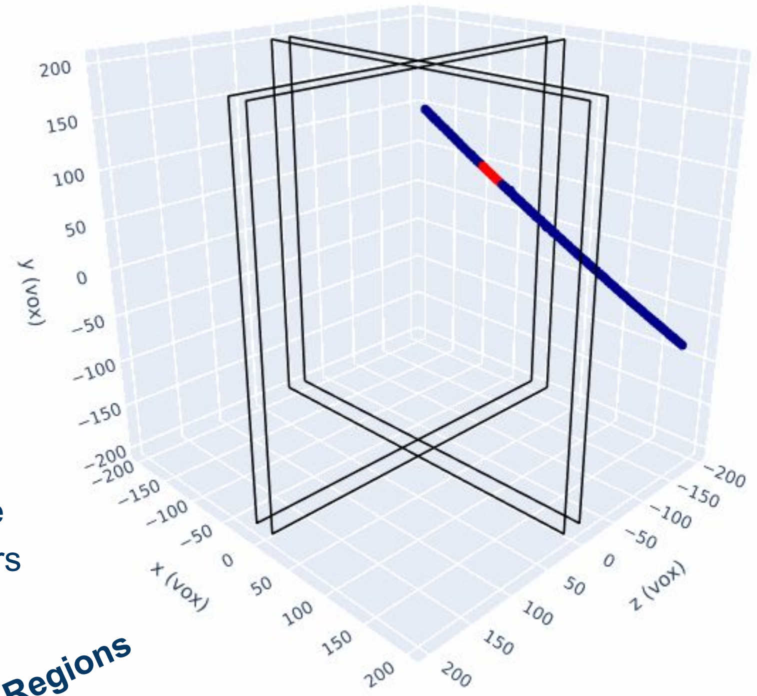


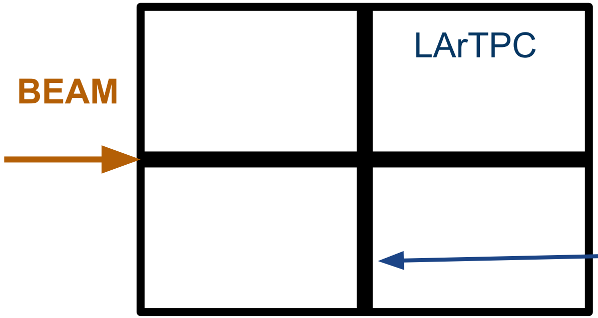
Inactive Regions in the 2×2 Demonstrator

Liquid Argon Time Projection Chamber (LArTPC)

- A ~5 cm dead gap splits the detector into four modules
- Can a neural network infer hits across the gap?
- We train a voxel-completion network to bridge the gap

Active **Inactive**



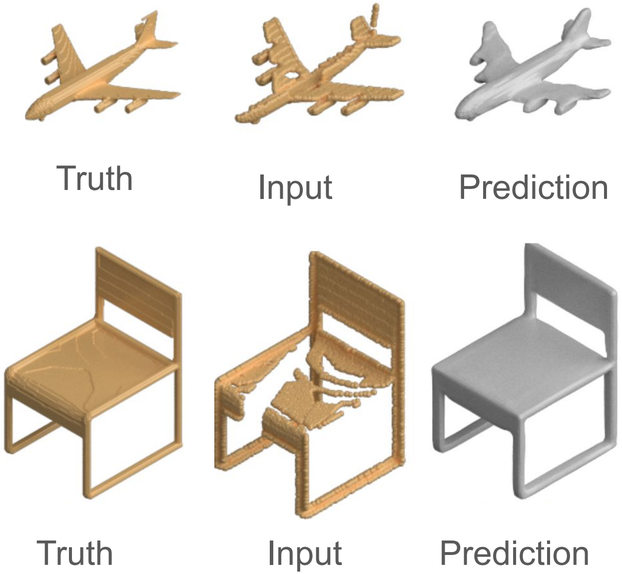


2 Liquid Argon Time Projection Chambers (LArTPCs) in each module

Inactive Regions

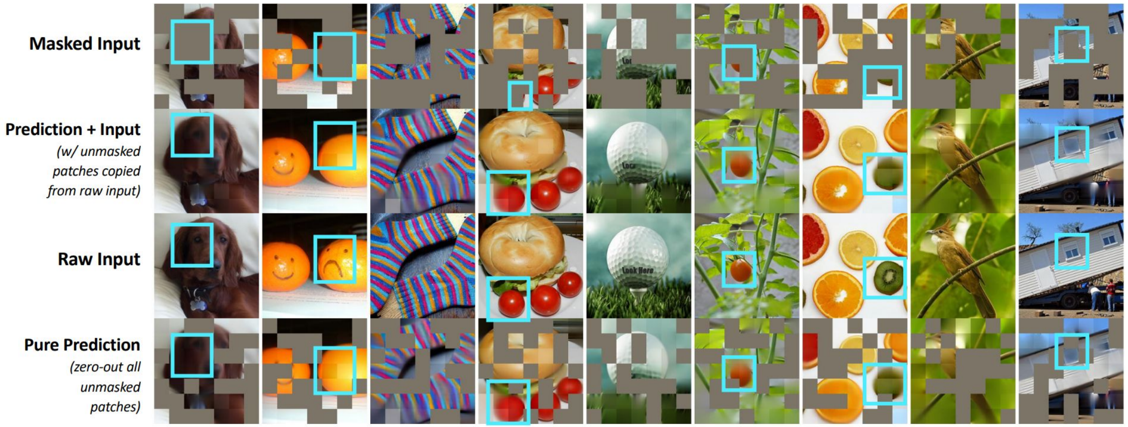
Top Down View

Which Architecture?



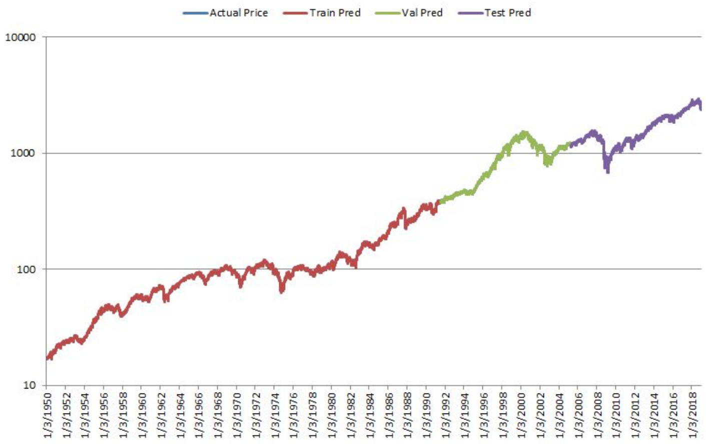
Vision Transformer

https://shapeformer.github.io/static/ShapeFormer.pdf



https://arxiv.org/pdf/2301.03580

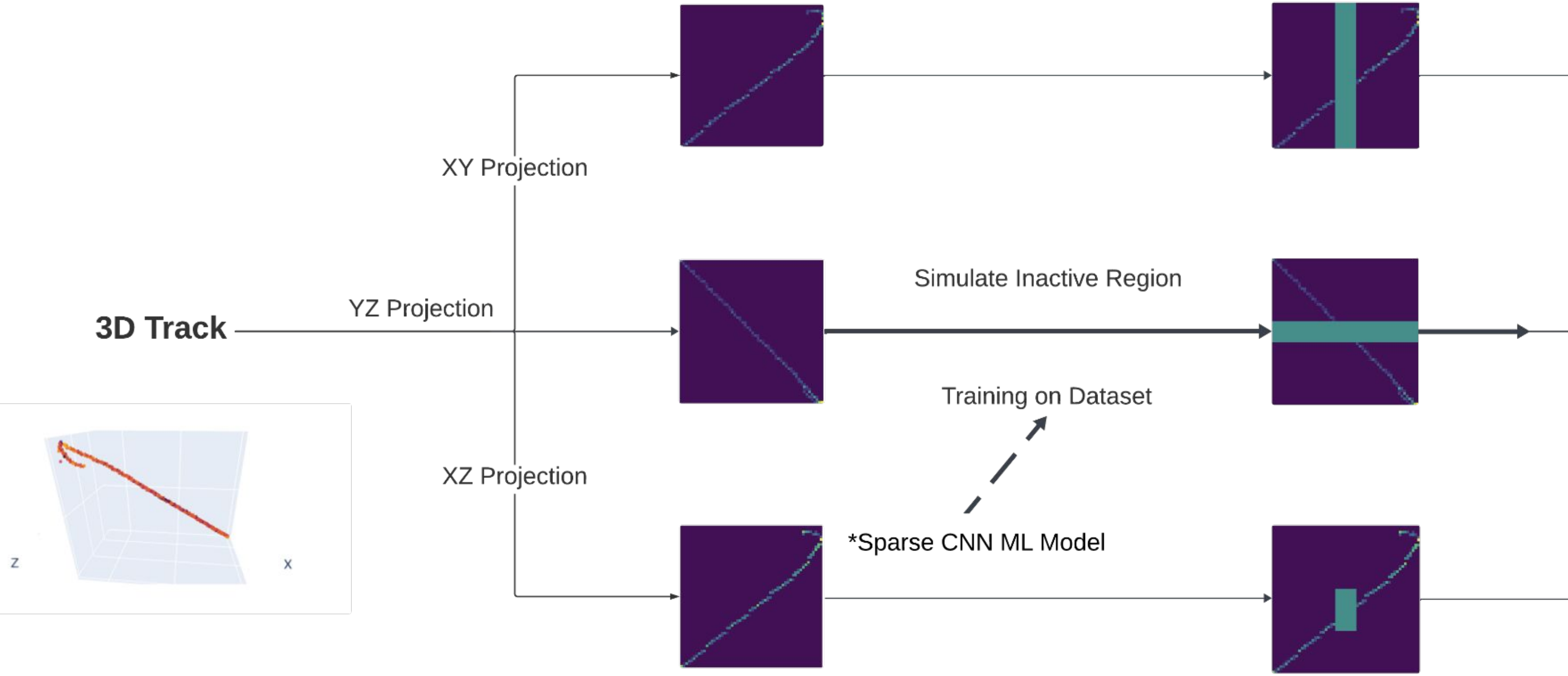
Masked AutoEncoder



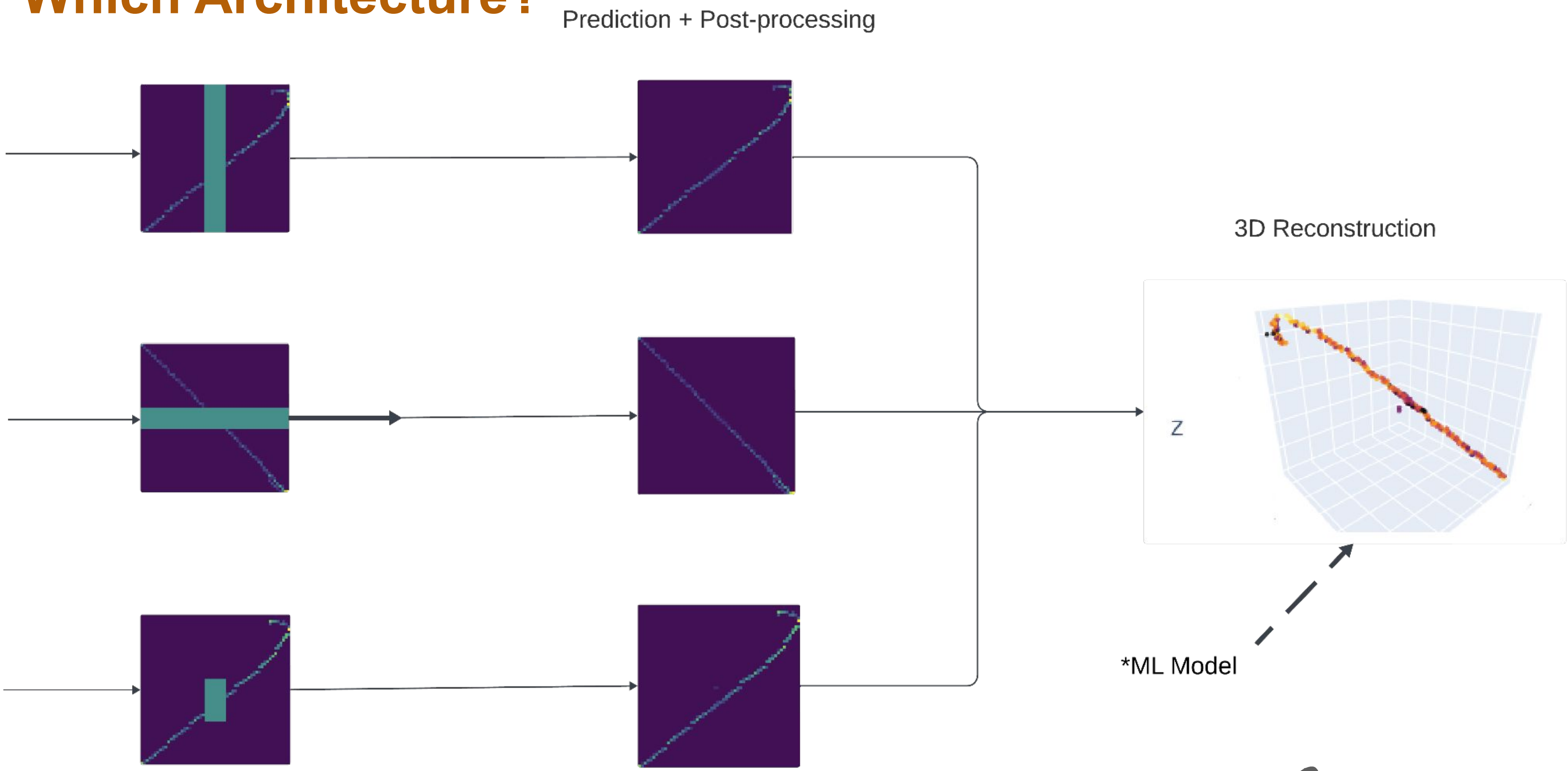
https://www.blueskycapitalmanagement.com/machine-learning-in-finance-why-you-should-not-use-lstms-to-predict-the-stock-market/

Long-short-term Memory (RNN)

Which Architecture?



Which Architecture?



Which Architecture?

Architecture	Main limitation
3D Vision Transformer	Required 3D depth
Masked Autoencoder (MAE)	Worked reasonably well in 2D, but going from 2D to 3D produced many ghost points.
LSTM/RNN	Requires a well-defined *time axis*. Difficult for particle tracks due to curving, branching, prongs.
GNN	Requires explicit nodes and edges. Generative GNNs would not transfer well to particle tracks.

https://shapeformer.github.io/static/ShapeFormer.pdf

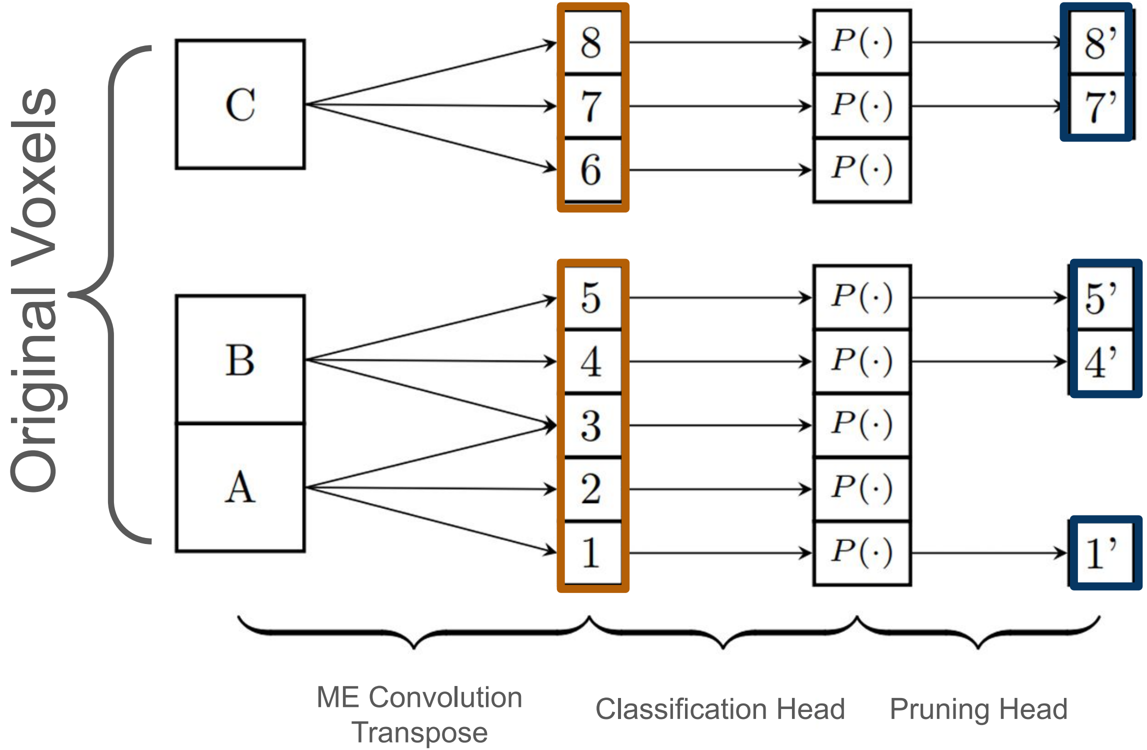
https://arxiv.org/pdf/2301.03580

https://arxiv.org/pdf/1801.02143

https://arxiv.org/pdf/2012.07397

+ Others

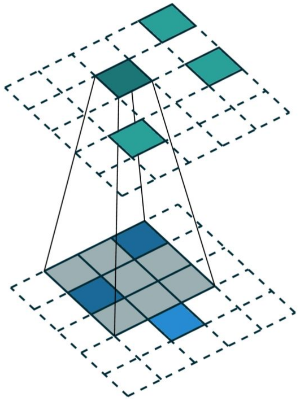
Multi-Scale Candidate Generation & Pruning



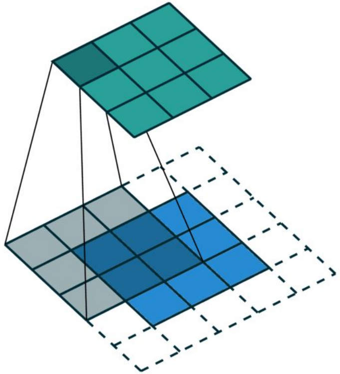
Generated candidate voxels based on kernel shape and neighborhood

Voxels that survive past pruning and start the process again

Sparse Convolution



Dense Convolution



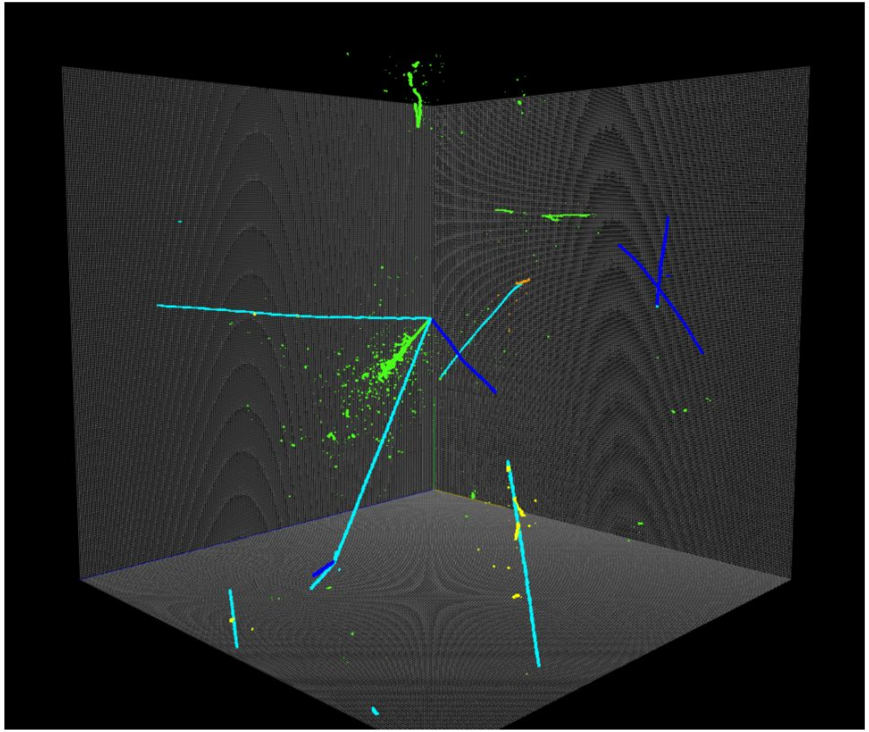
The Minkowski Engine’s sparse convolutions lets us slide the kernel over only the voxels we care about

https://github.com/NVIDIA/MinkowskiEngine

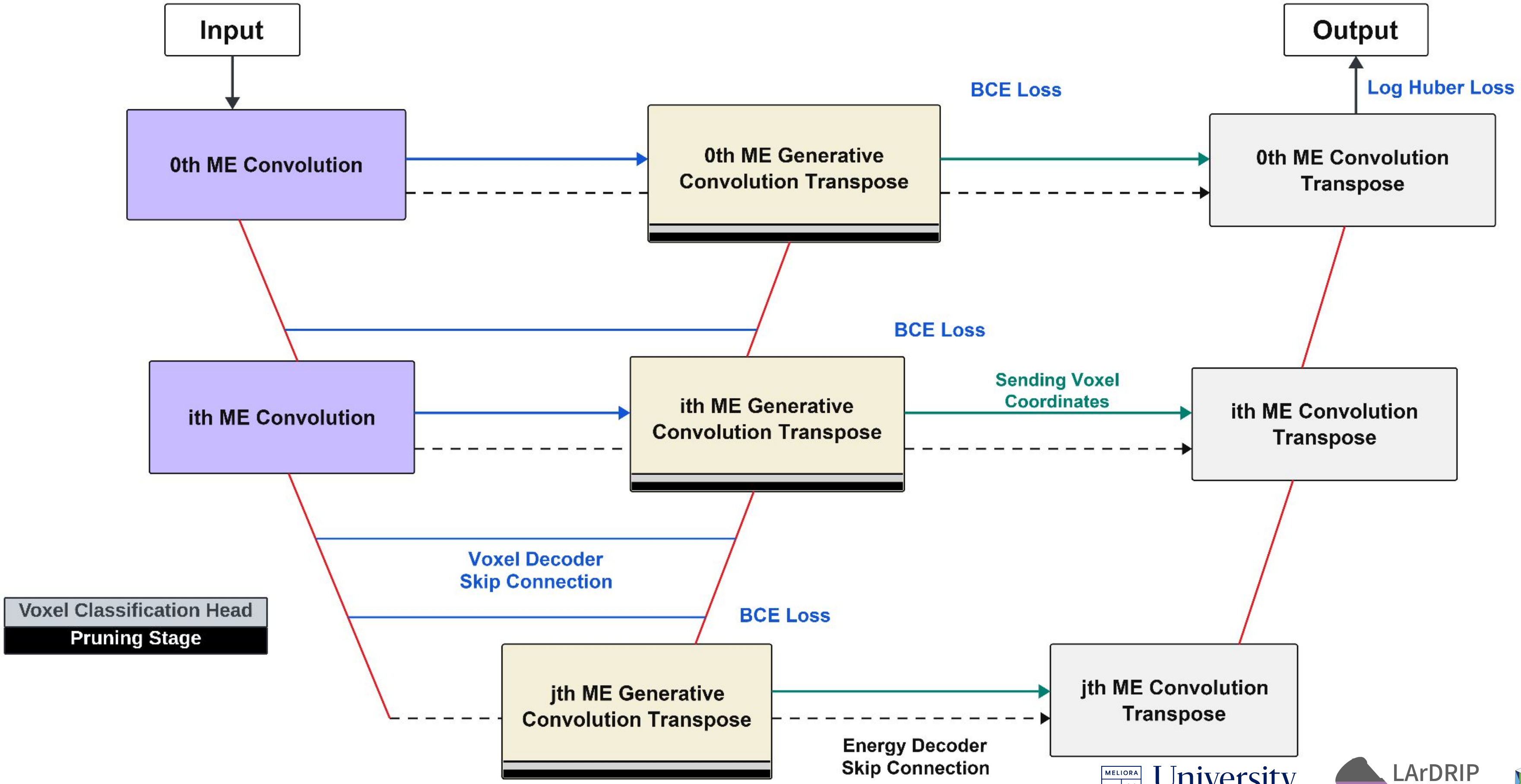
PILArNet Dataset

PILArNet: Public Dataset for Particle Imaging Liquid Argon Detectors in High Energy Physics

C. Adams,[1] K. Terao,[2] and T. Wongjirad[3]



LArDRIP: A Multi-Scale Generative Sparse 3D UNet



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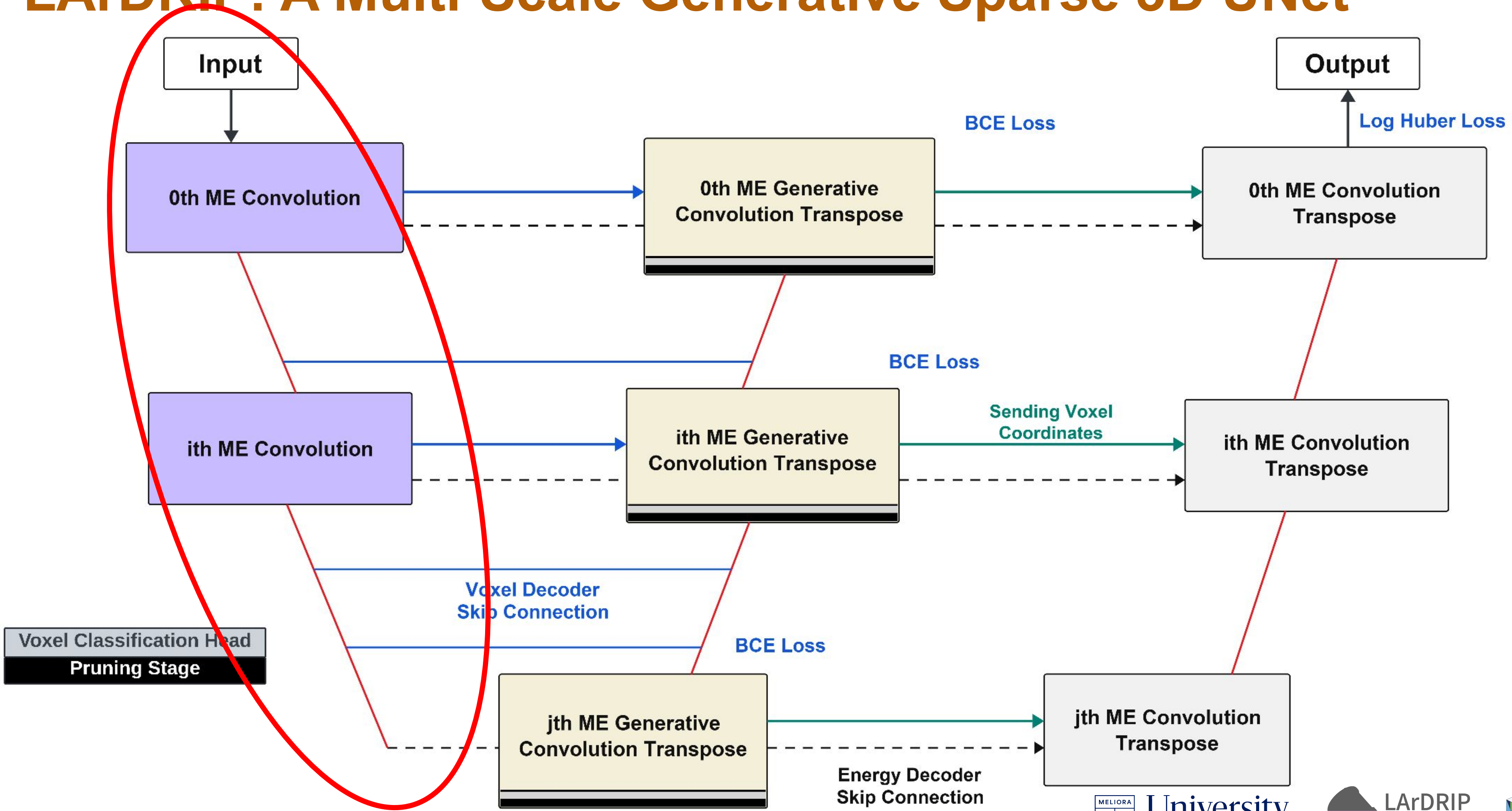

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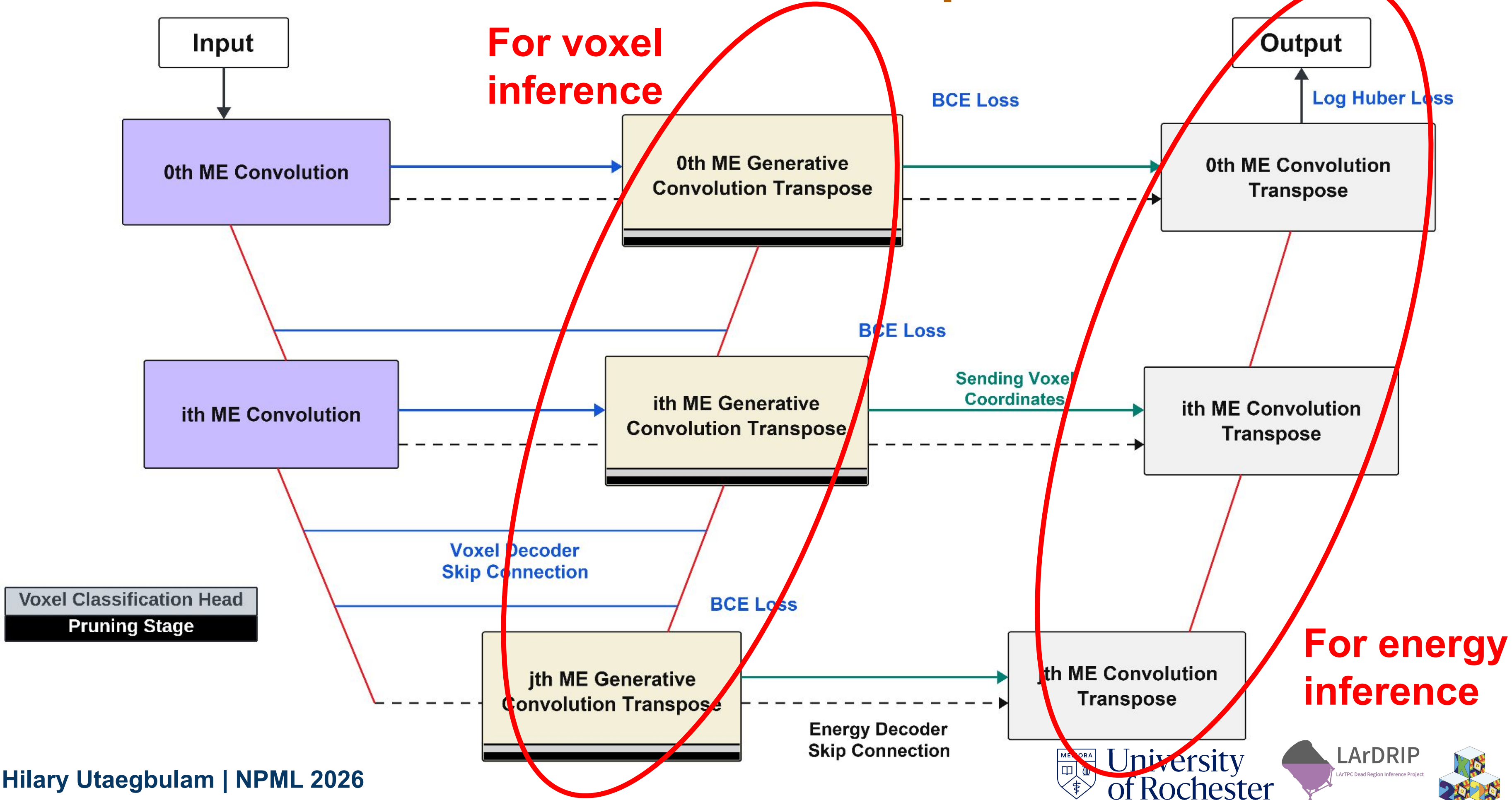

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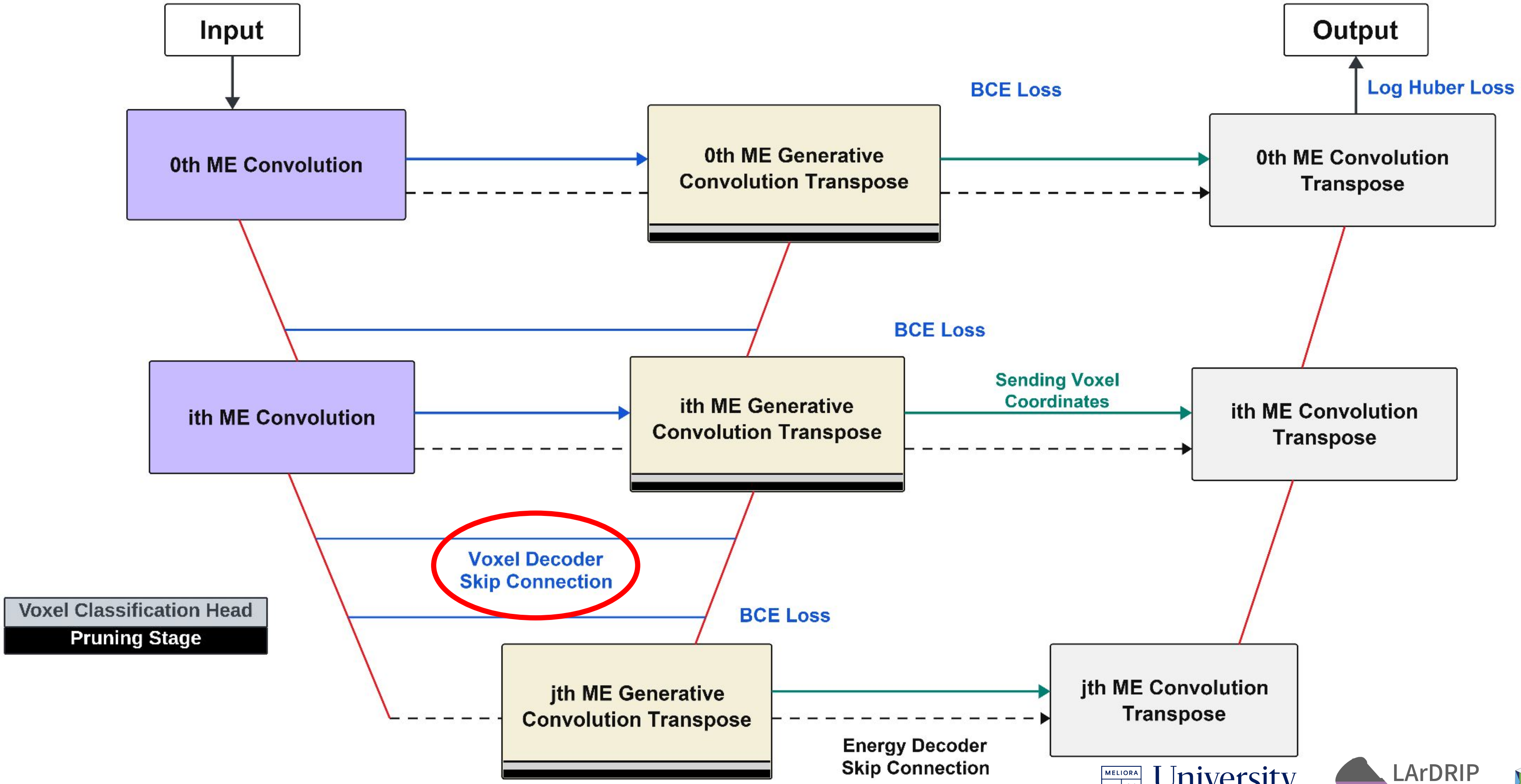

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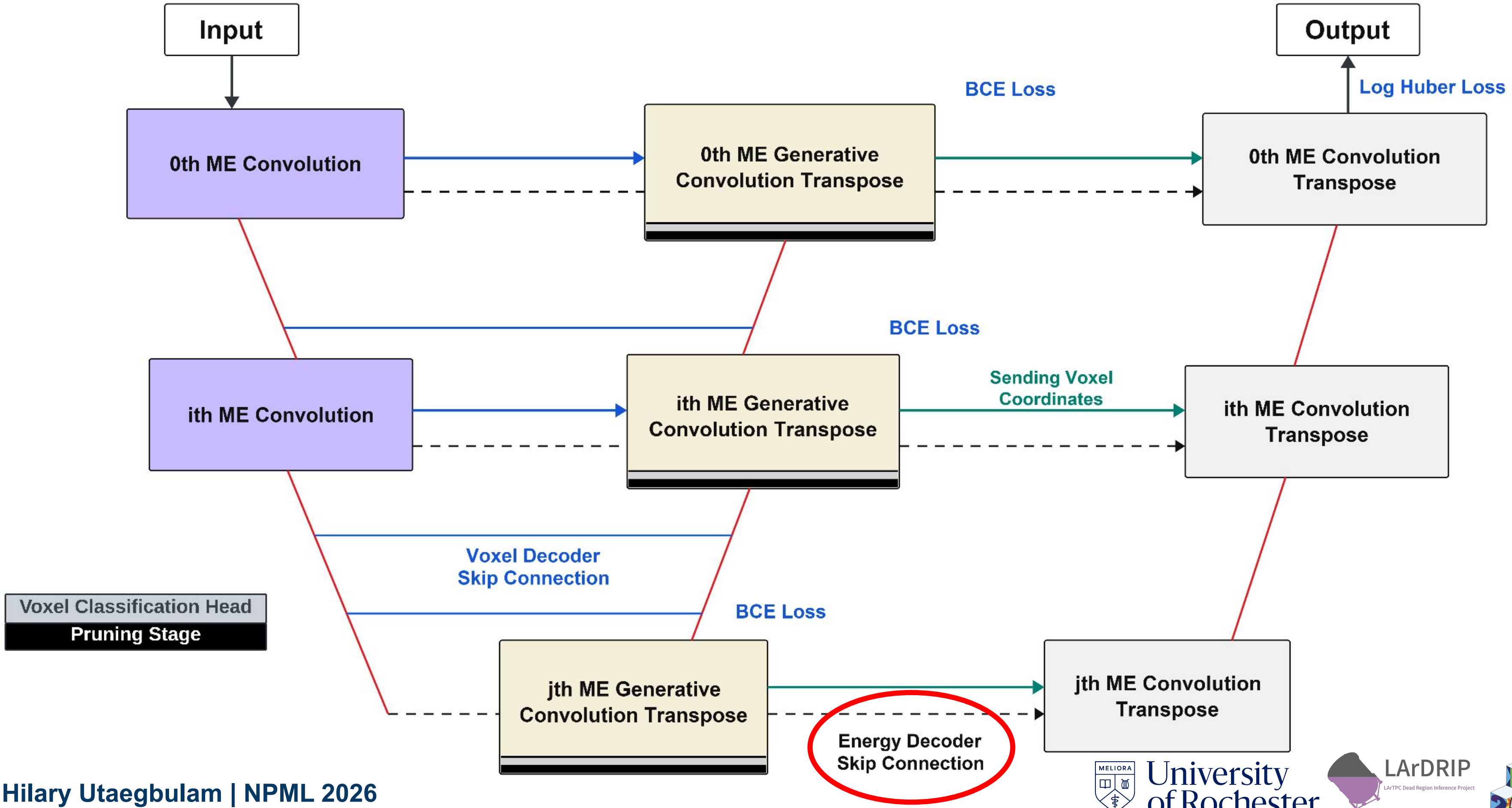

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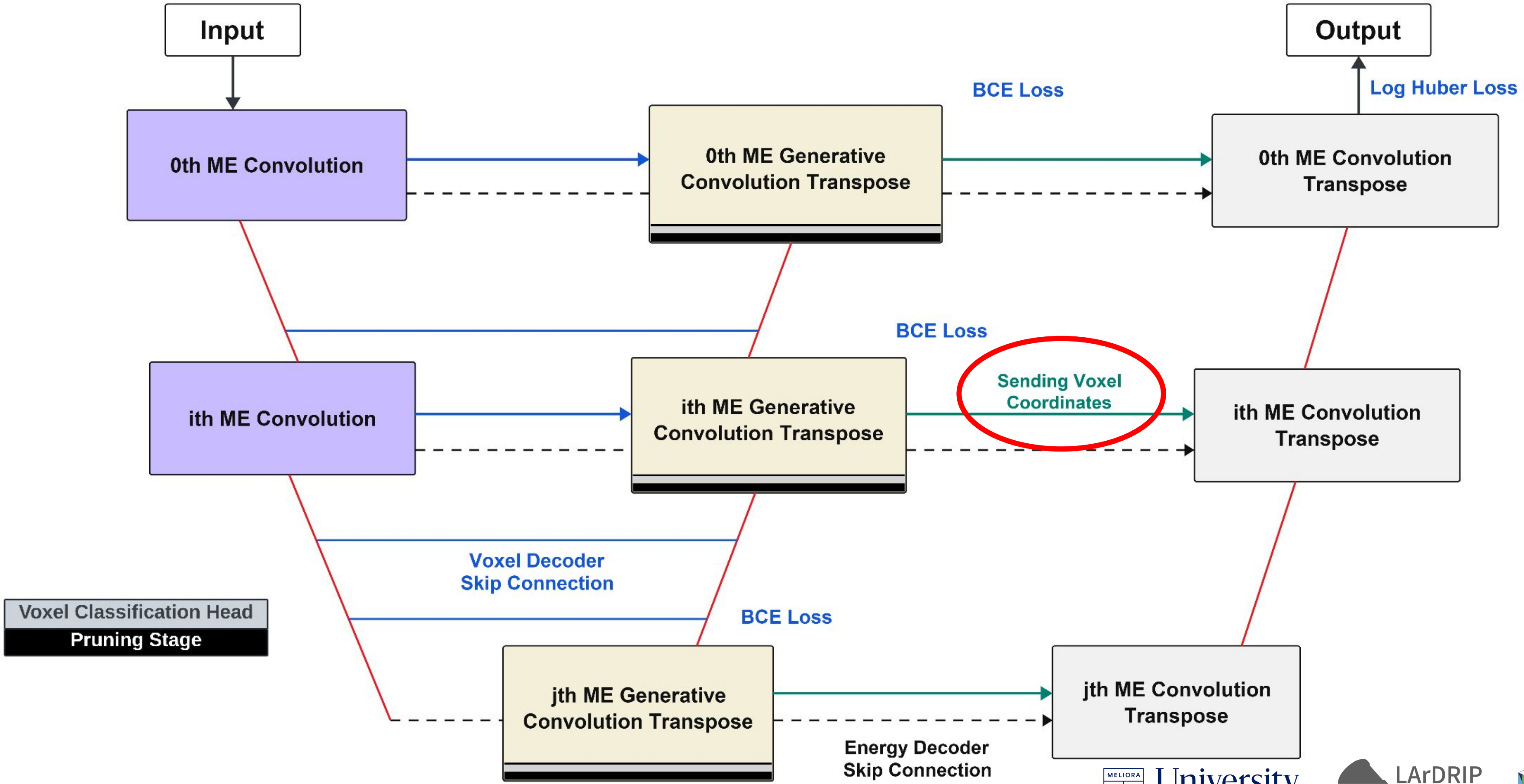

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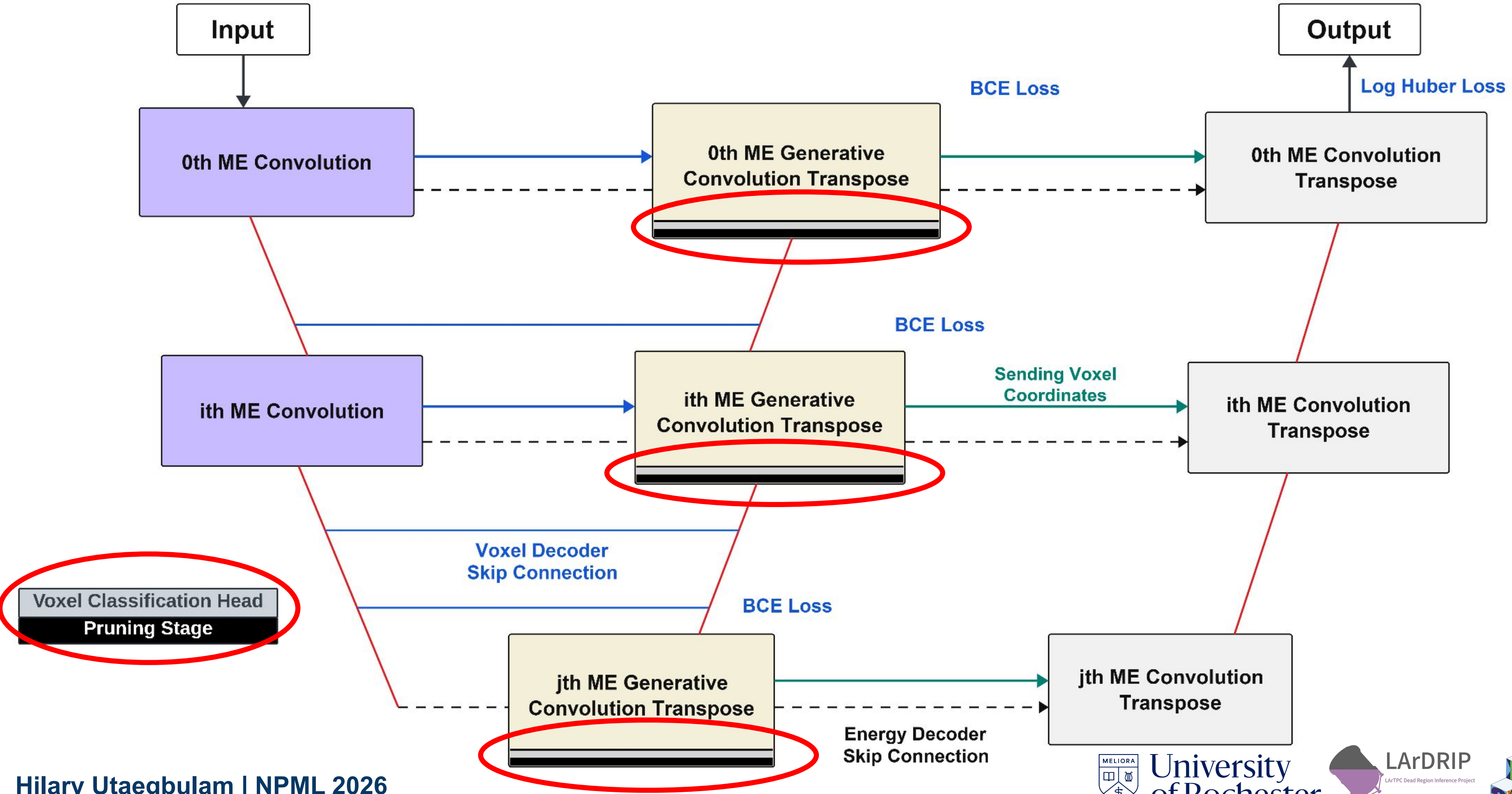

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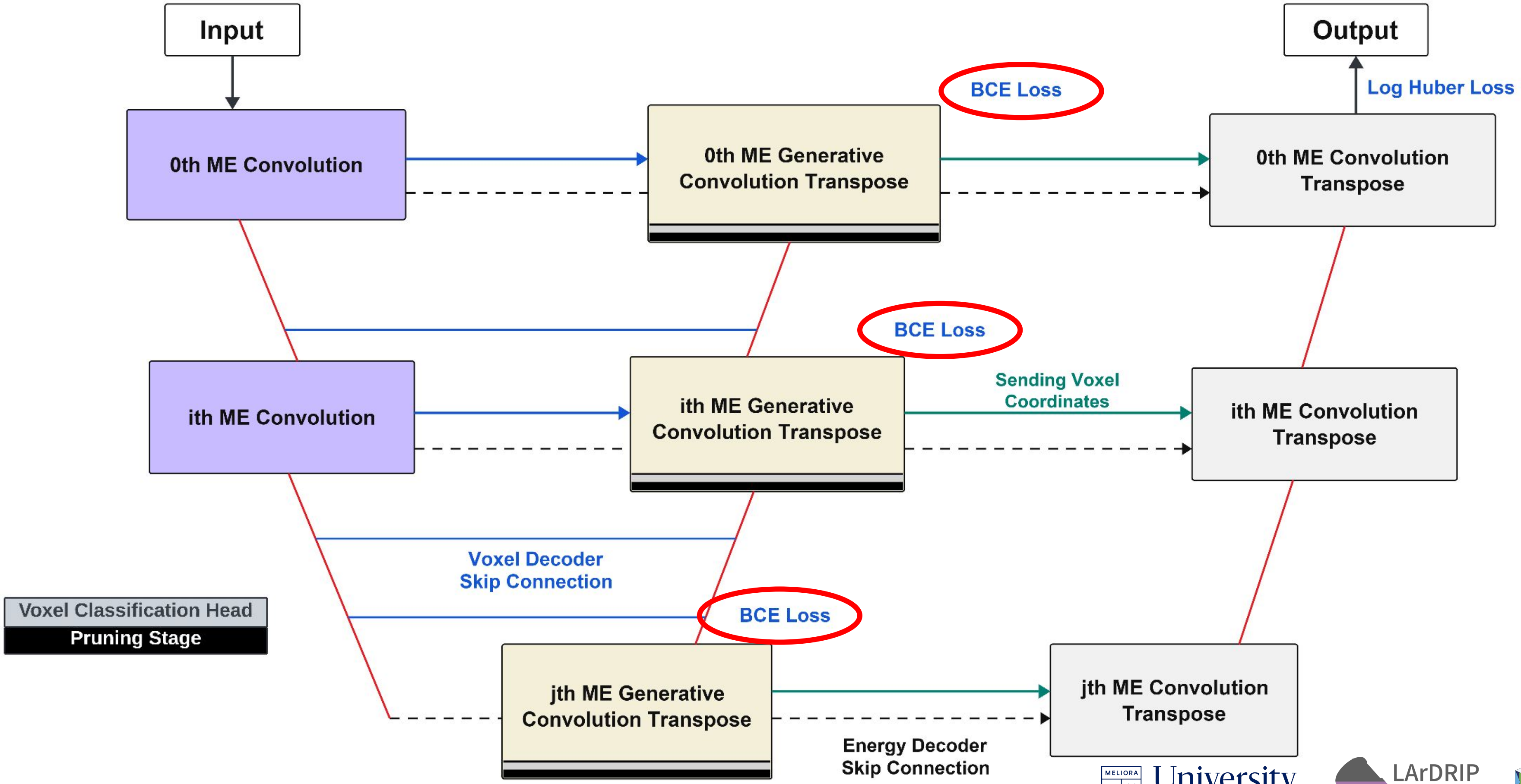

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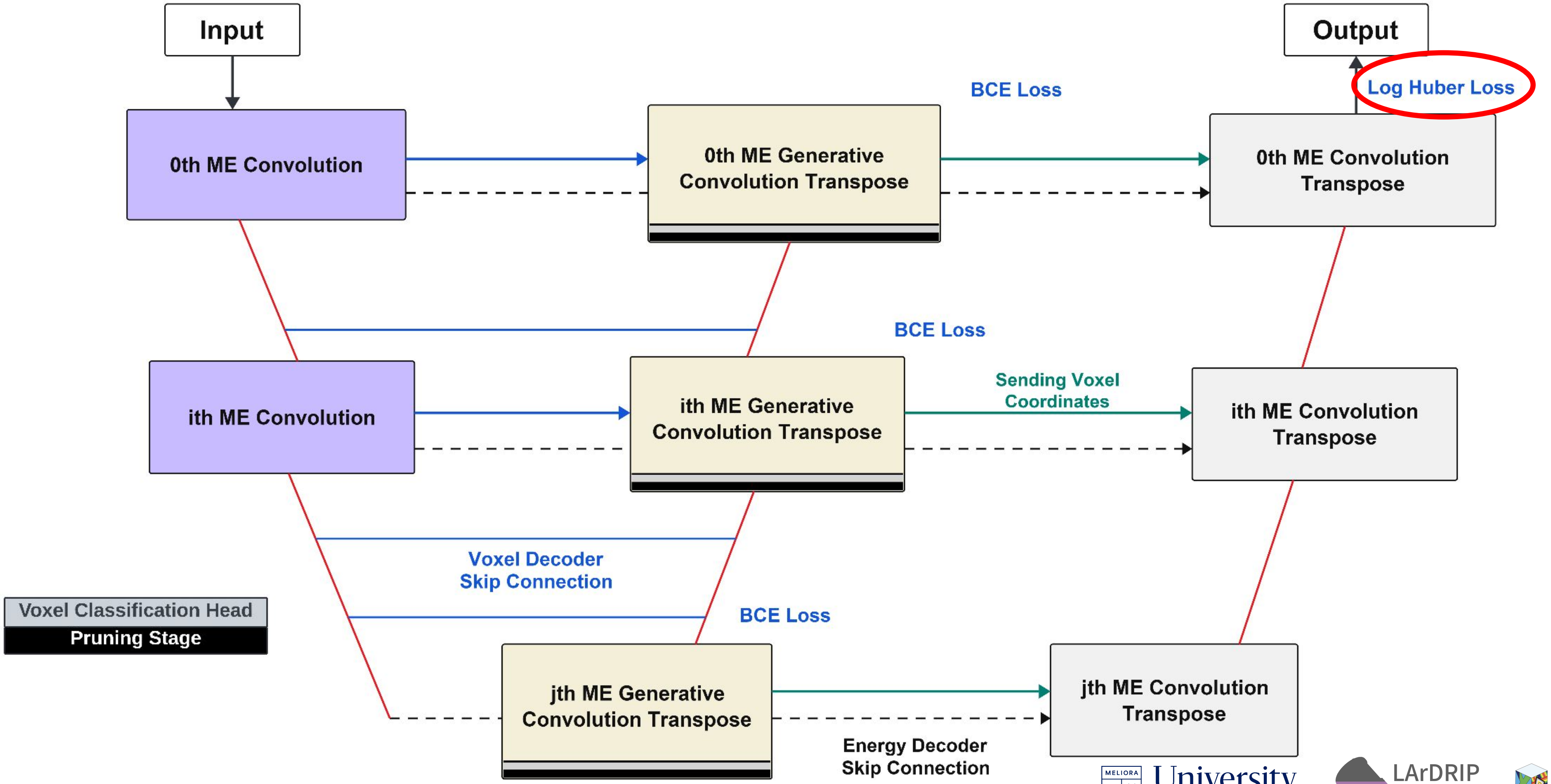

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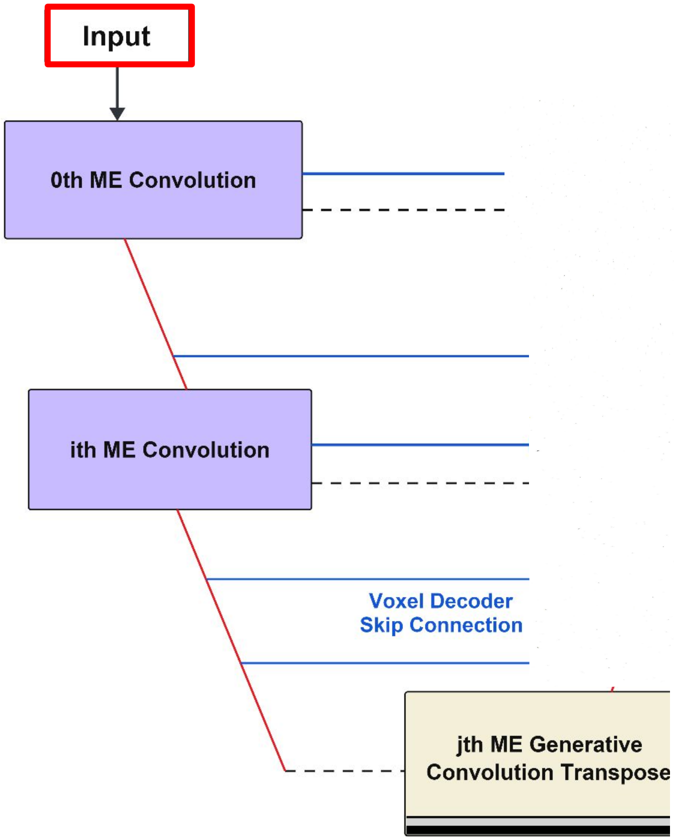
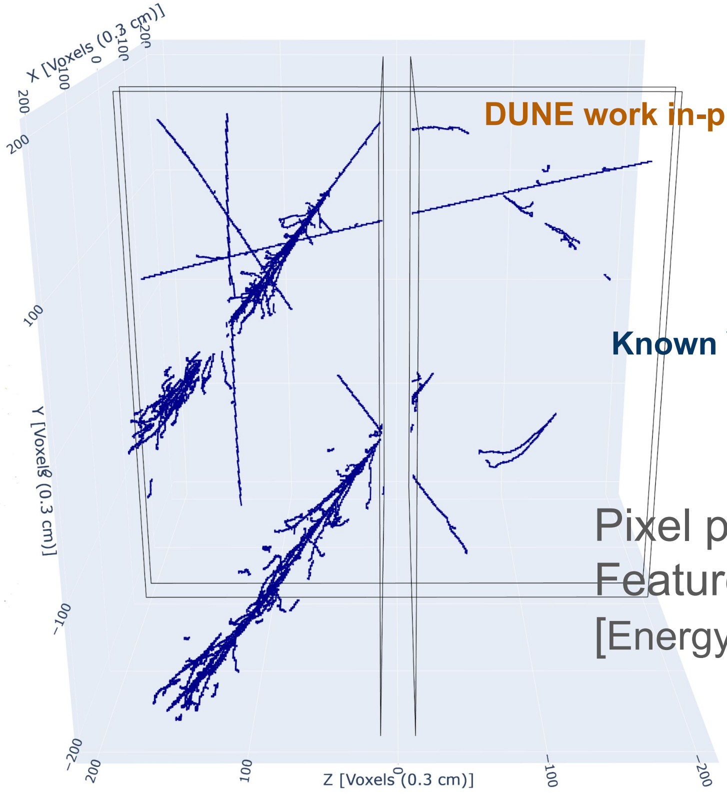


Input



Pixel pitch: 0.3 cm
Feature size: 2
[Energy, Semantic Type]

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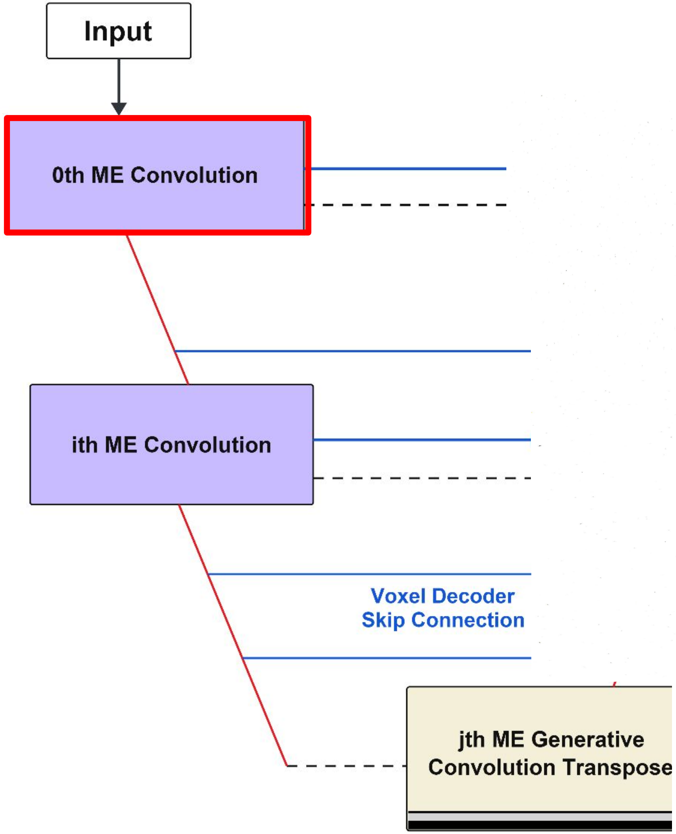
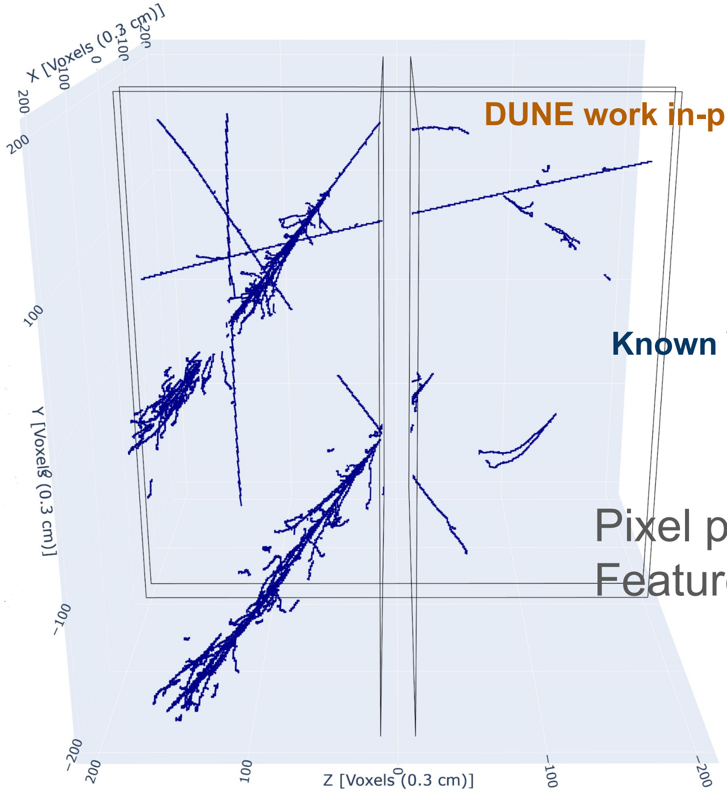
LArDRIP LArTPC Dead Region Inference Project

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





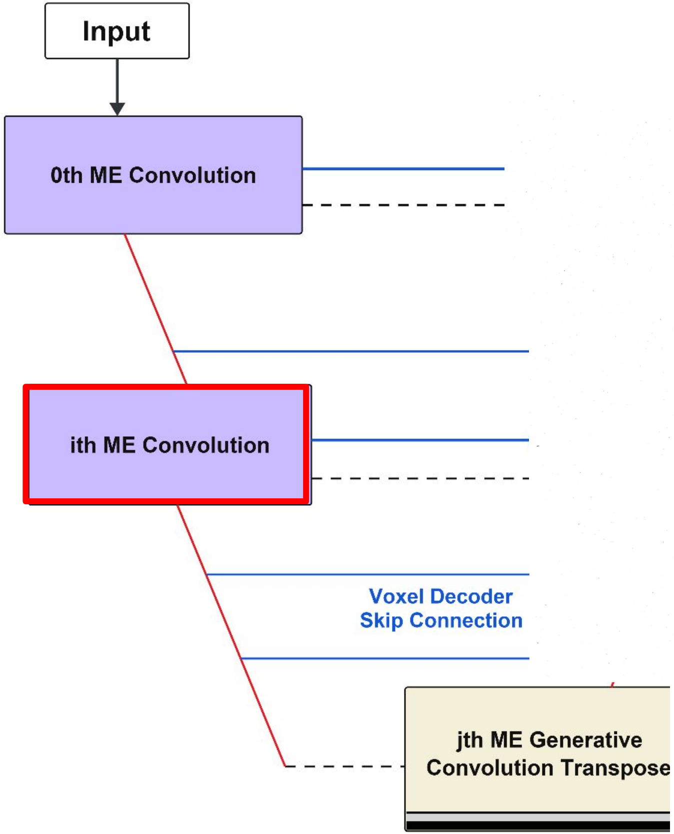
x1

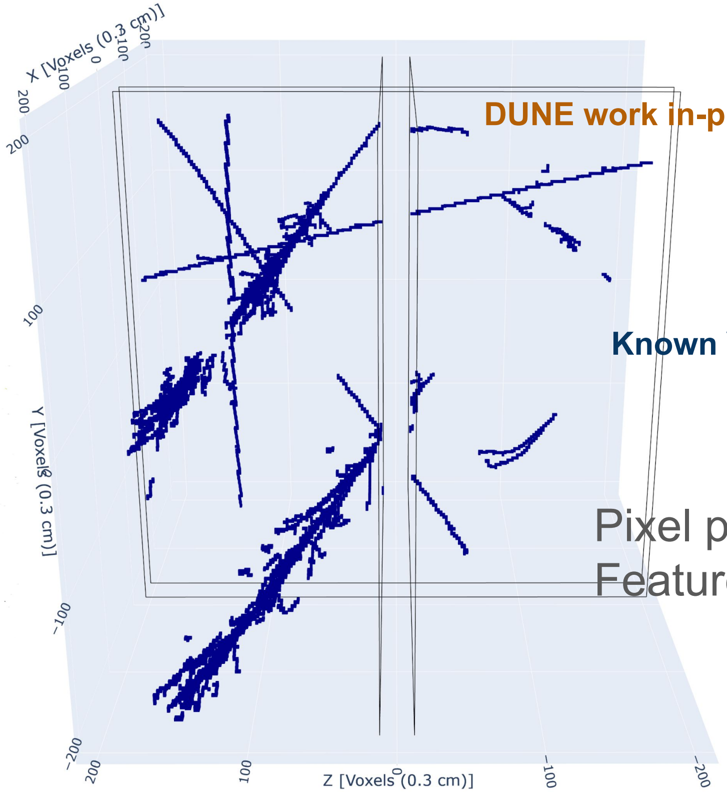
DUNE work in-progress

Known Voxels

Pixel pitch: 0.3 cm
Feature size: 22

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





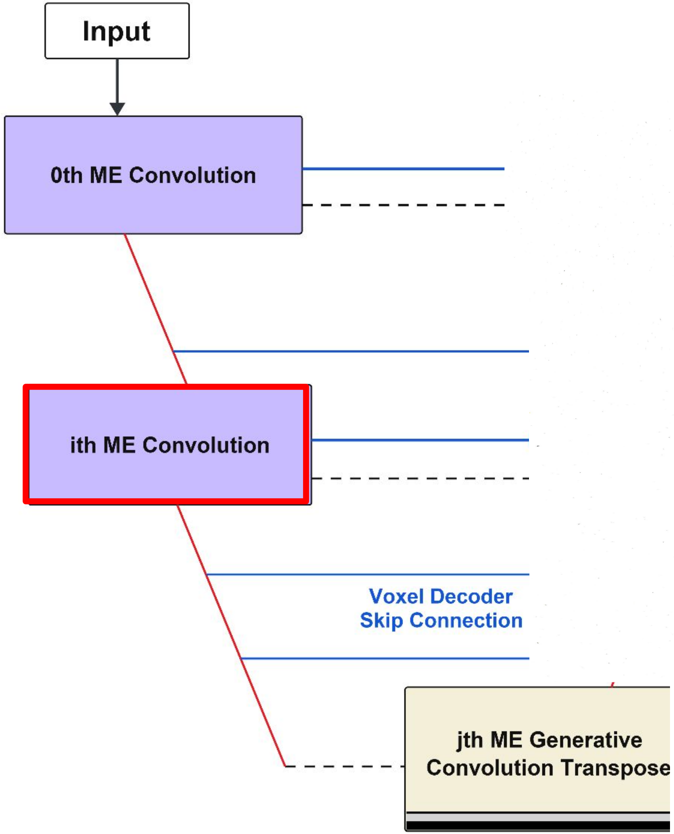
x2

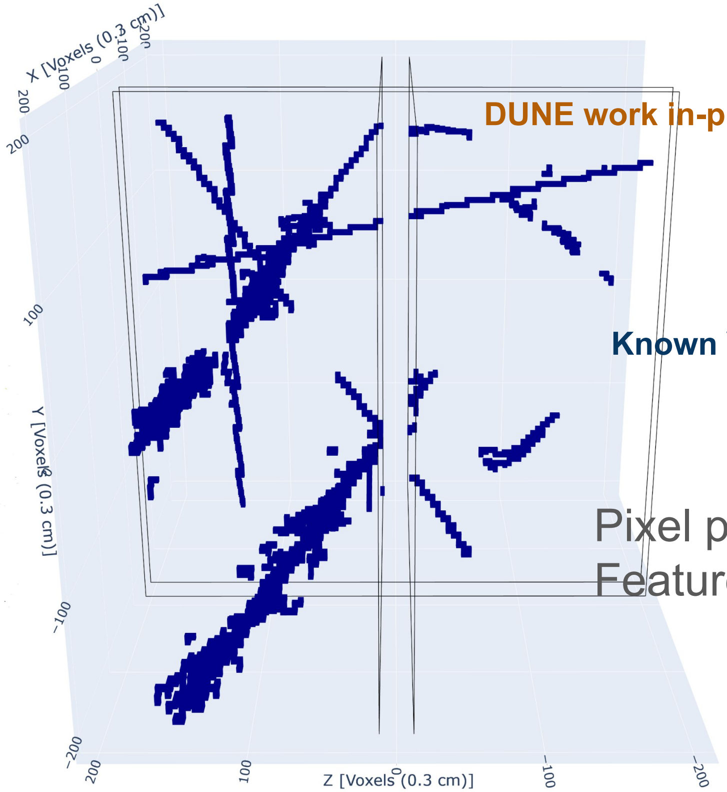
DUNE work in-progress

Known Voxels

Pixel pitch: 0.6 cm
Feature size: 58

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





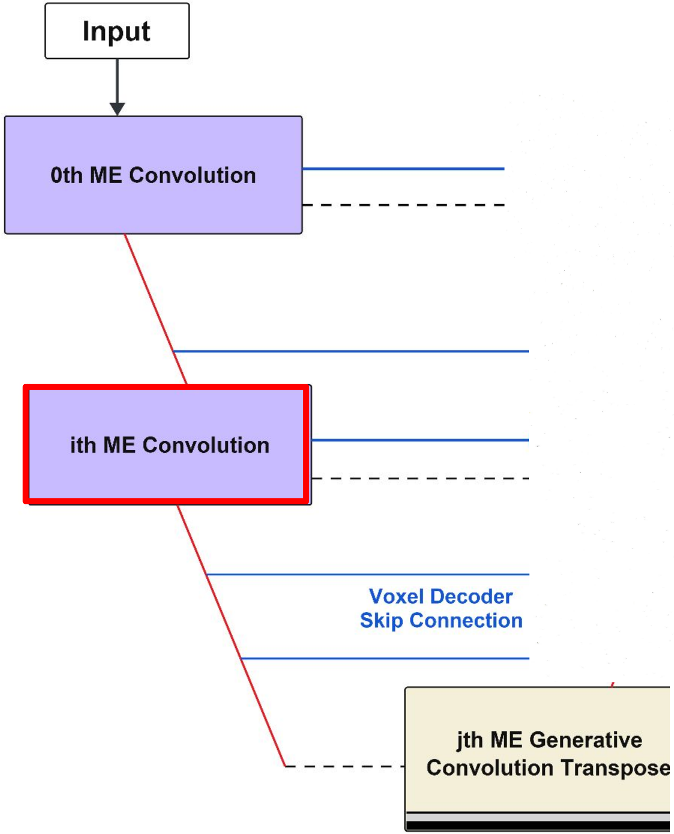
x4

DUNE work in-progress

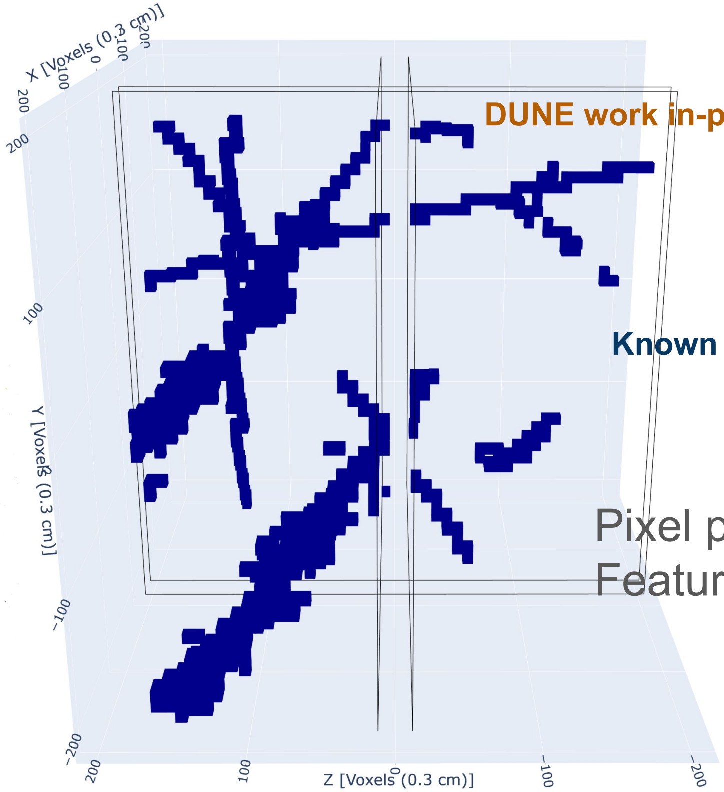
Known Voxels

Pixel pitch: 1.2 cm
Feature size: 84

LArDRIP: A Multi-Scale Generative Sparse 3D UNet



x8

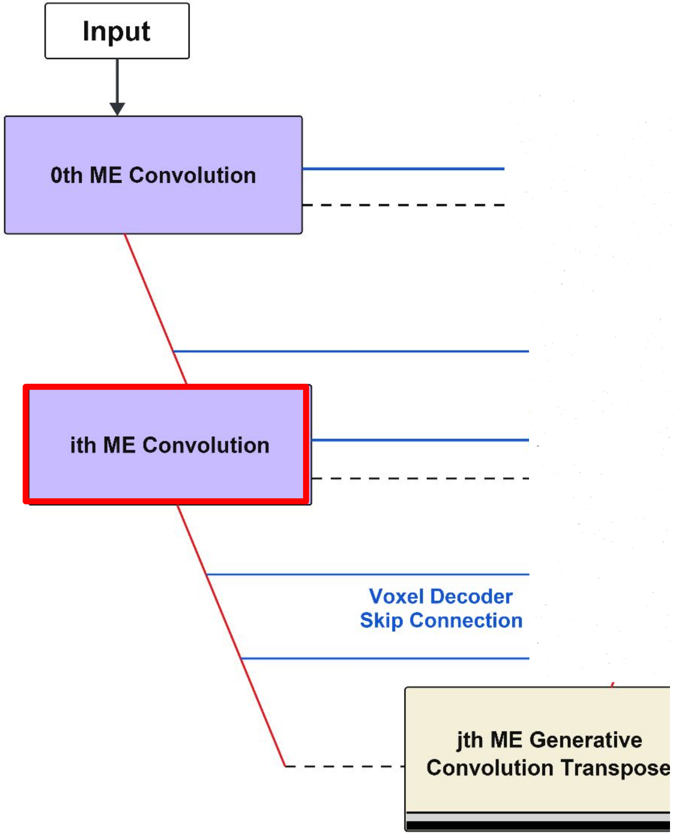


DUNE work in-progress

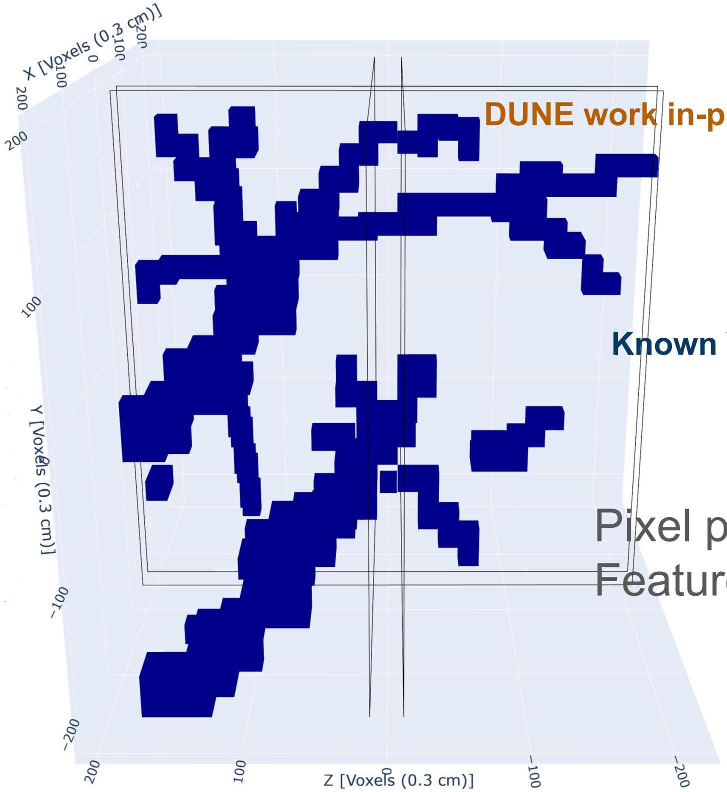
Known Voxels

Pixel pitch: 2.4 cm
Feature size: 158

LArDRIP: A Multi-Scale Generative Sparse 3D UNet



x16

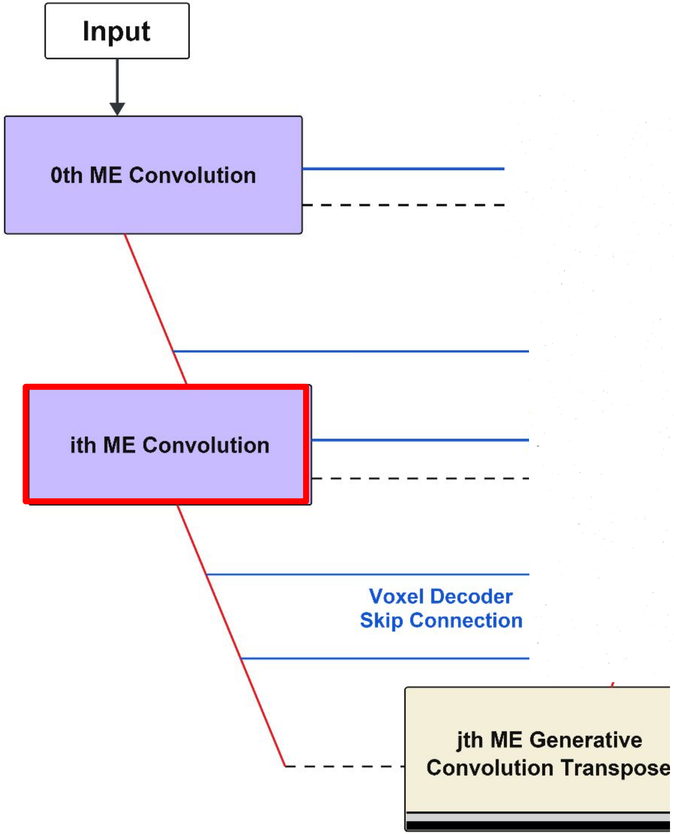


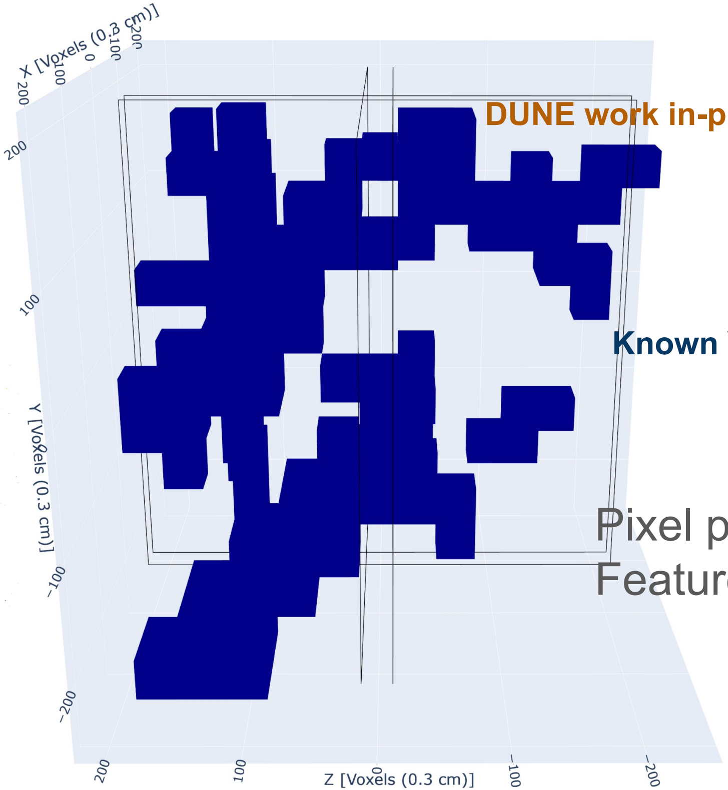
DUNE work in-progress

Known Voxels

Pixel pitch: 4.8 cm
Feature size: 312

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





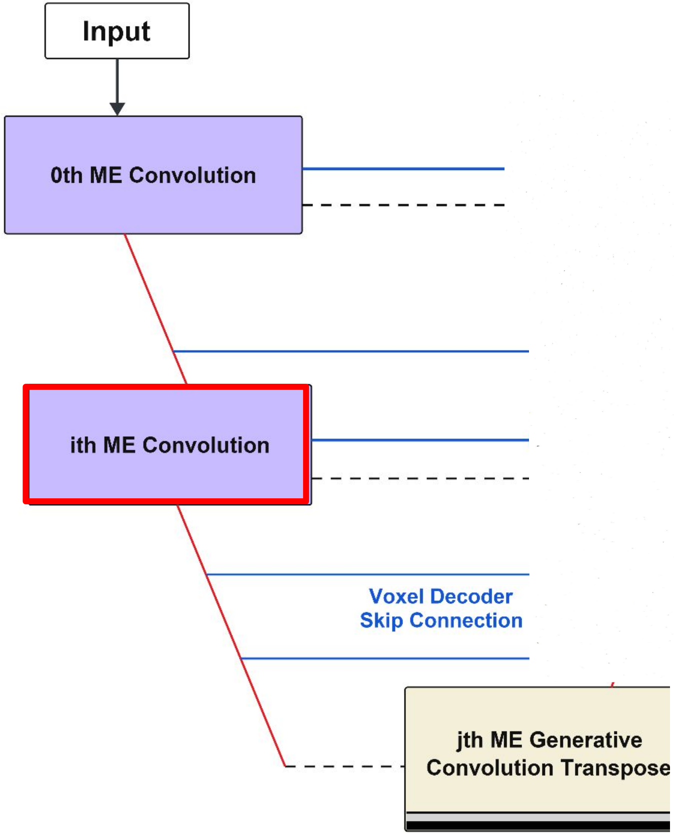
x32

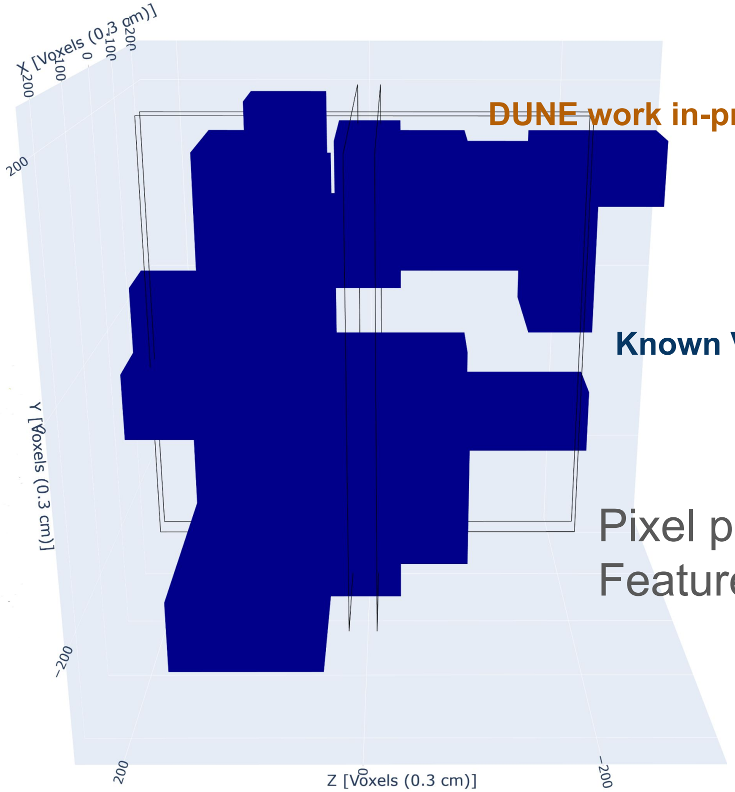
DUNE work in-progress

Known Voxels

Pixel pitch: 9.6 cm
Feature size: 676

LArDRIP: A Multi-Scale Generative Sparse 3D UNet

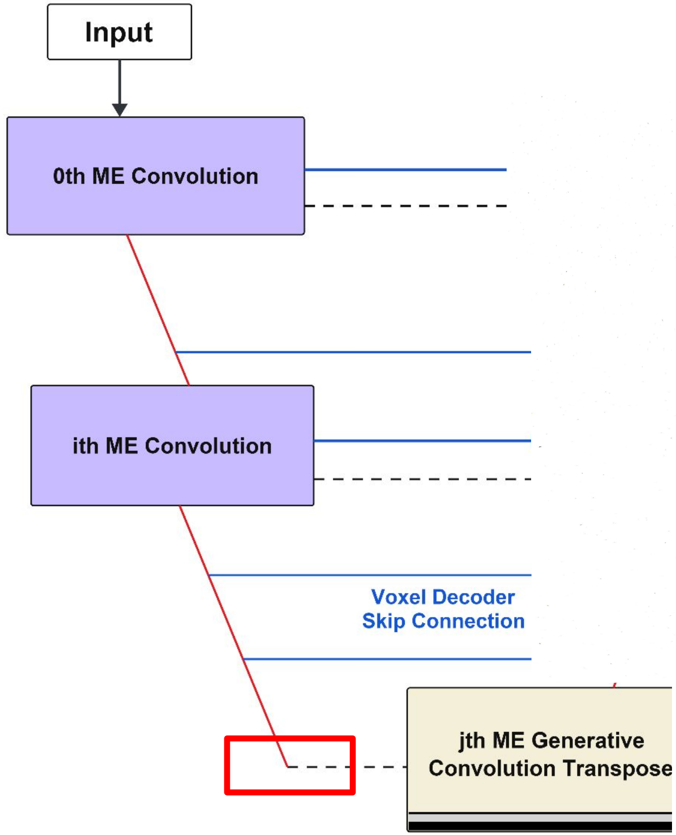




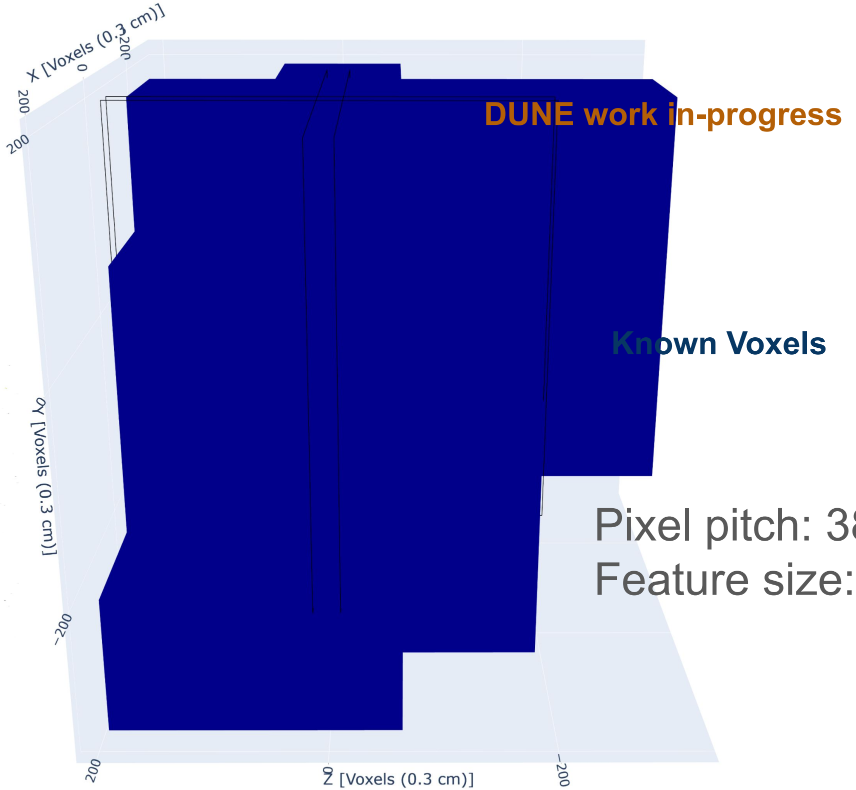
x64

Pixel pitch: 19.2 cm
Feature size: 1024

LArDRIP: A Multi-Scale Generative Sparse 3D UNet

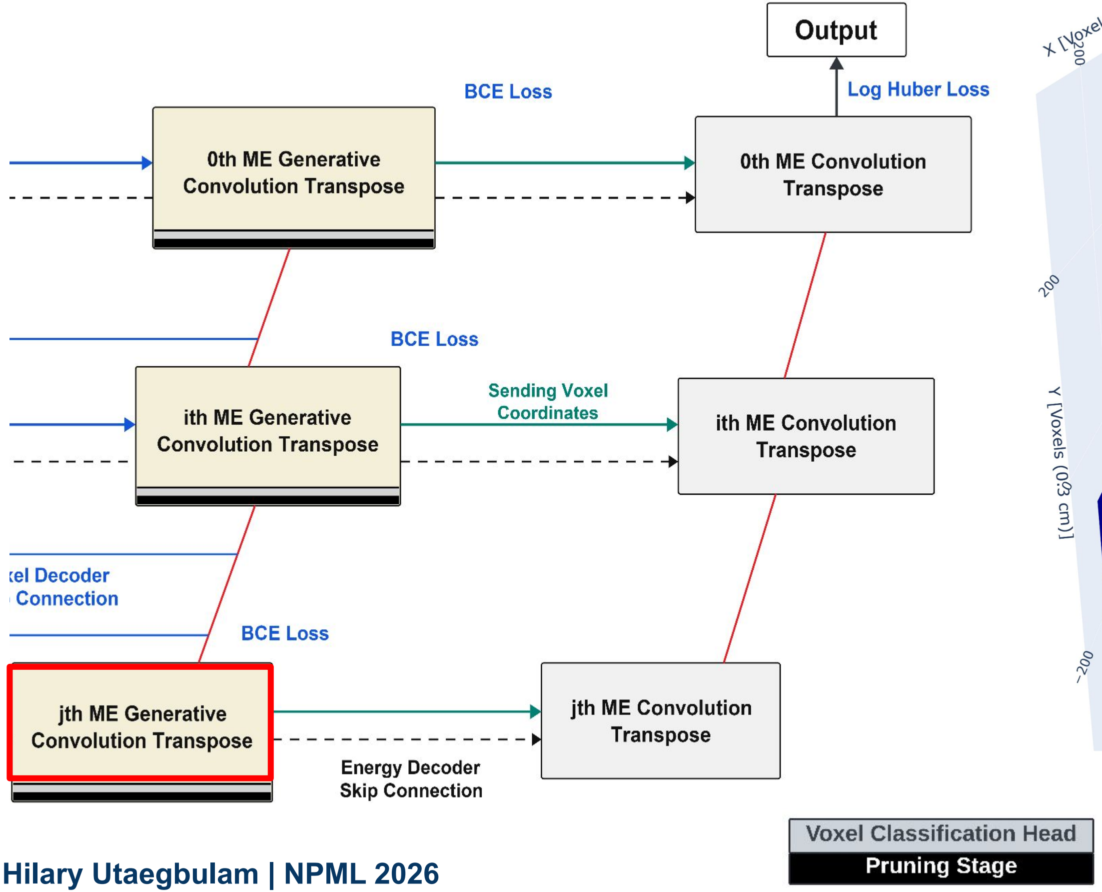


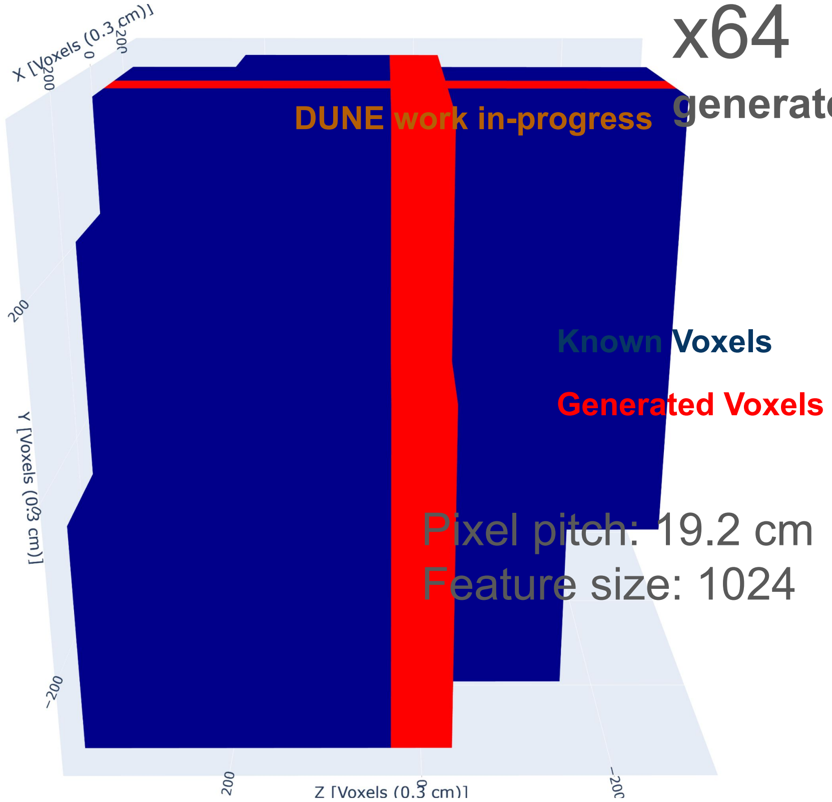
x128



Pixel pitch: 38.4 cm
Feature size: 1576

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





x64 generate

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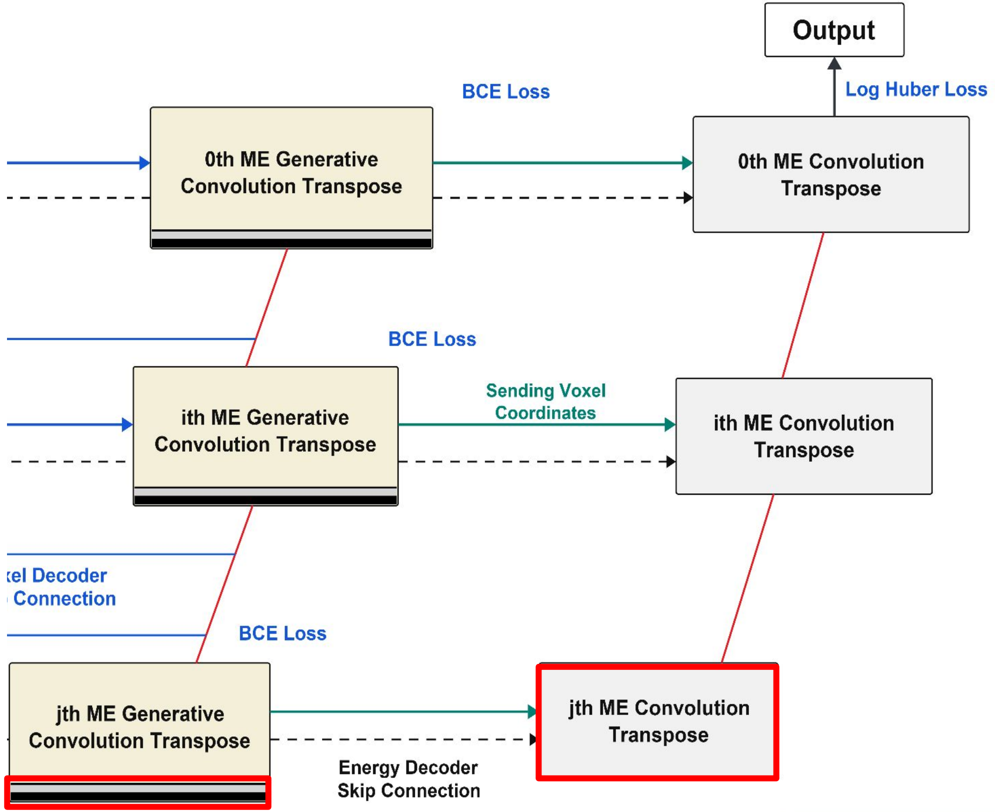
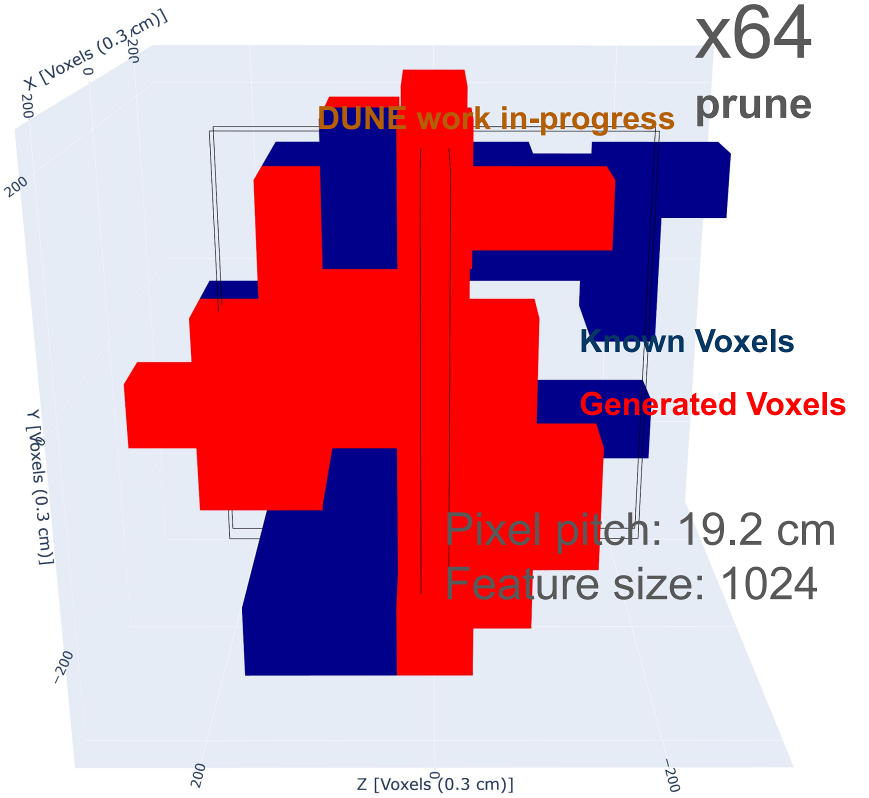
LArDRIP

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





Voxel Classification Head
Pruning Stage

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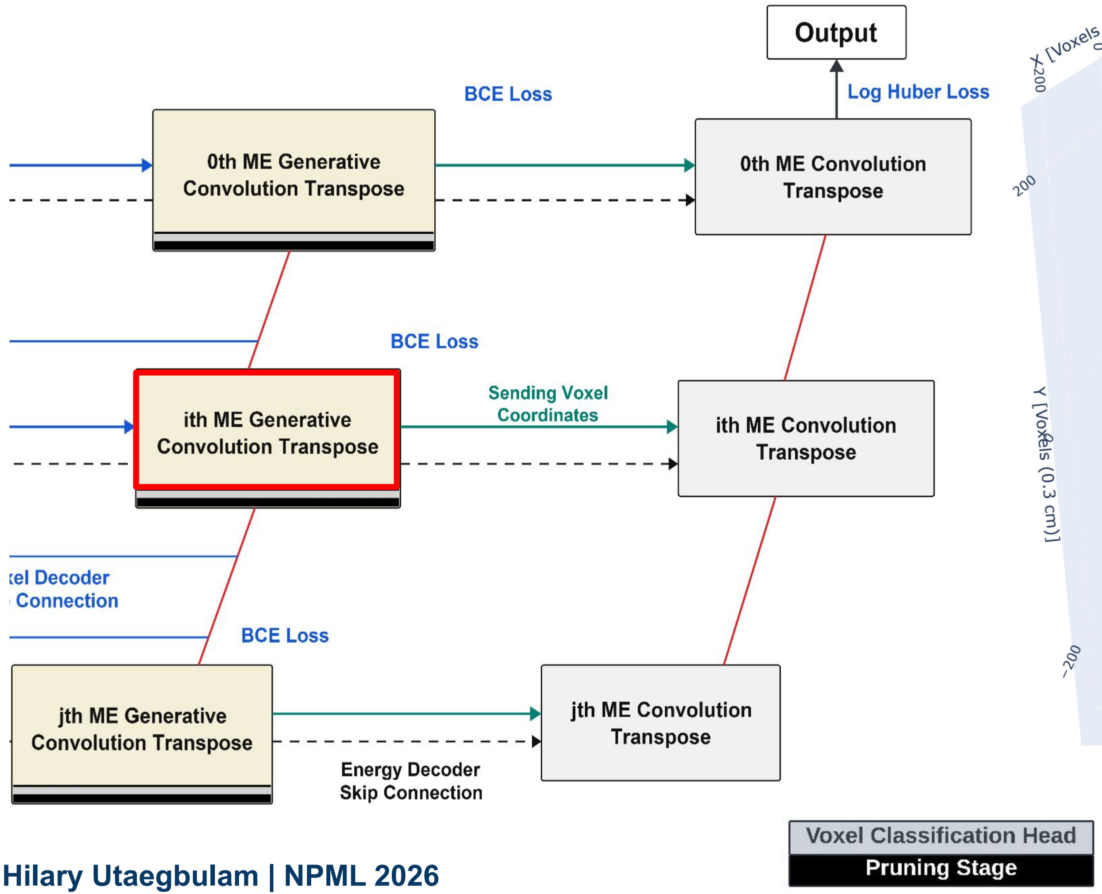
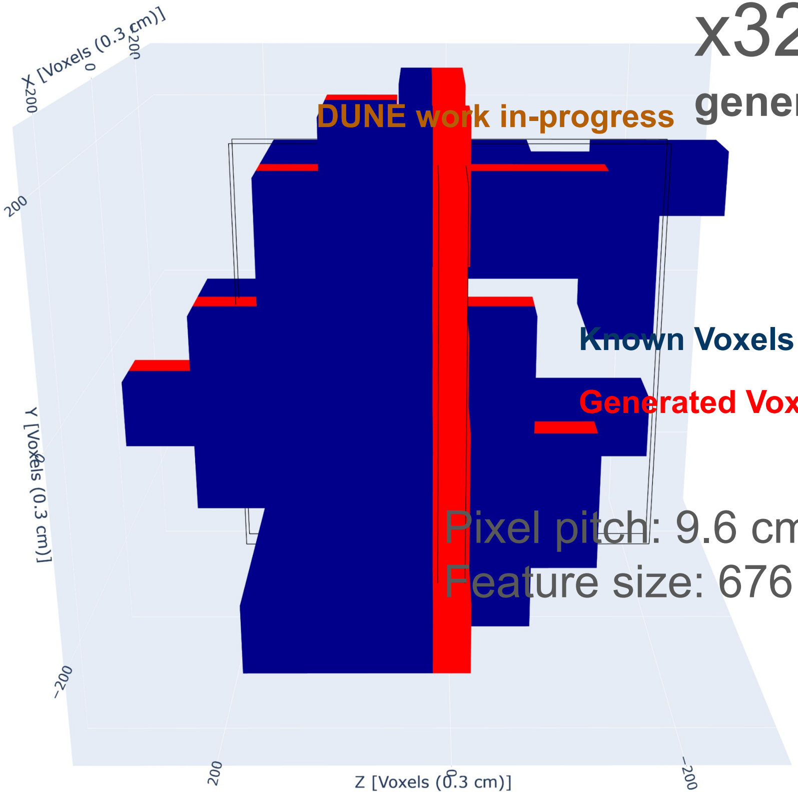
LArDRIP

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





x32
generate

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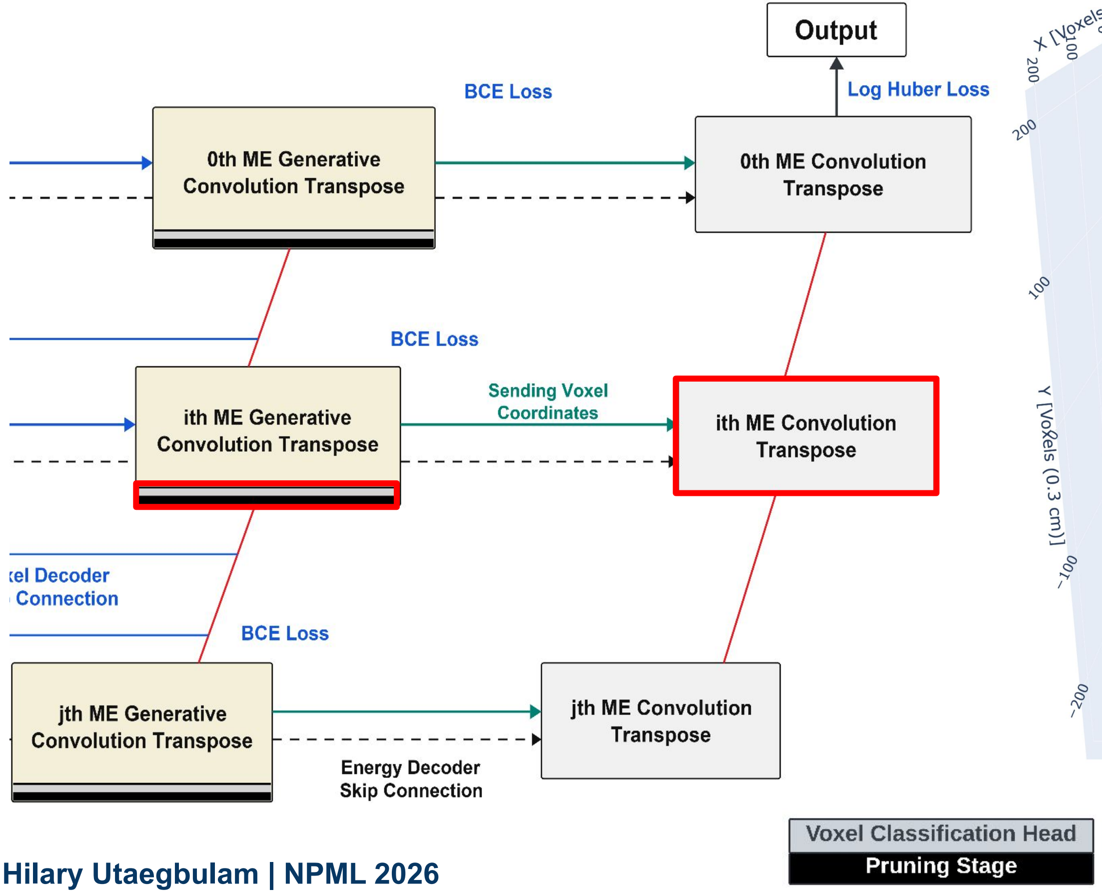
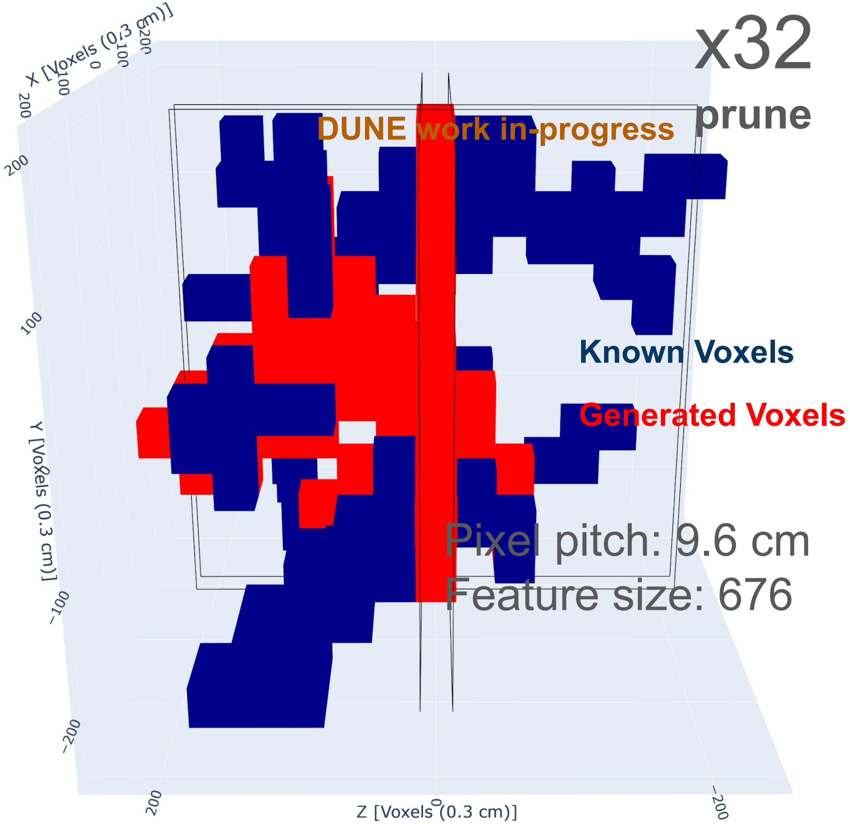
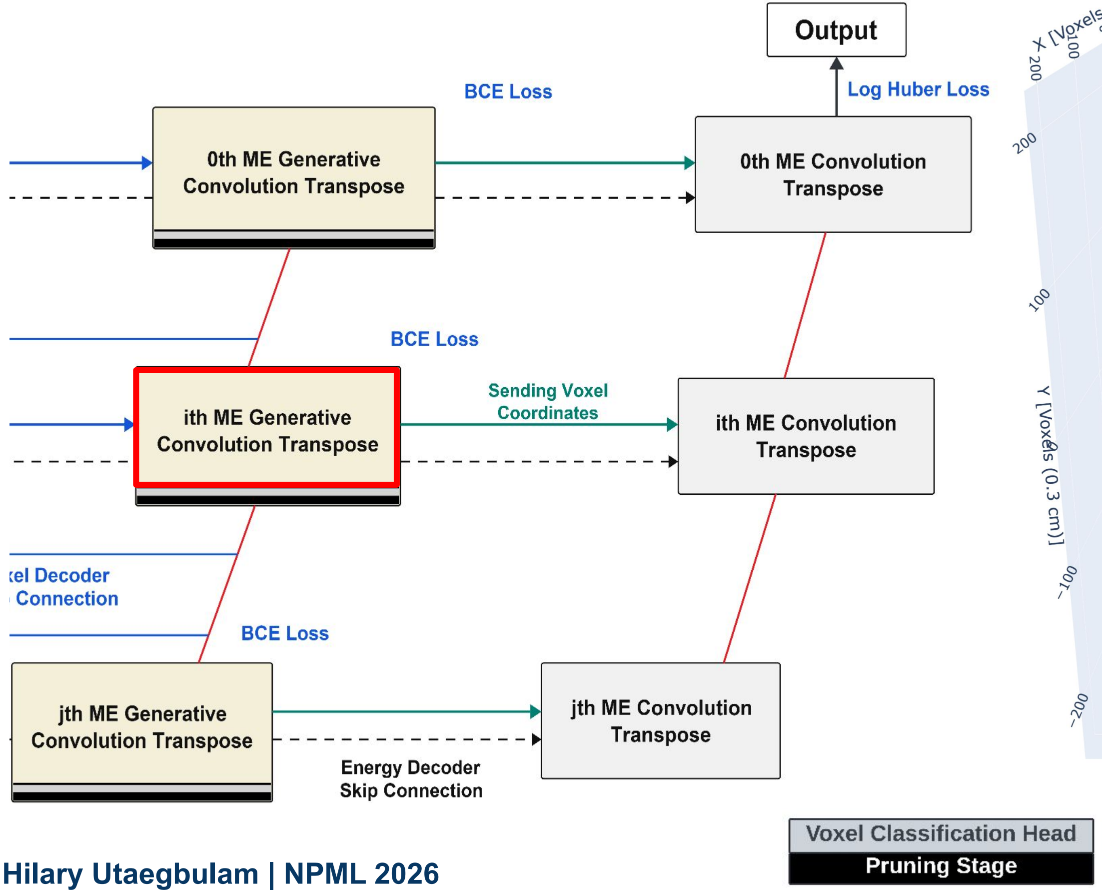
LArDRIP
LArTPC Dead Region Inference Project

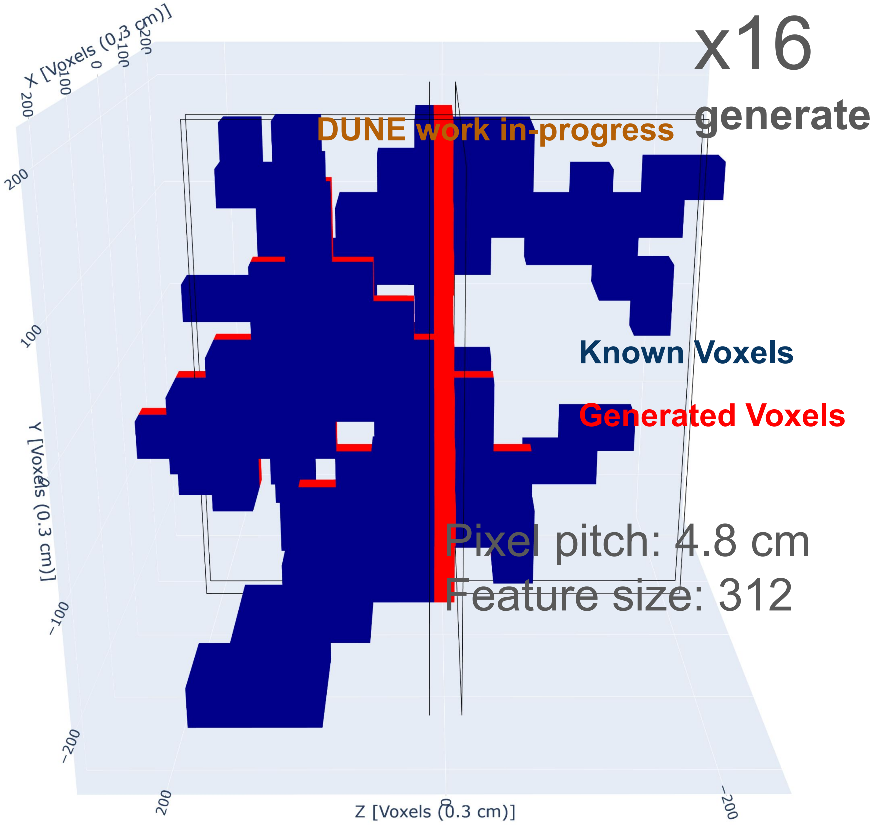
LArDRIP: A Multi-Scale Generative Sparse 3D UNet





LArDRIP: A Multi-Scale Generative Sparse 3D UNet





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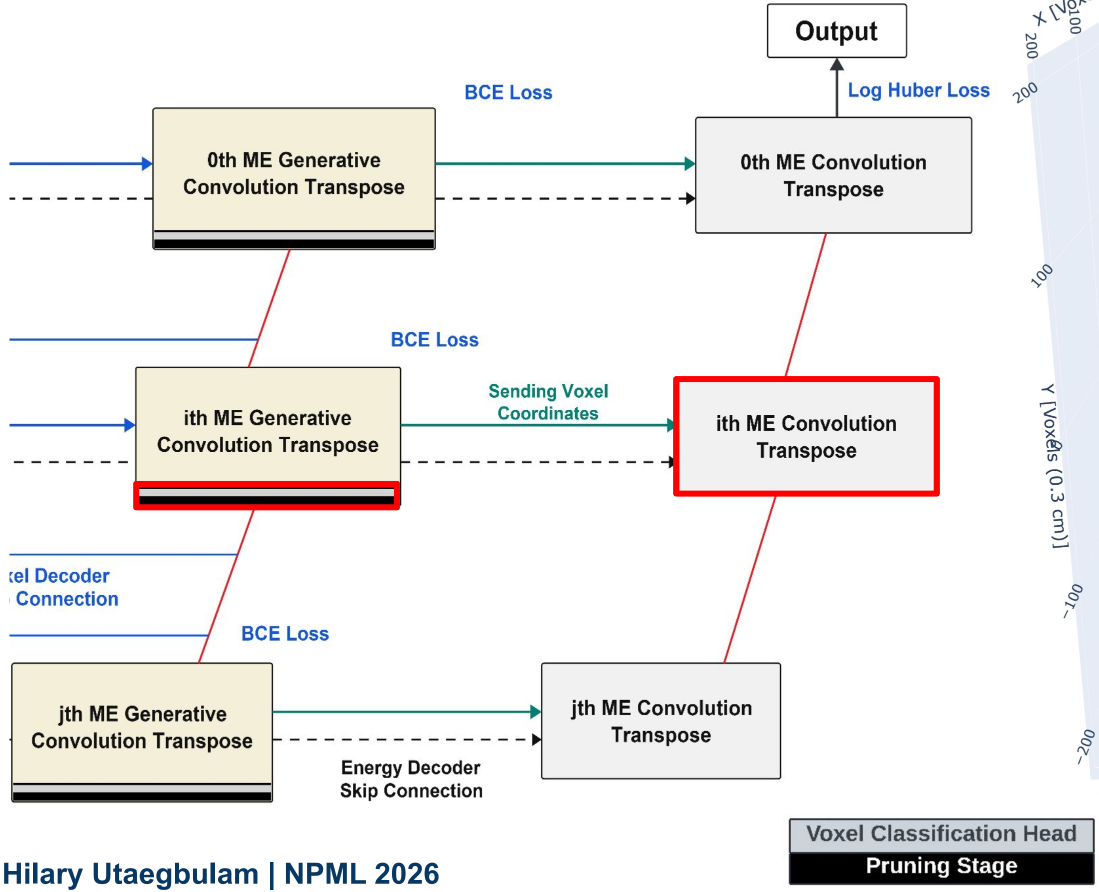
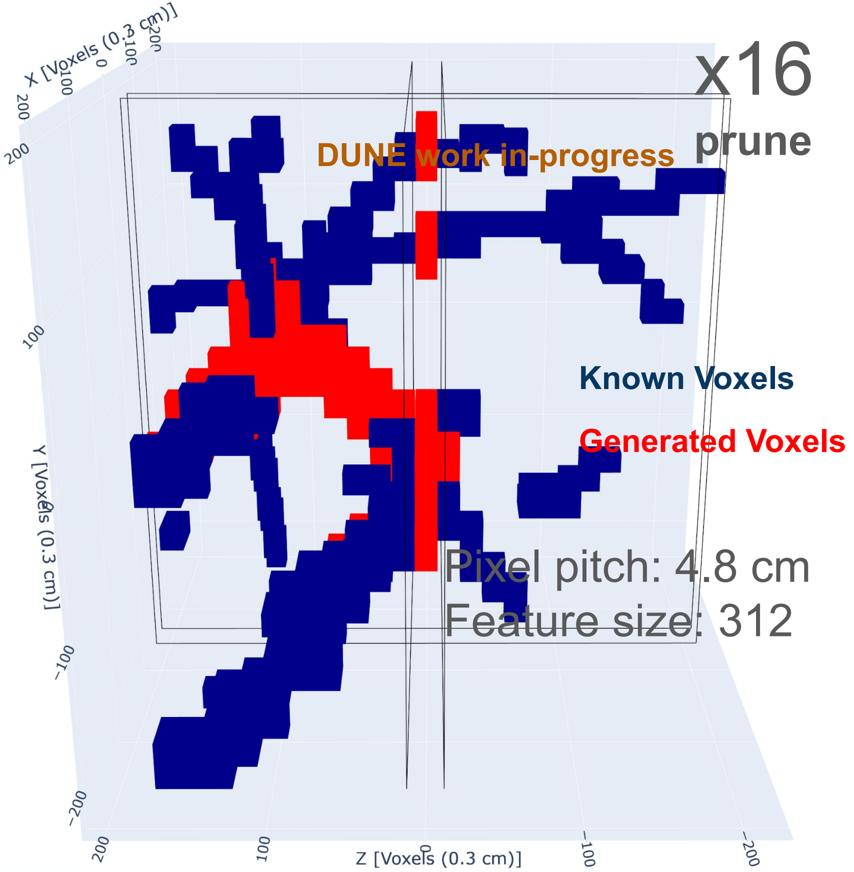
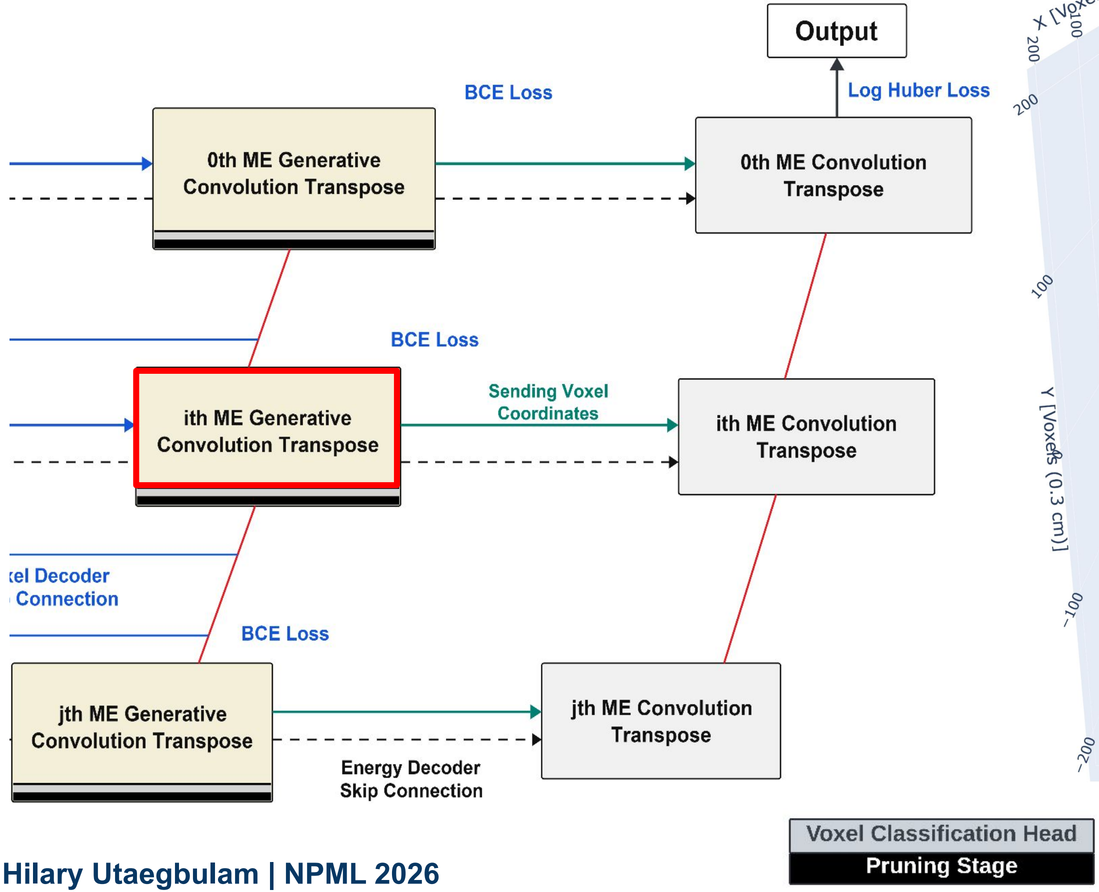


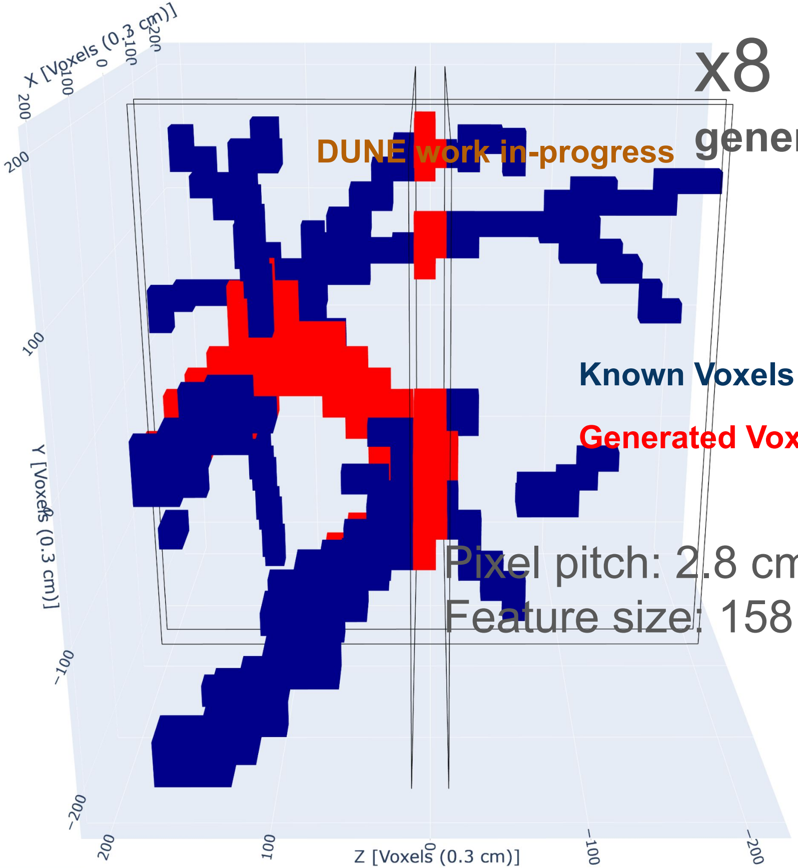


x16
prune

Pixel pitch: 4.8 cm
Feature size: 312

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





x8 generate

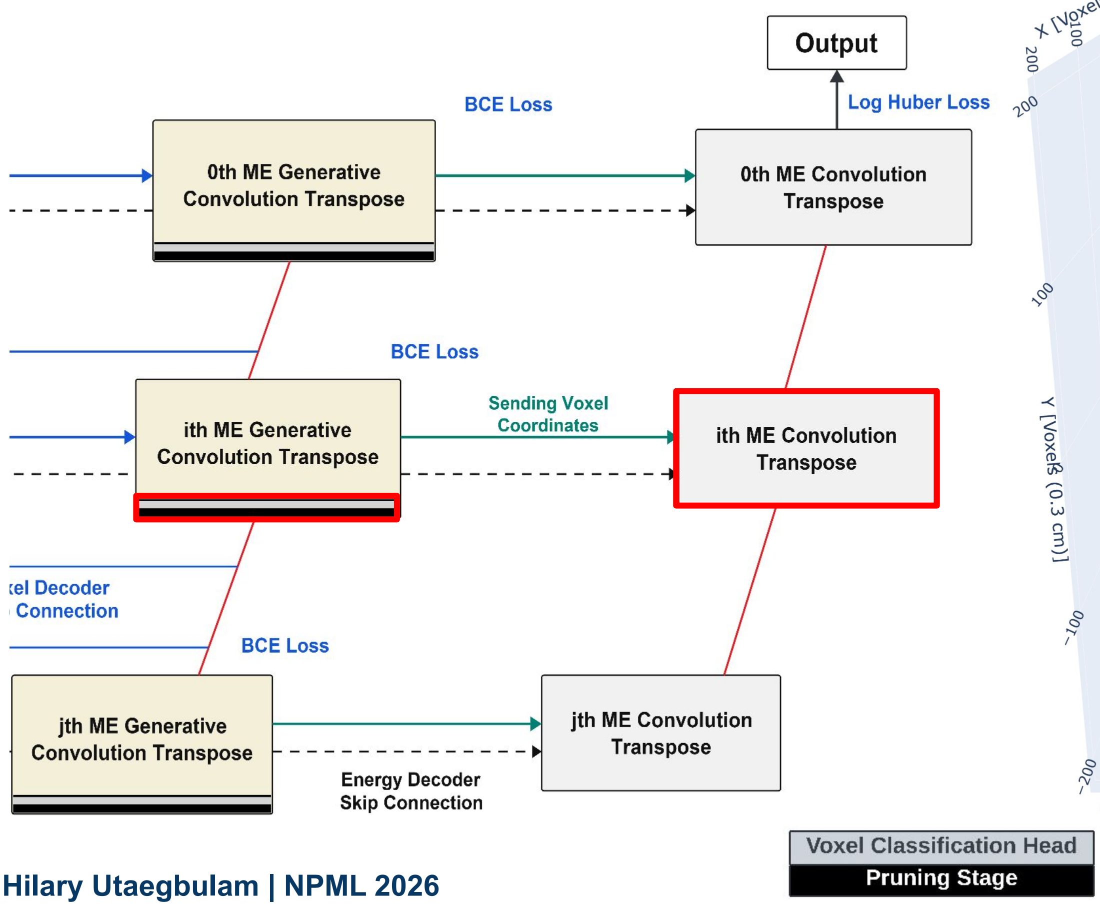
Known Voxels

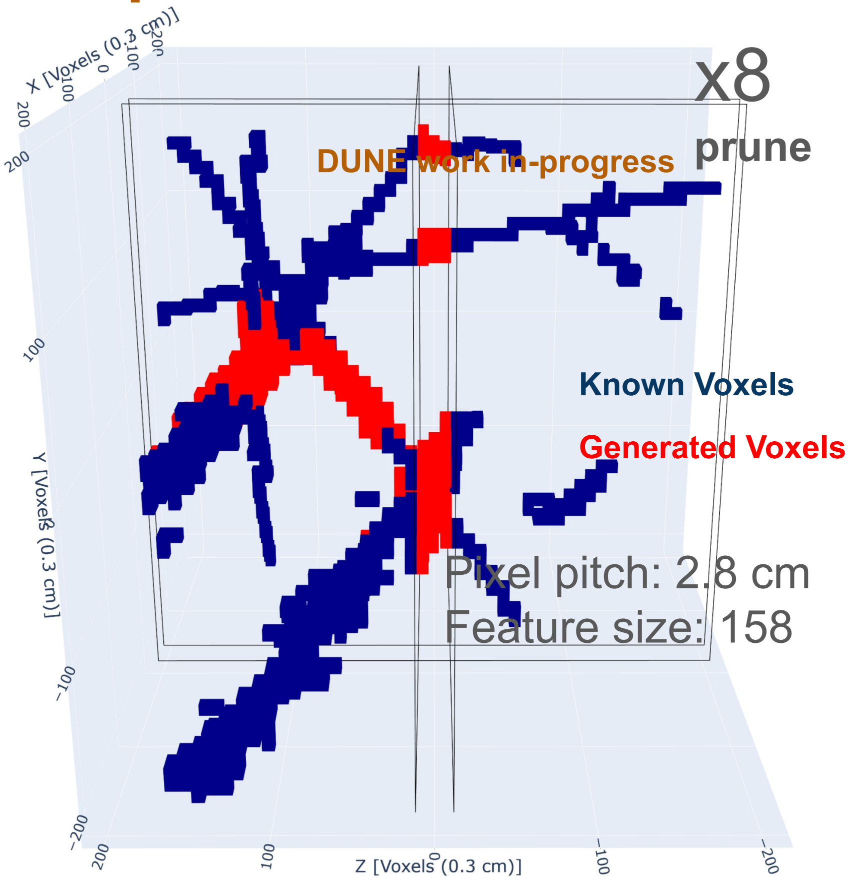
Generated Voxels

Pixel pitch: 2.8 cm

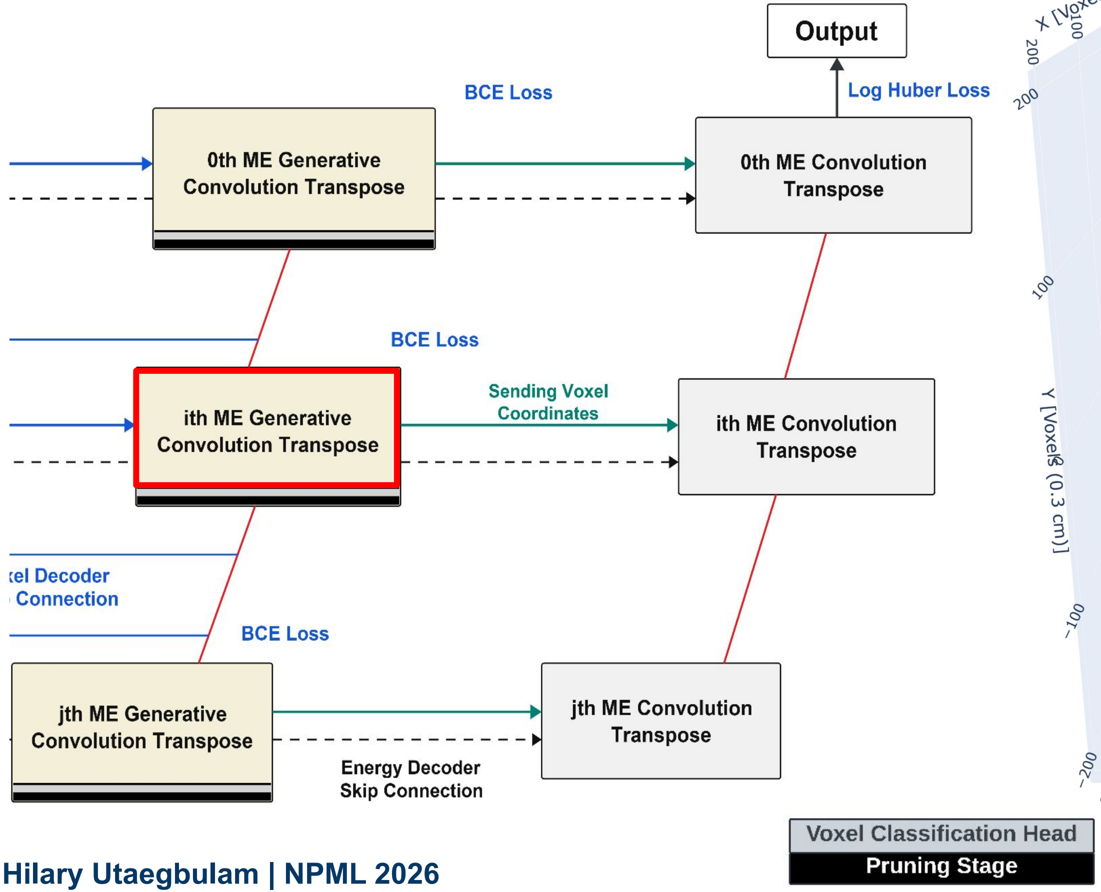
Feature size: 158

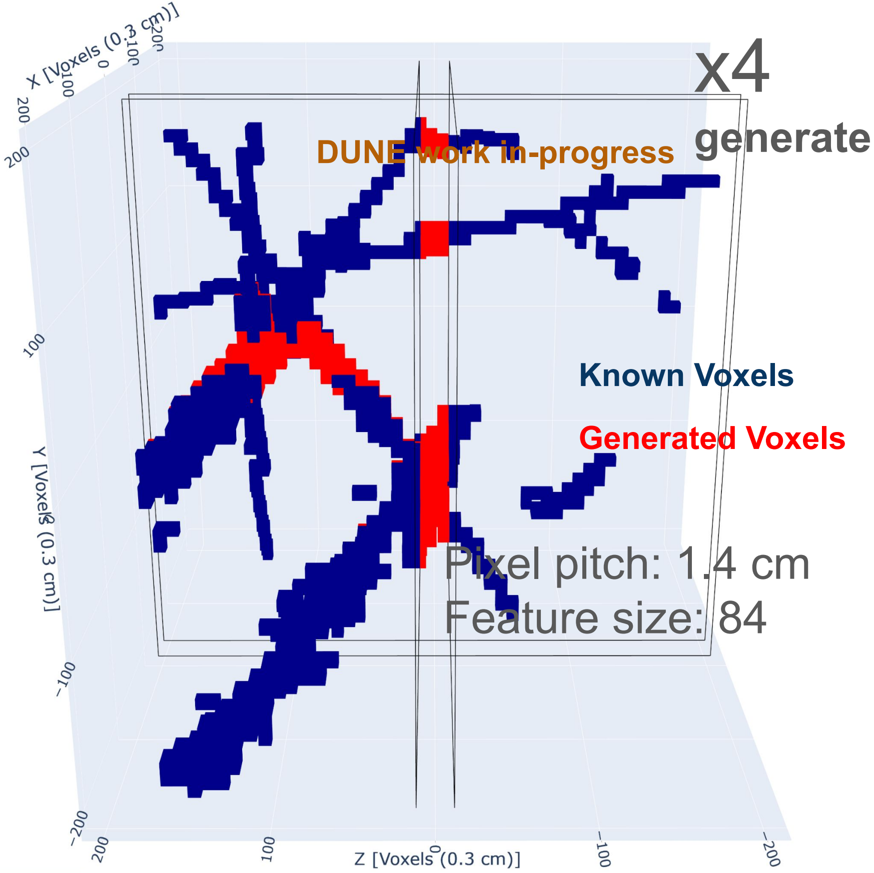
LArDRIP: A Multi-Scale Generative Sparse 3D UNet



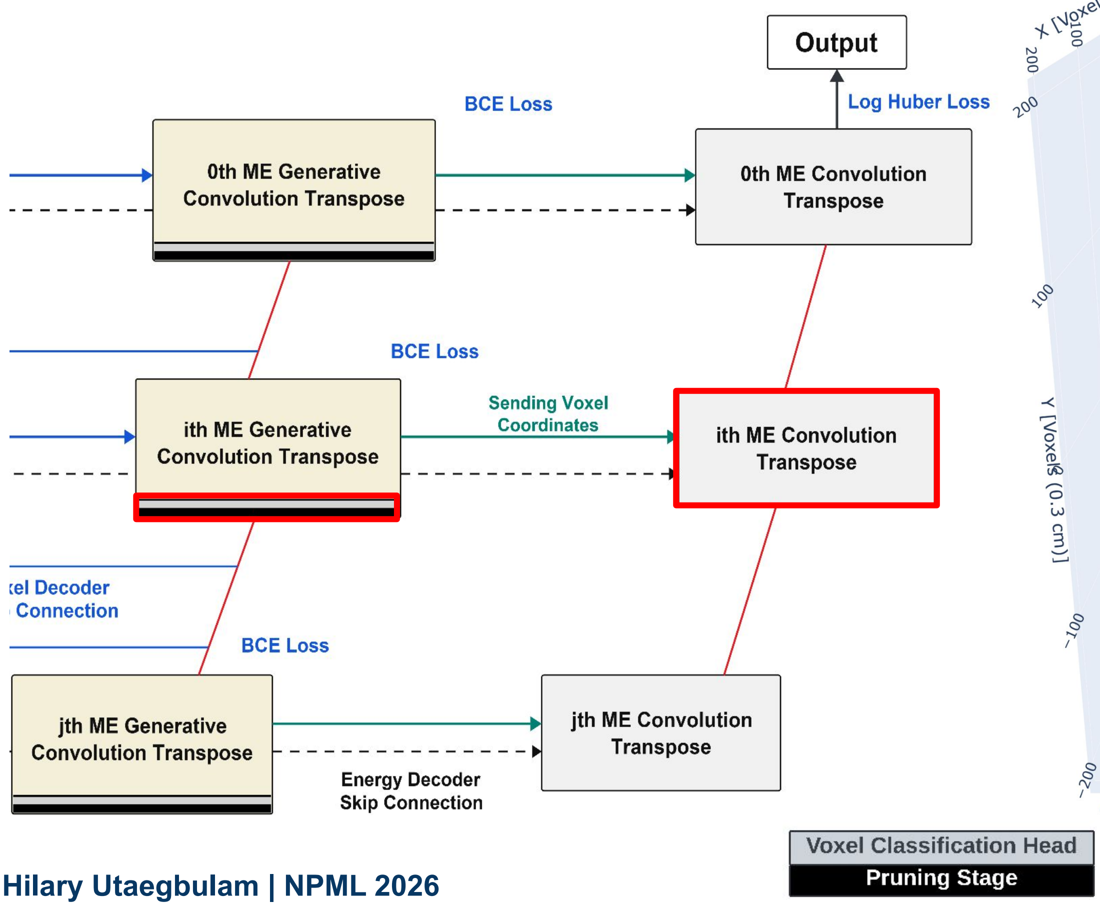


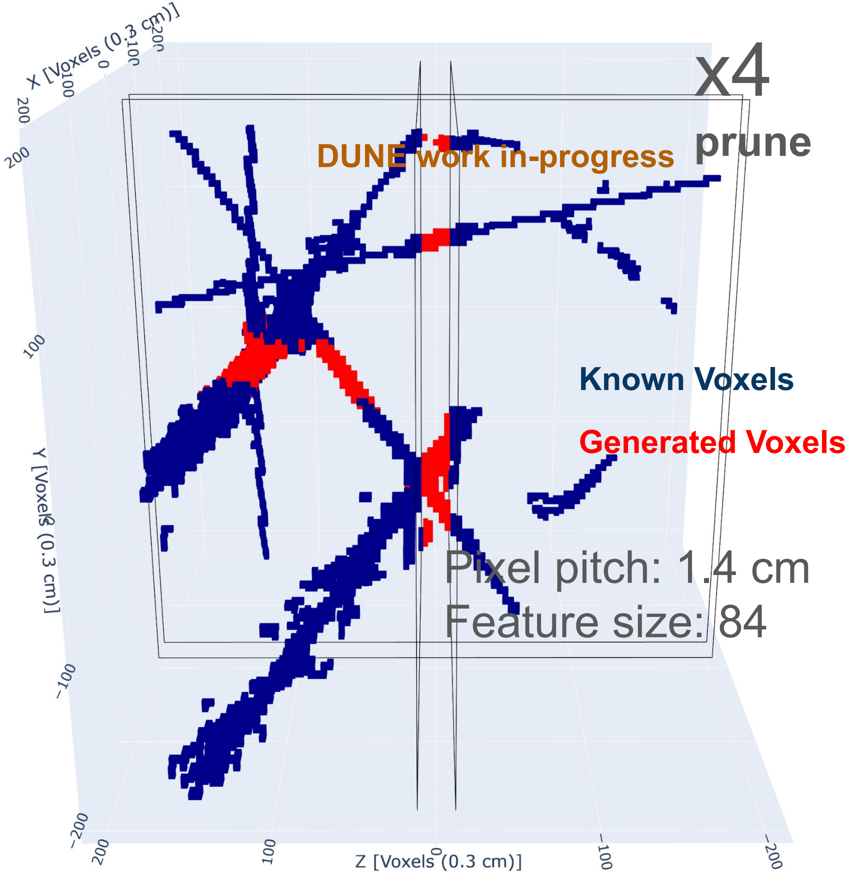
LArDRIP: A Multi-Scale Generative Sparse 3D UNet



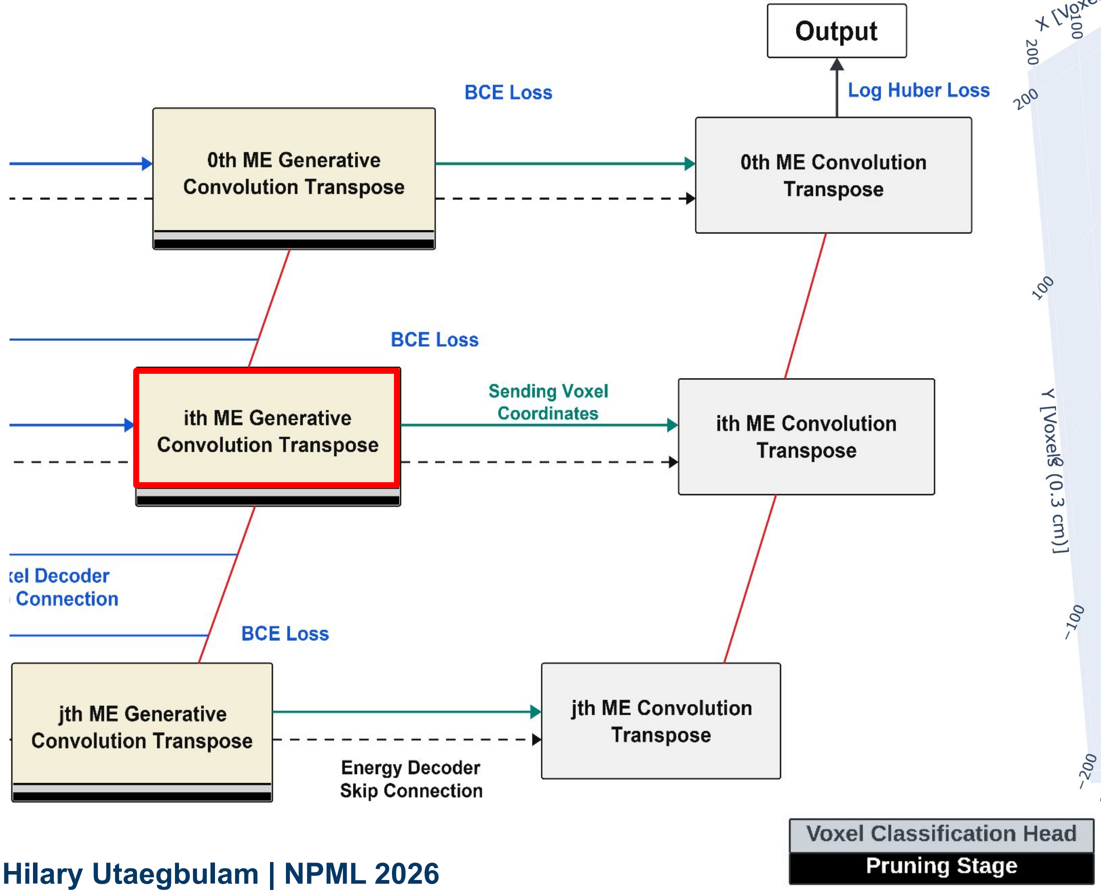


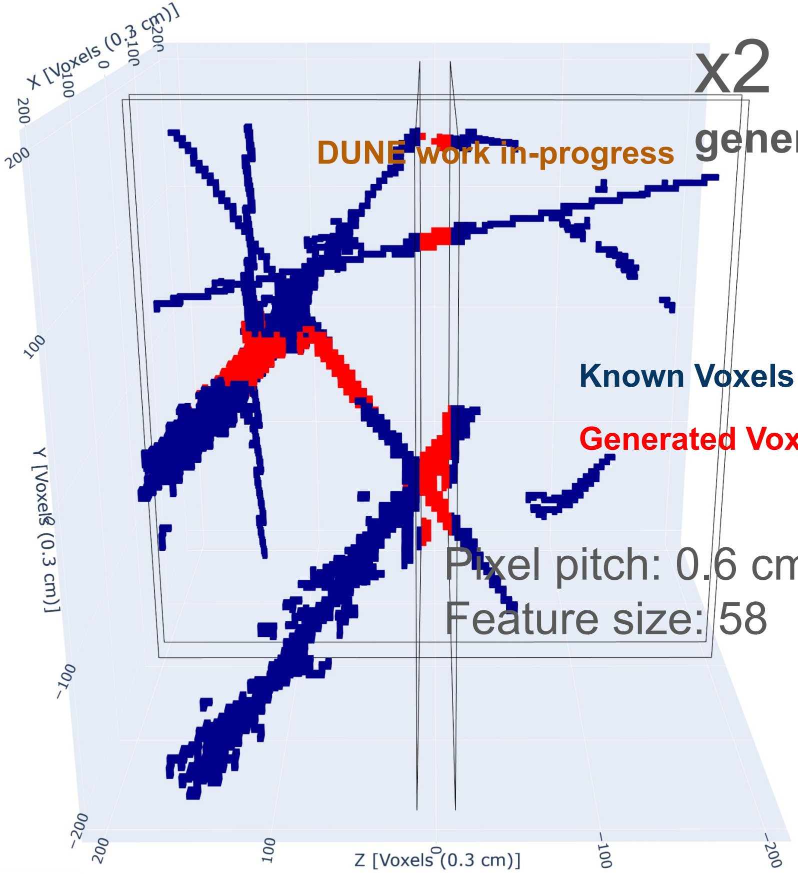
LArDRIP: A Multi-Scale Generative Sparse 3D UNet





LArDRIP: A Multi-Scale Generative Sparse 3D UNet





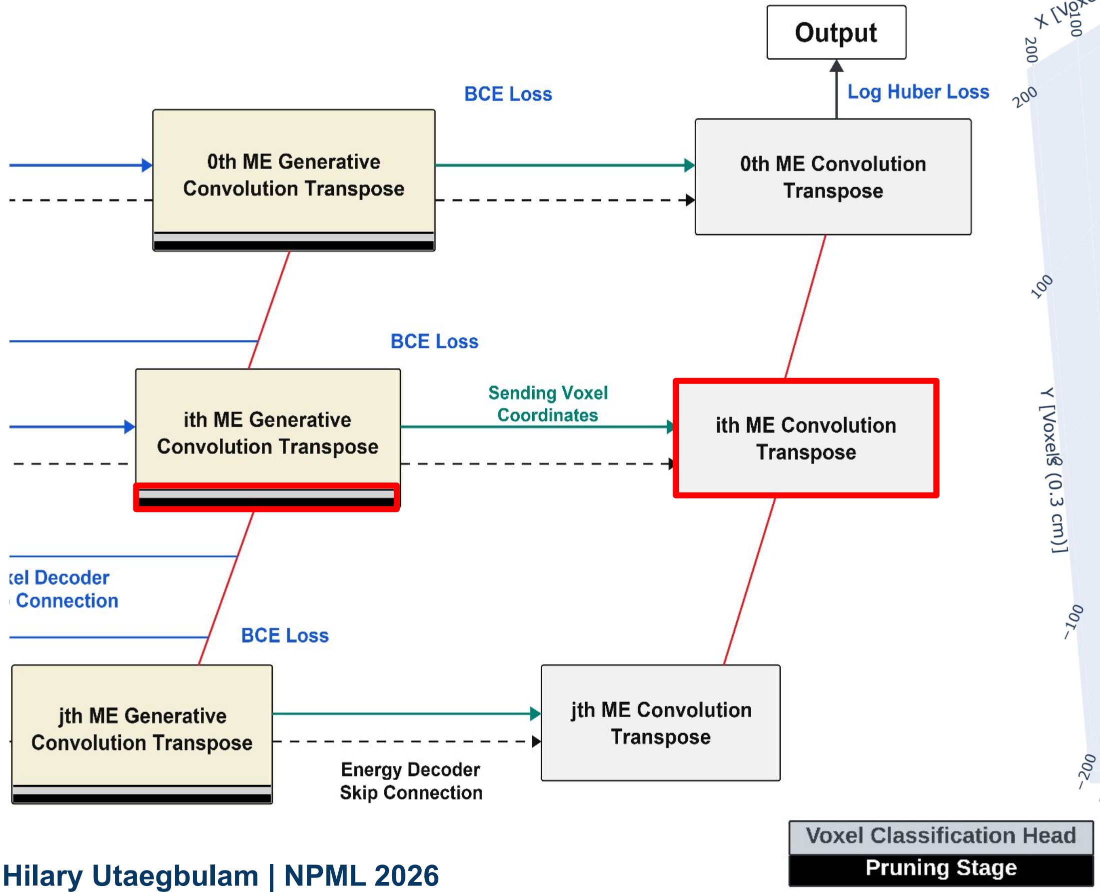
x2 generate

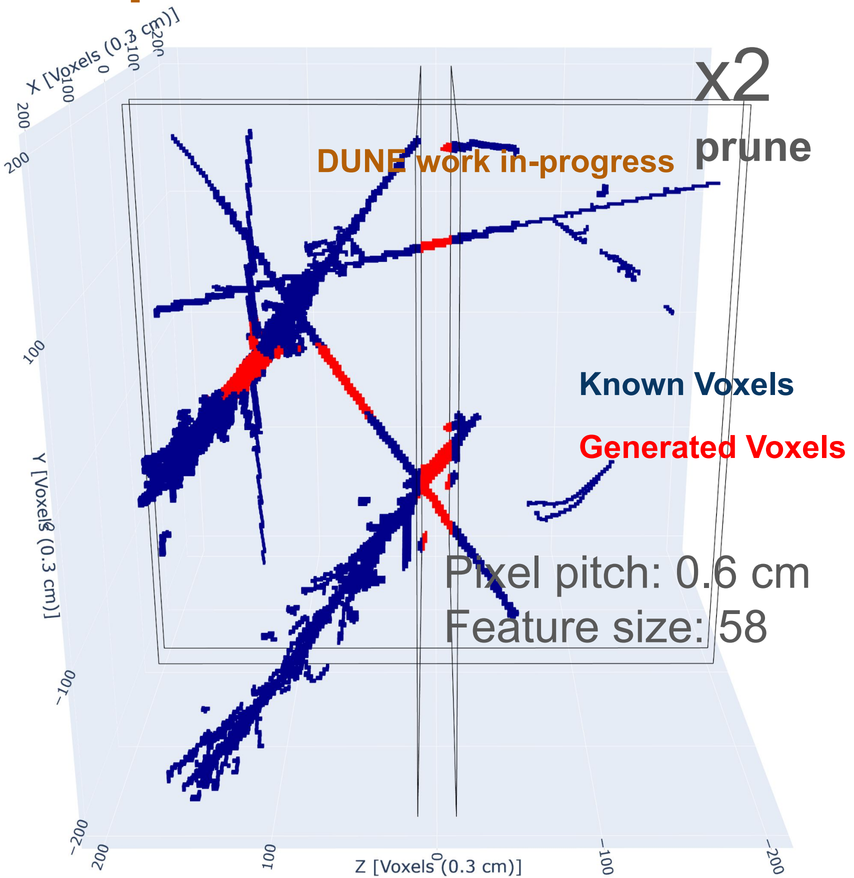
Known Voxels

Generated Voxels

Pixel pitch: 0.6 cm
Feature size: 58

LArDRIP: A Multi-Scale Generative Sparse 3D UNet





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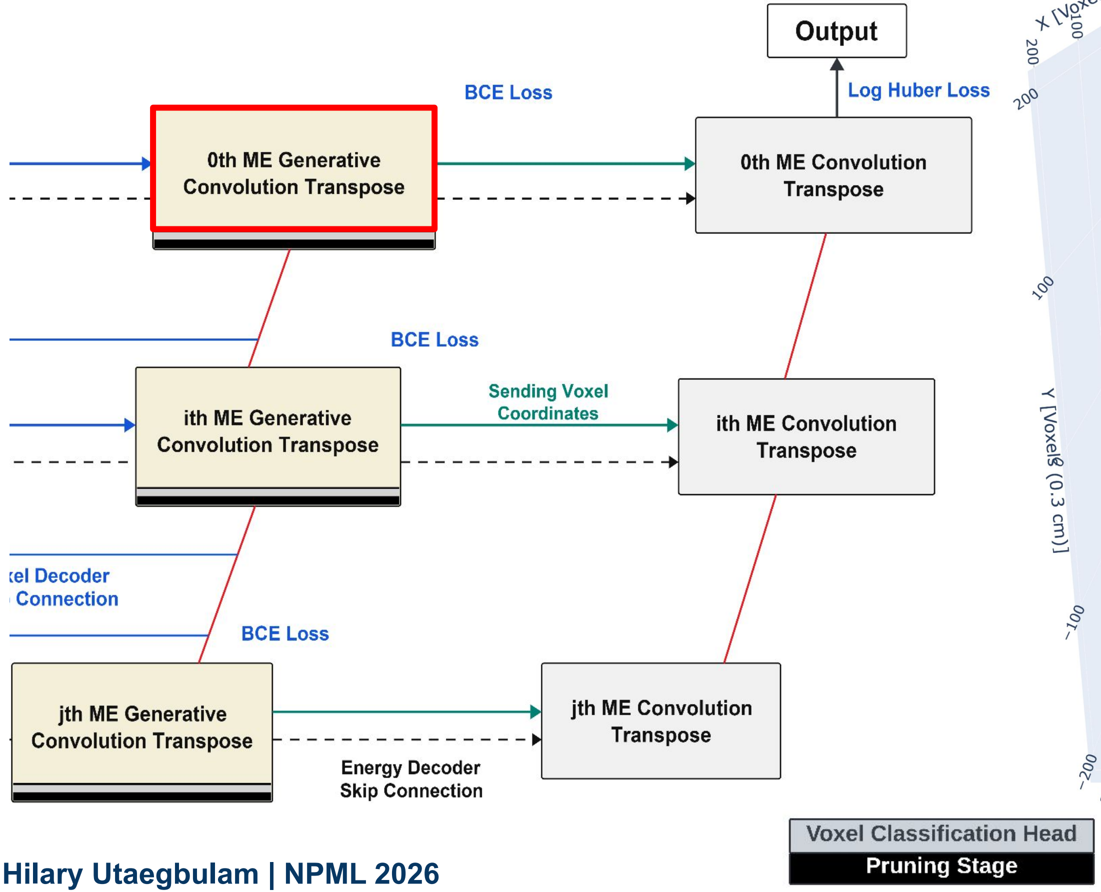
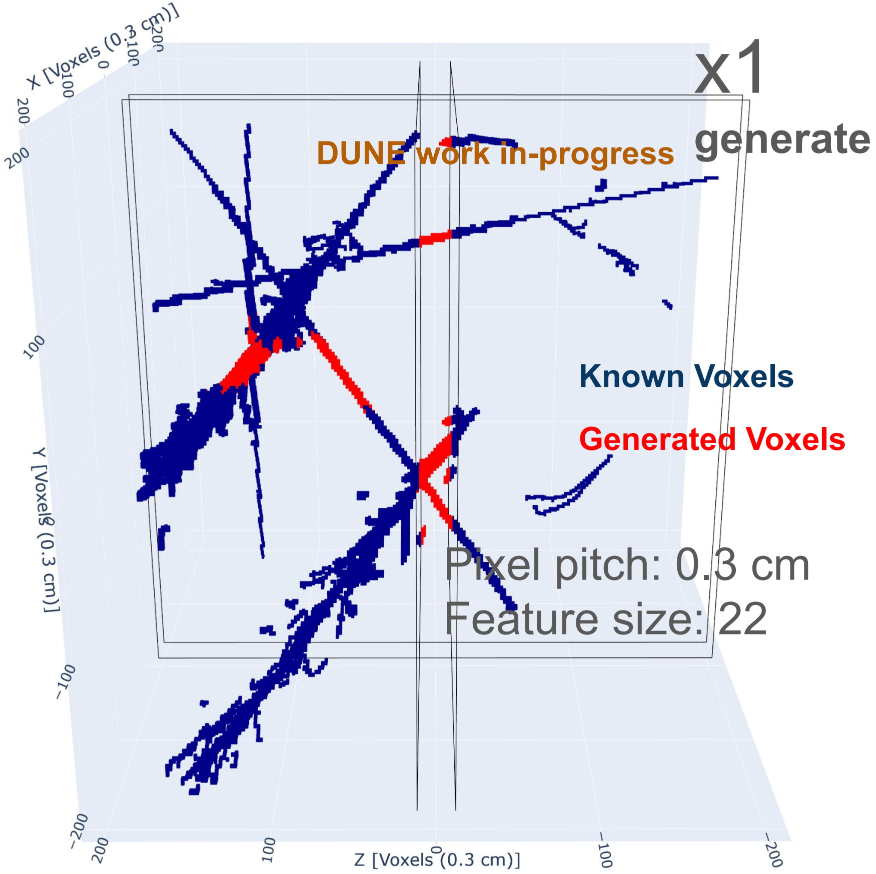
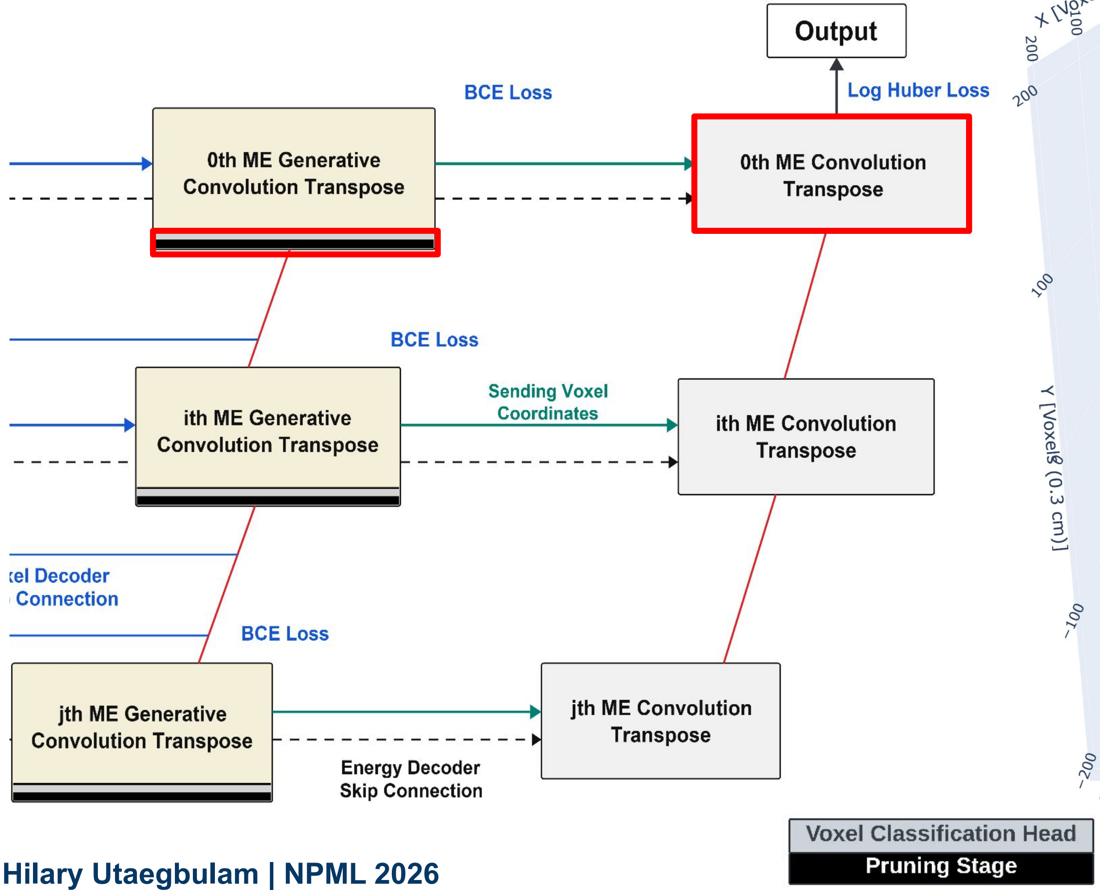
LArDRIP
LArTPC Dead Region Inference Project

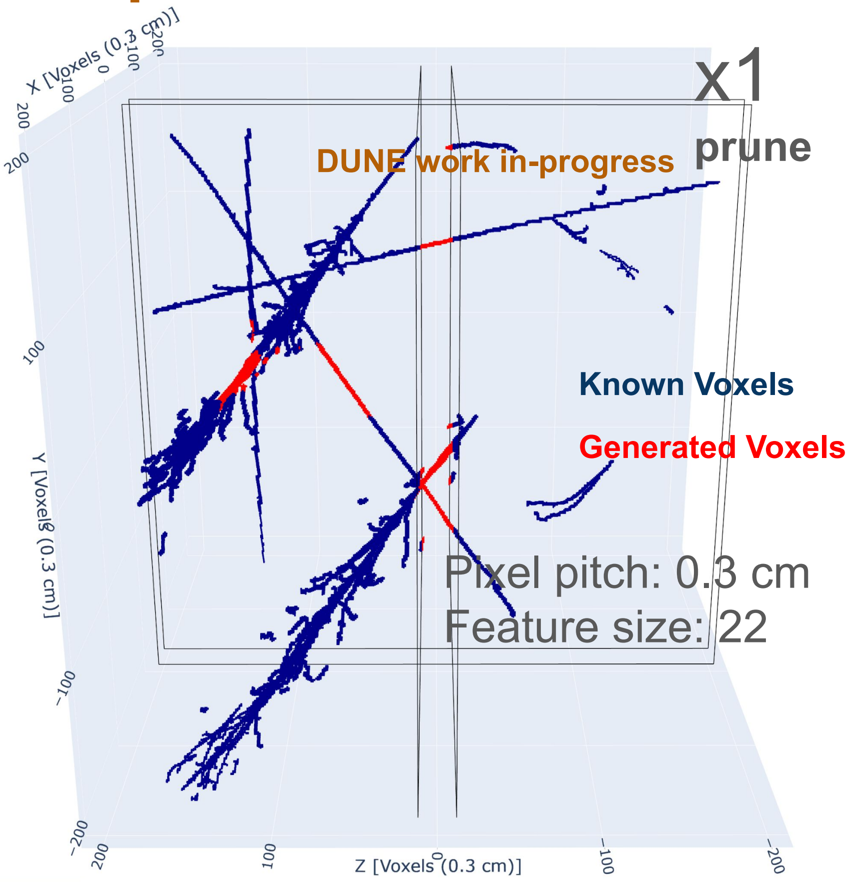
LArDRIP: A Multi-Scale Generative Sparse 3D UNet



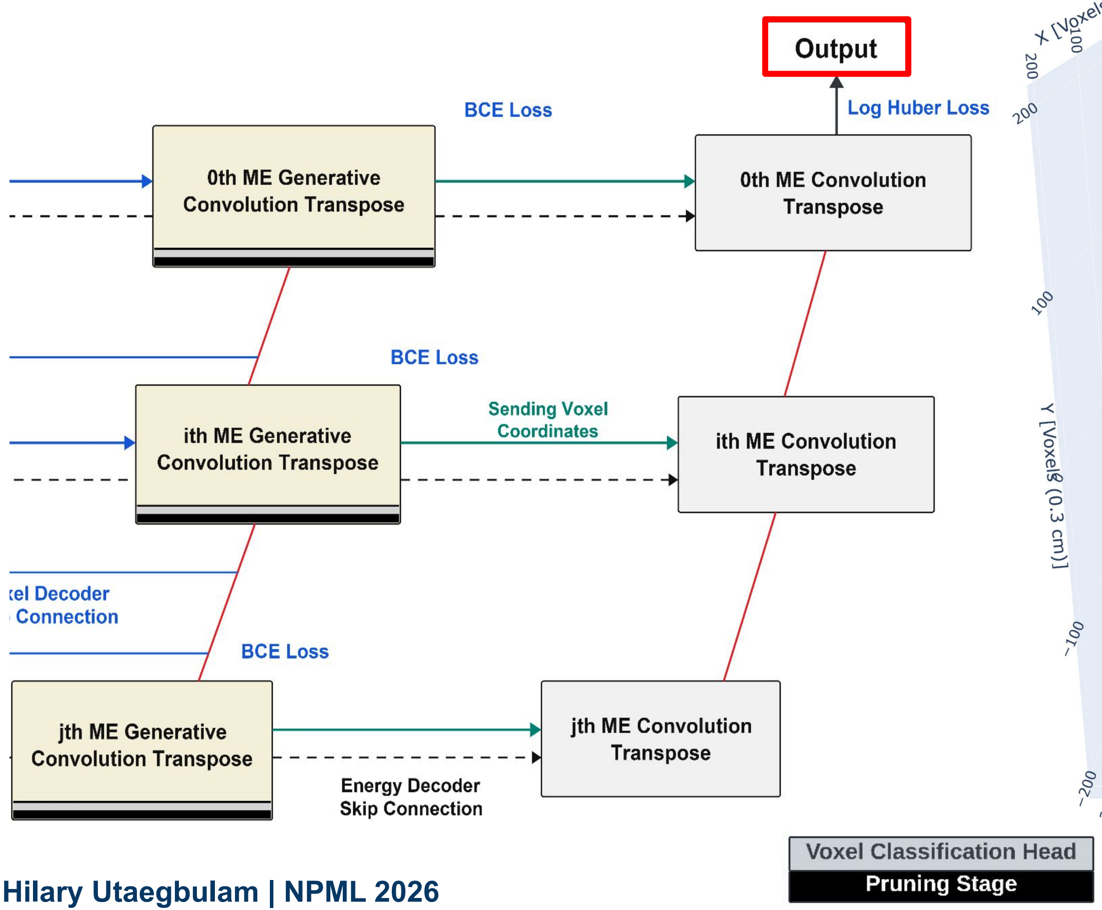


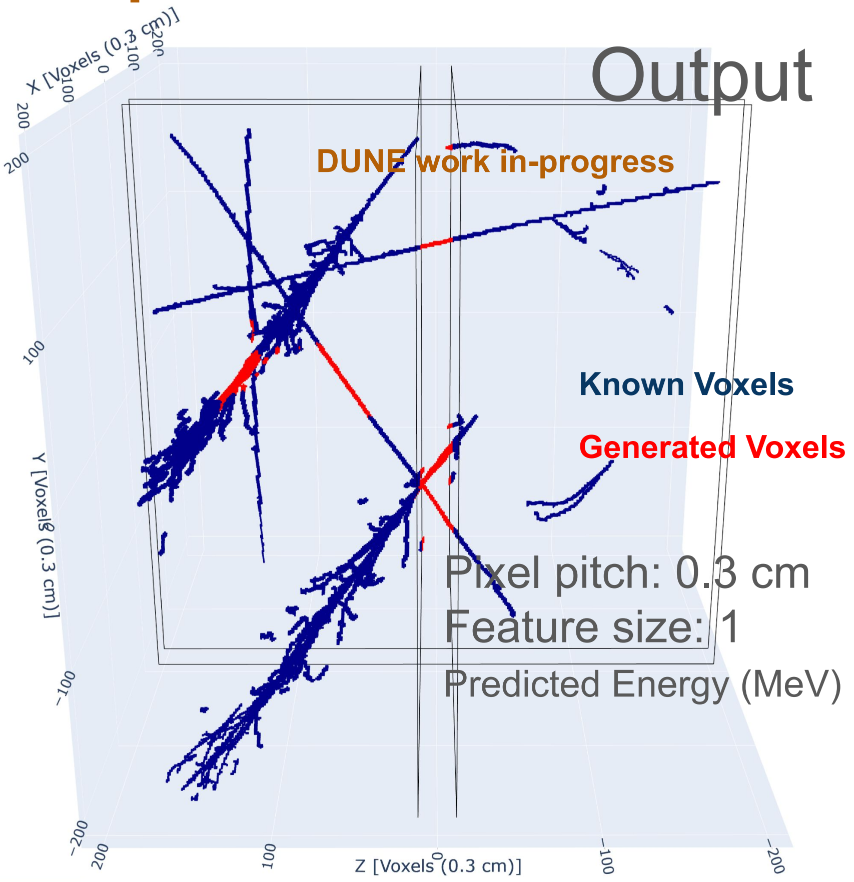
LArDRIP: A Multi-Scale Generative Sparse 3D UNet





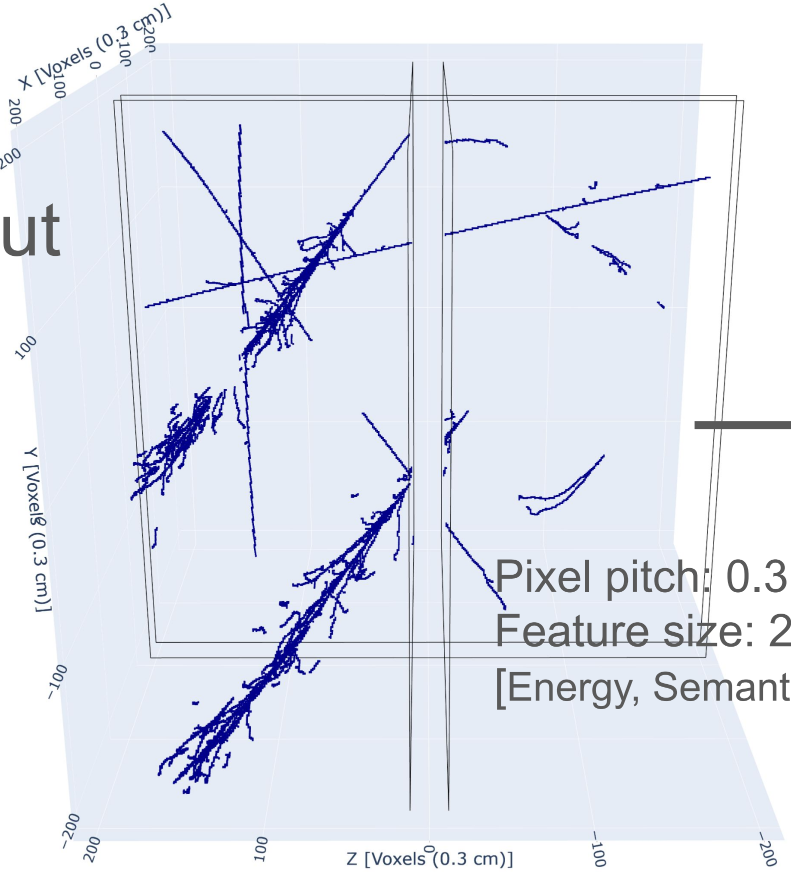
LArDRIP: A Multi-Scale Generative Sparse 3D UNet





LArDRIP: A Multi-Scale Generative Sparse 3D UNet

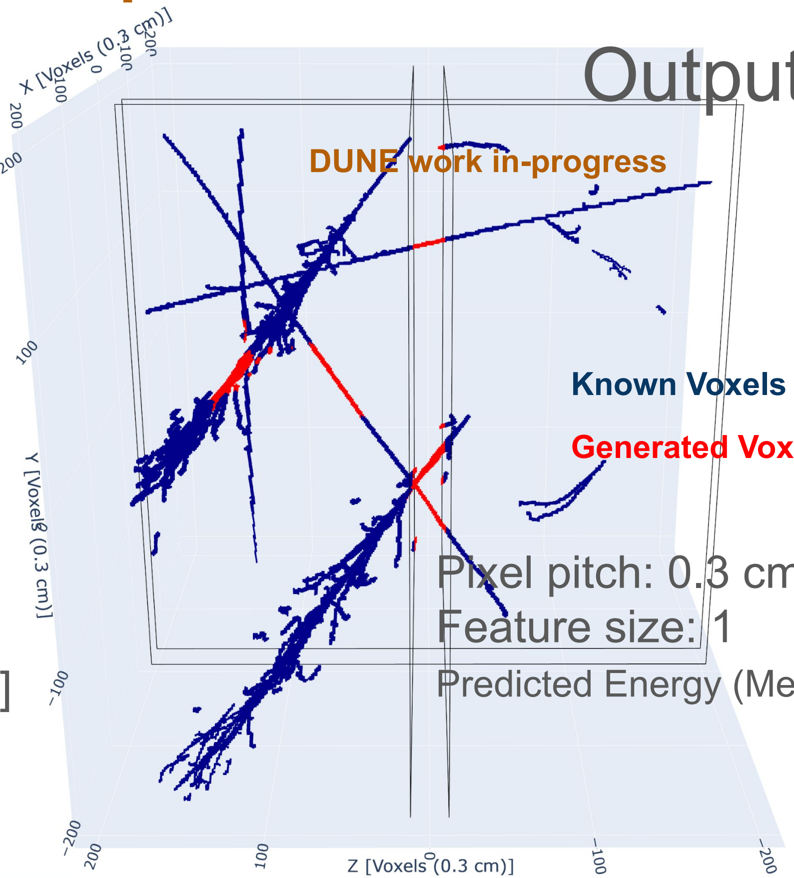
Input



Pixel pitch: 0.3 cm
Feature size: 2
[Energy, Semantic Type]

Output

DUNE work in-progress



Known Voxels

Generated Voxels

Pixel pitch: 0.3 cm
Feature size: 1
Predicted Energy (MeV)

Voxel Classification Head

Pruning Stage

BCE Loss

Too Strict!

$$\mathcal{L}_{\mathrm{BCE},i} = -\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

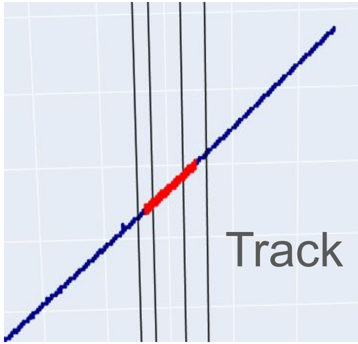
- BCE loss is too strict for showers!
- It must be tuned to allow for some tolerance. The model is correct, if it's close enough.

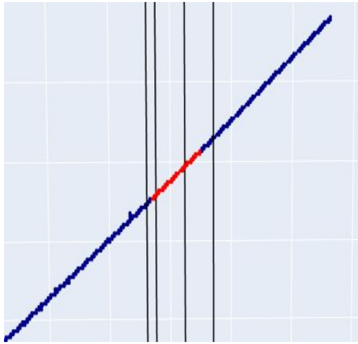
BCE Loss

Too Strict!

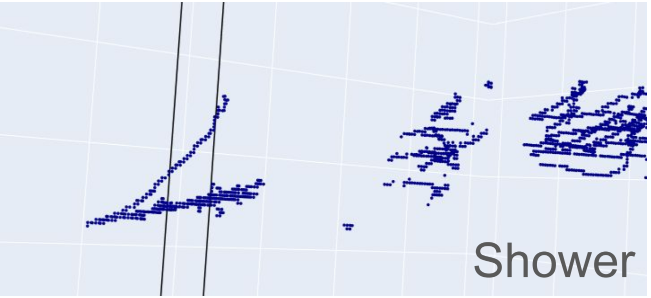
$$\mathcal{L}_{\mathrm{BCE},i} = -\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

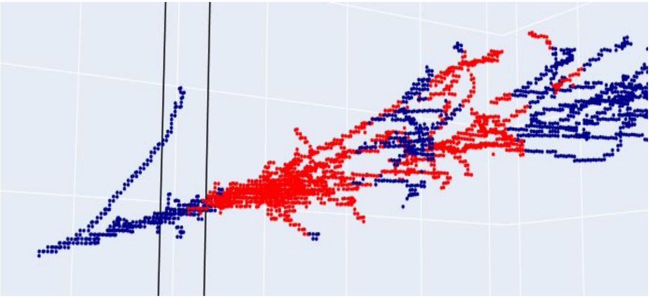
Prediction





Truth





BCE Loss

Soft Labels!

$$\mathcal{L}_{\mathrm{BCE},i} = -\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

- BCE loss is too strict
- It must be tuned to allow for some tolerance. The model is correct, if it's close enough.
- Different loss function for the inactive regions. Usual BCE loss for the active regions.

Fix! "less binary, binary cross entropy"

0.9 to 1-ish for **PilarNet**

Hyperparameters

In the inactive regions

$$y_i = \begin{cases} \mathbf{0.9} & \text{if voxel } i \text{ is within a Chebyshev radius-1 neighborhood of a target voxel} \\ 0 & \text{otherwise} \end{cases}$$

BCE Loss

Soft Labels!

$$\mathcal{L}_{\mathrm{BCE},i} = \quad 1 - p_i)]$$

- BCE loss is too
- It must be tuned n.
- Different loss fu

Fix! "less

0.9 to 1-ish for e regions
PilarNet

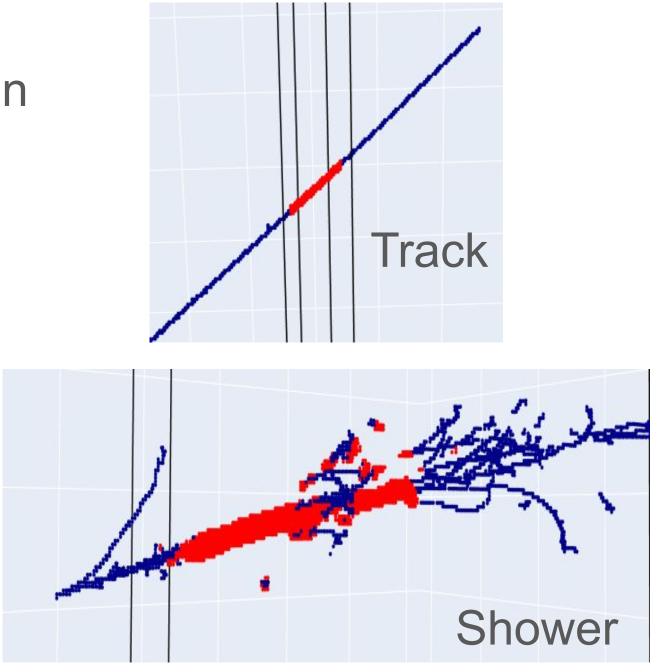
$$y_i = \begin{cases} \mathbf{0.9} & \text{if vo} \quad \text{a target voxel} \\ 0 & \text{otherwise} \end{cases}$$

BCE Loss

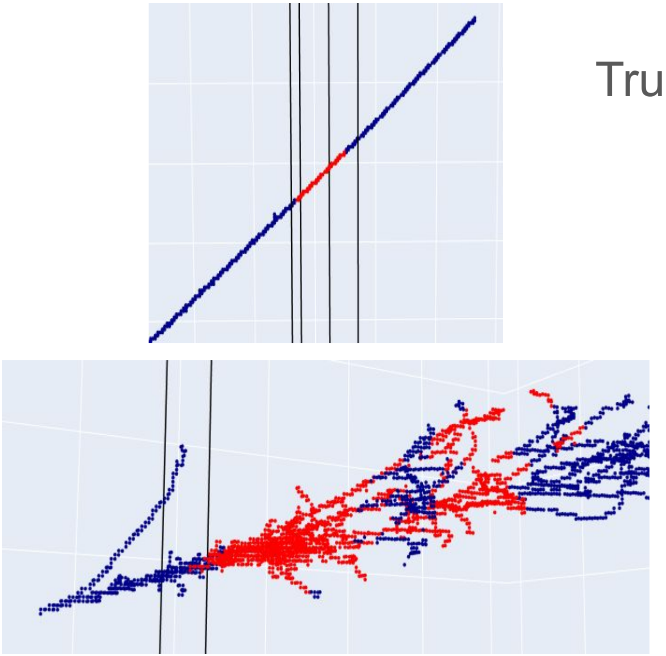
Soft Labels!

$$\mathcal{L}_{\mathrm{BCE},i} = -\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right]$$

Prediction



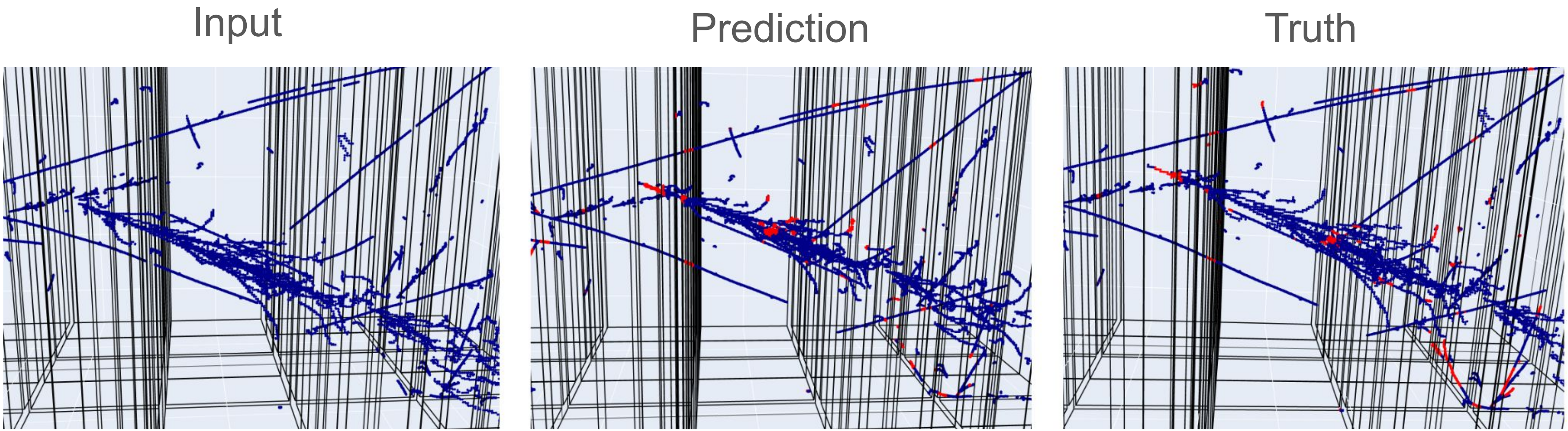
Truth



BCE Loss

Soft Labels!

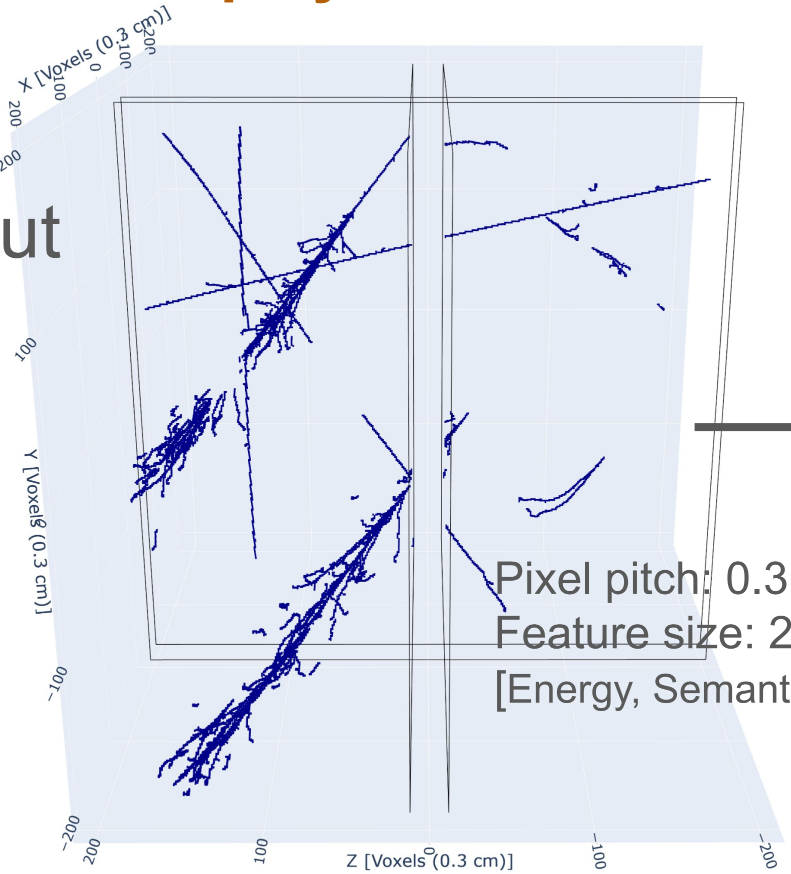
$$\mathcal{L}_{\mathrm{BCE},i} = -\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$



- Spread energy predictions into generated voxels in such a way that the sum total in the inactive region approaches the truth!

Event Displays

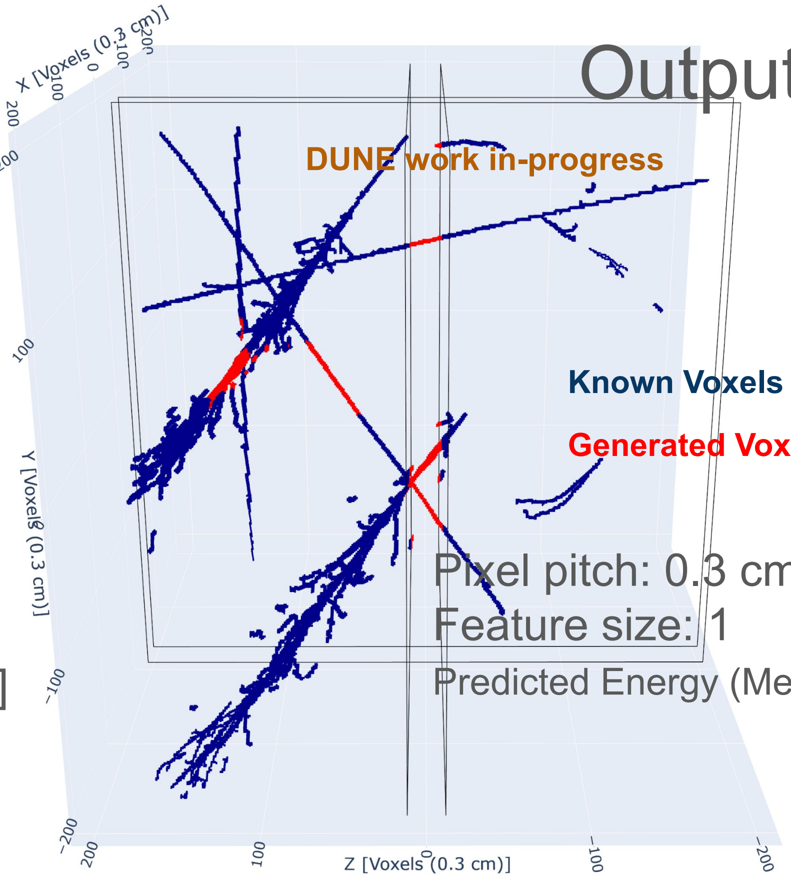
Input



Pixel pitch: 0.3 cm
Feature size: 2
[Energy, Semantic Type]

Output

DUNE work in-progress



Known Voxels

Generated Voxels

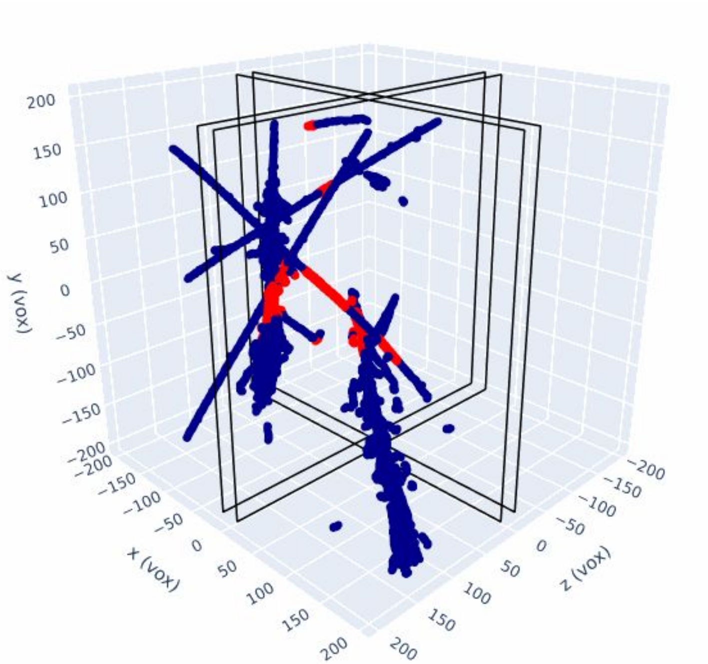
Pixel pitch: 0.3 cm
Feature size: 1
Predicted Energy (MeV)

Voxel Classification Head
Pruning Stage

Event Displays

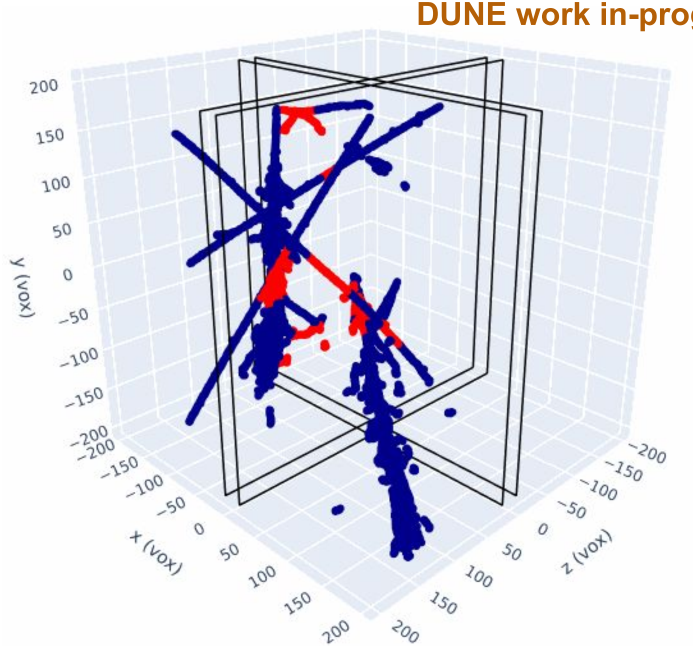
Predicted Inactive Energy: 987 MeV
True Inactive Energy: 900.6 MeV
Inactive Energy Resolution: 0.095

Prediction



Truth

DUNE work in-progress

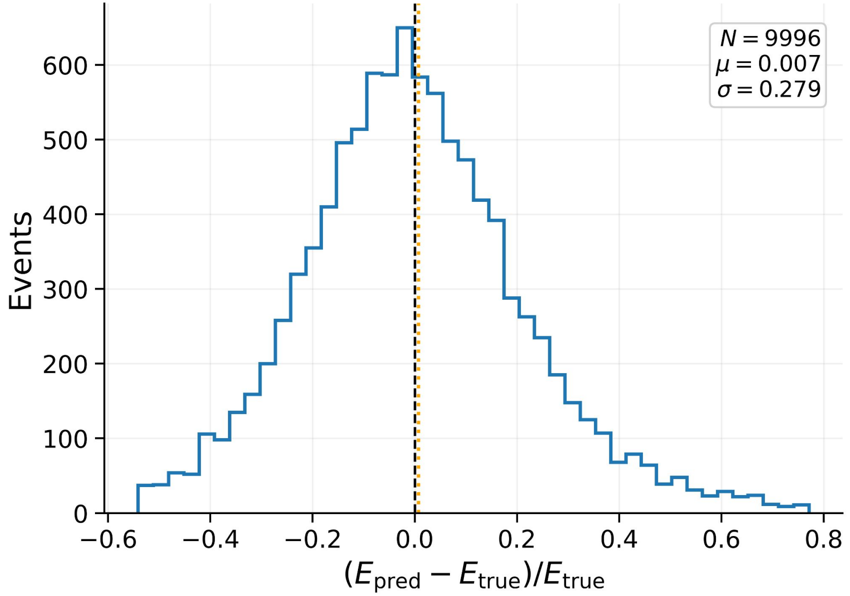


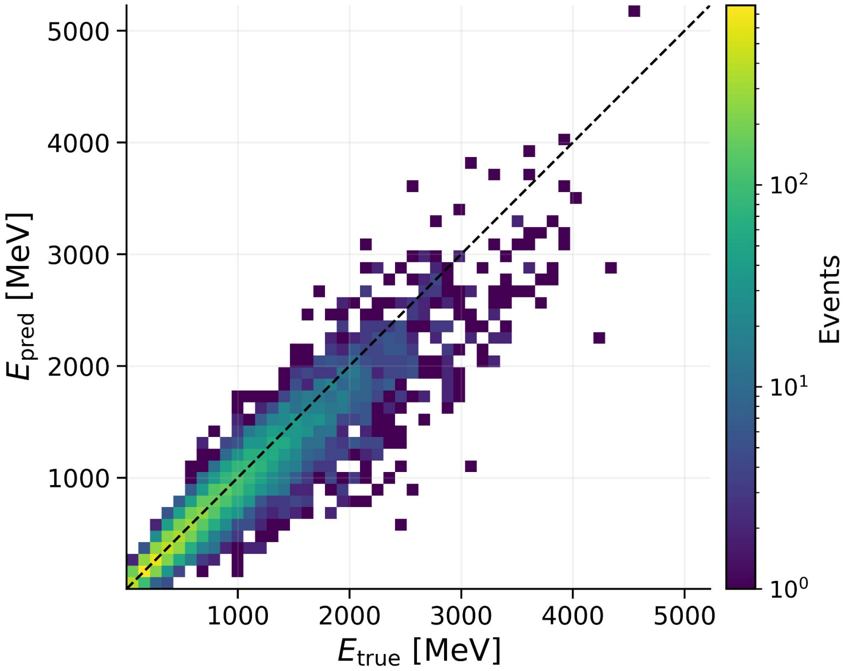
Active Region (Known)
Inactive Region (Predicted)

Predicted Total Energy: 6280 MeV
True Total Energy: 6200 MeV
Total Energy Resolution: 0.012

Active Region (Known)
Inactive Region (Truth)

Plots





- Using PILArNet Dataset
- https://arxiv.org/abs/2006.01993

- Particle Types:
  - Electron showers
  - Photon showers
  - Muon tracks
  - Pion tracks
  - Proton tracks

Many "spills" which contain all particle types

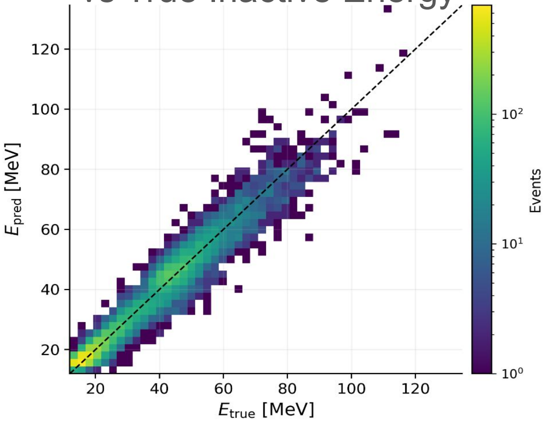
Plots

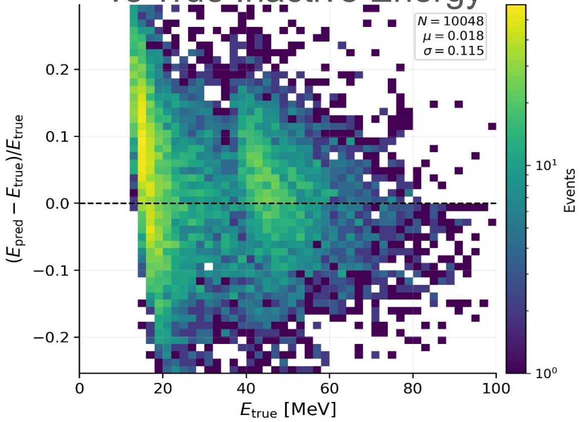
Inactive Region Energy

Predicted Inactive Energy vs True Inactive Energy

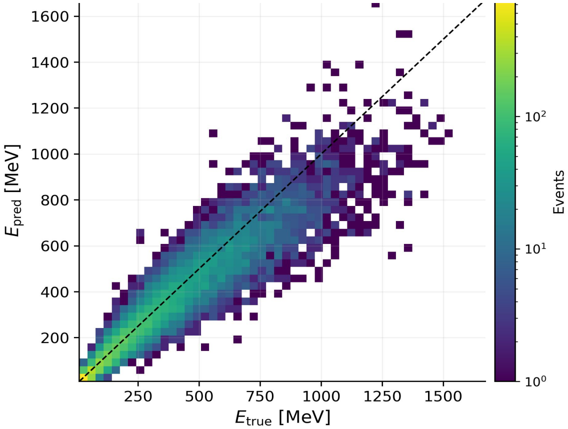
Inactive Energy Resolution vs True Inactive Energy

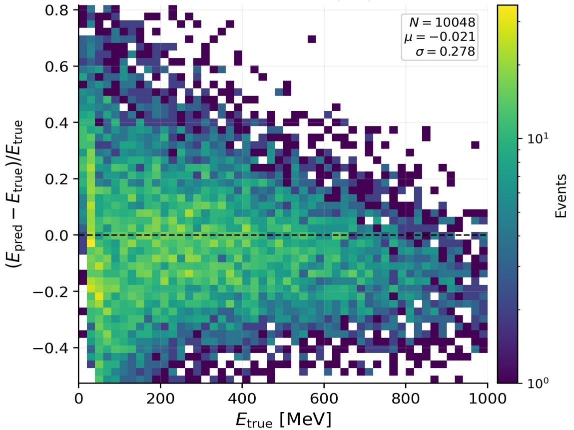
Muons





Electron showers





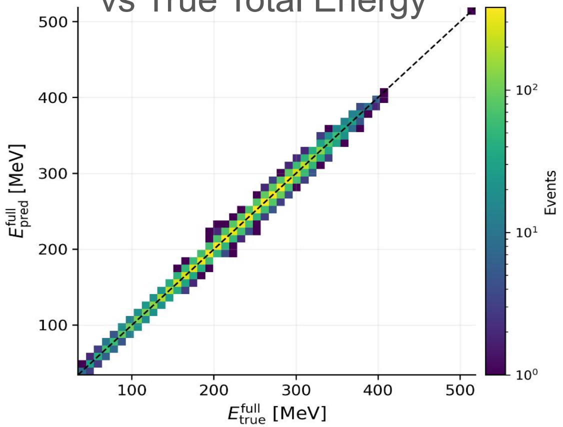
Plots

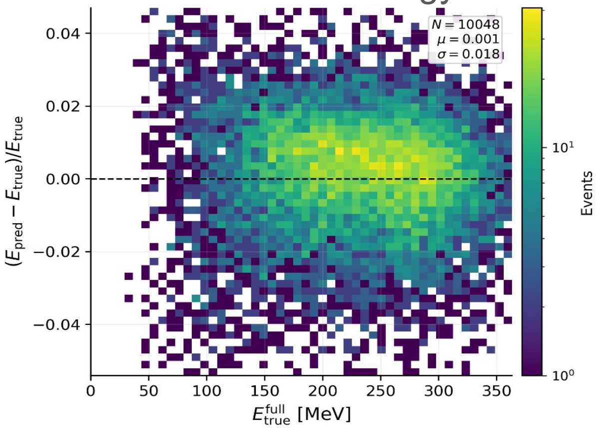
Total Energy

Predicted Total Energy vs True Total Energy

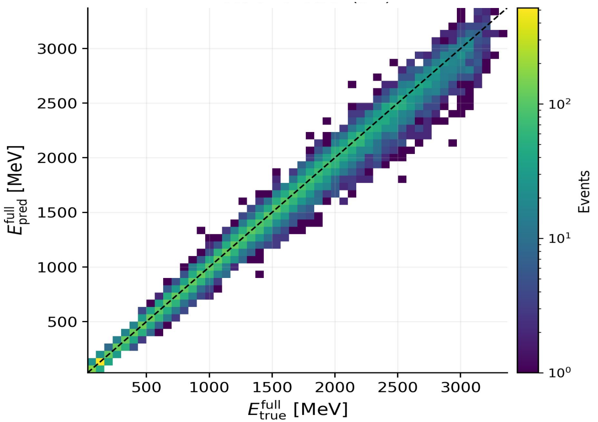
Total Energy Resolution vs True Total Energy

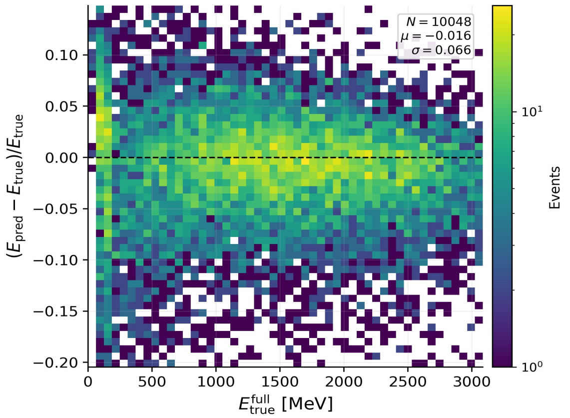
Muons





Electron showers





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Plots

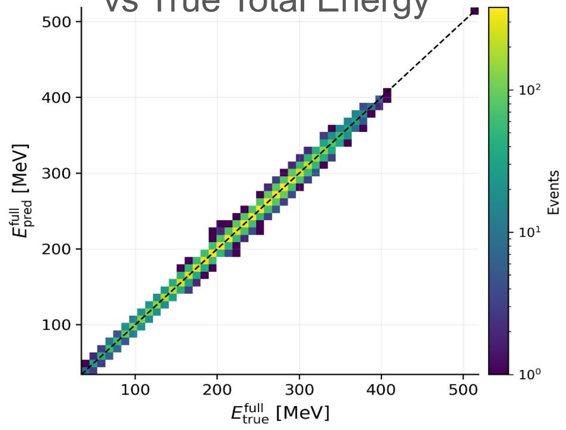
Total Energy

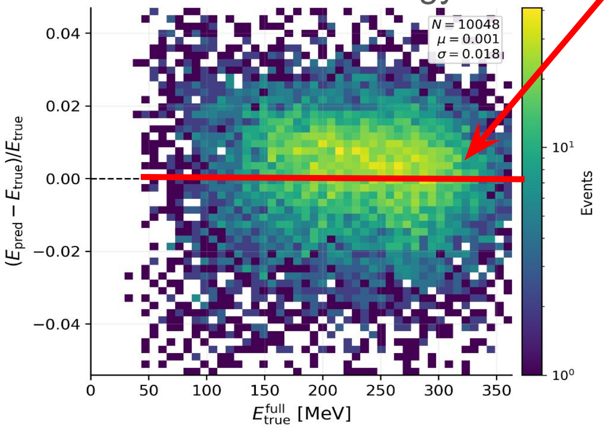
An *ideal* model would give resolution ~ 0

Predicted Total Energy vs True Total Energy

Total Energy Resolution vs True Total Energy

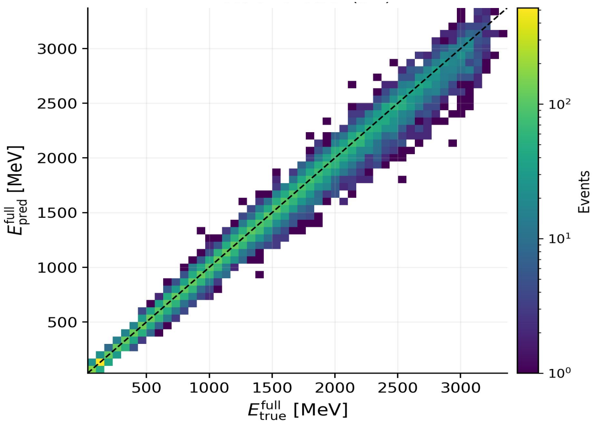
Muons

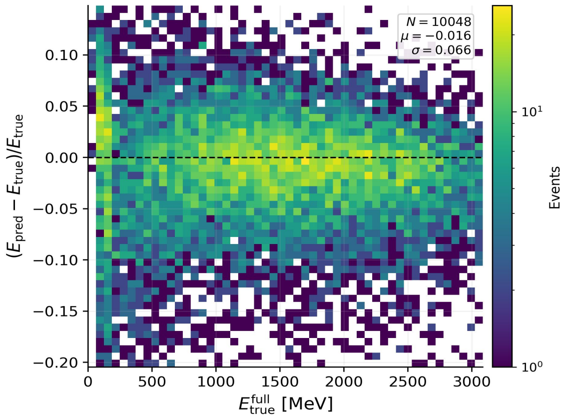




Deviation from an *ideal* model due to LArDRIP's prediction

Electron showers





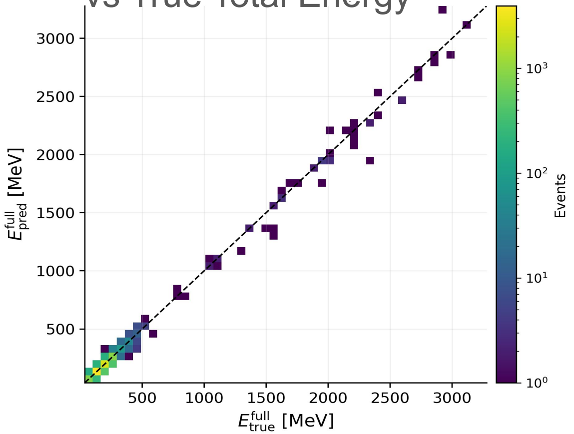
Plots

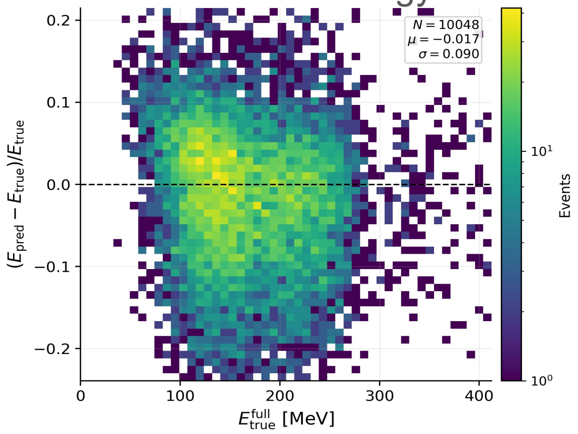
Total Energy

Predicted Total Energy vs True Total Energy

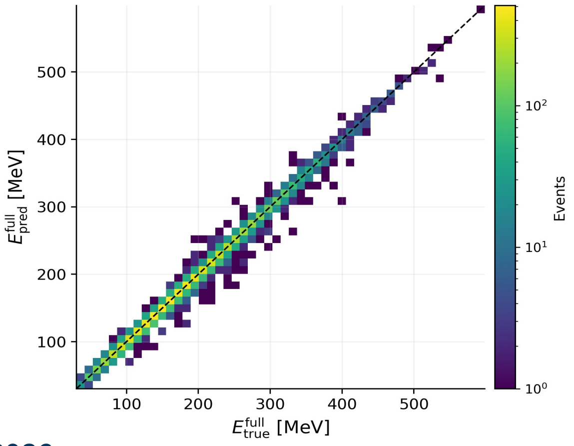
Total Energy Resolution vs True Total Energy

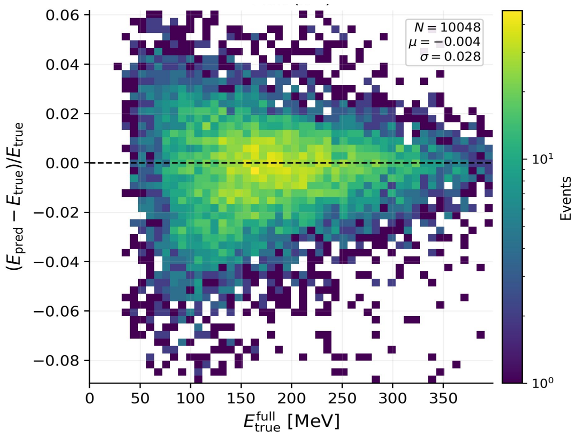
Photon Showers





Pions



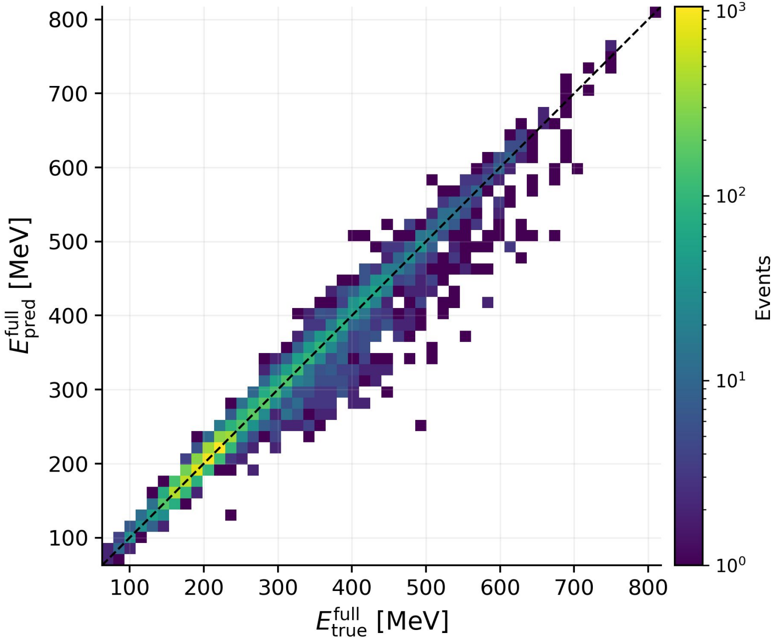


Plots

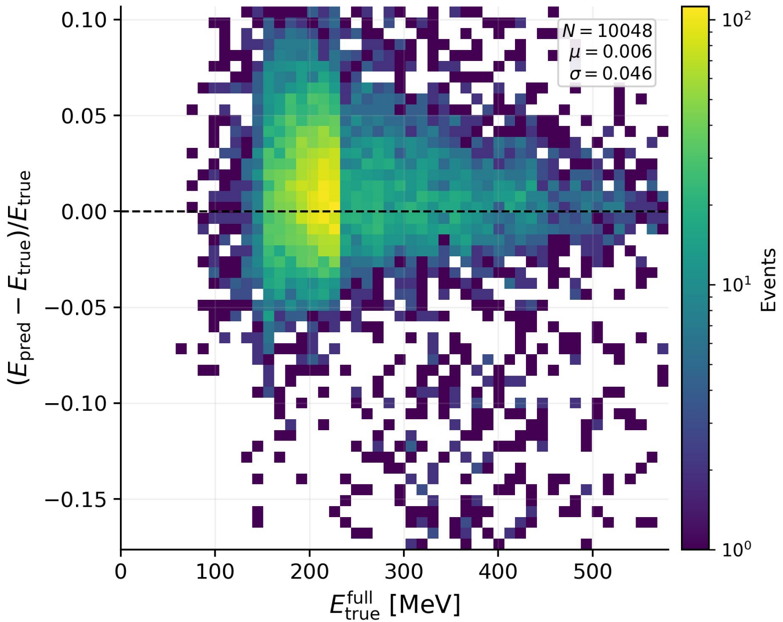
Total Energy

Protons

Predicted Total Energy
vs True Total Energy



Total Energy Resolution
vs True Total Energy



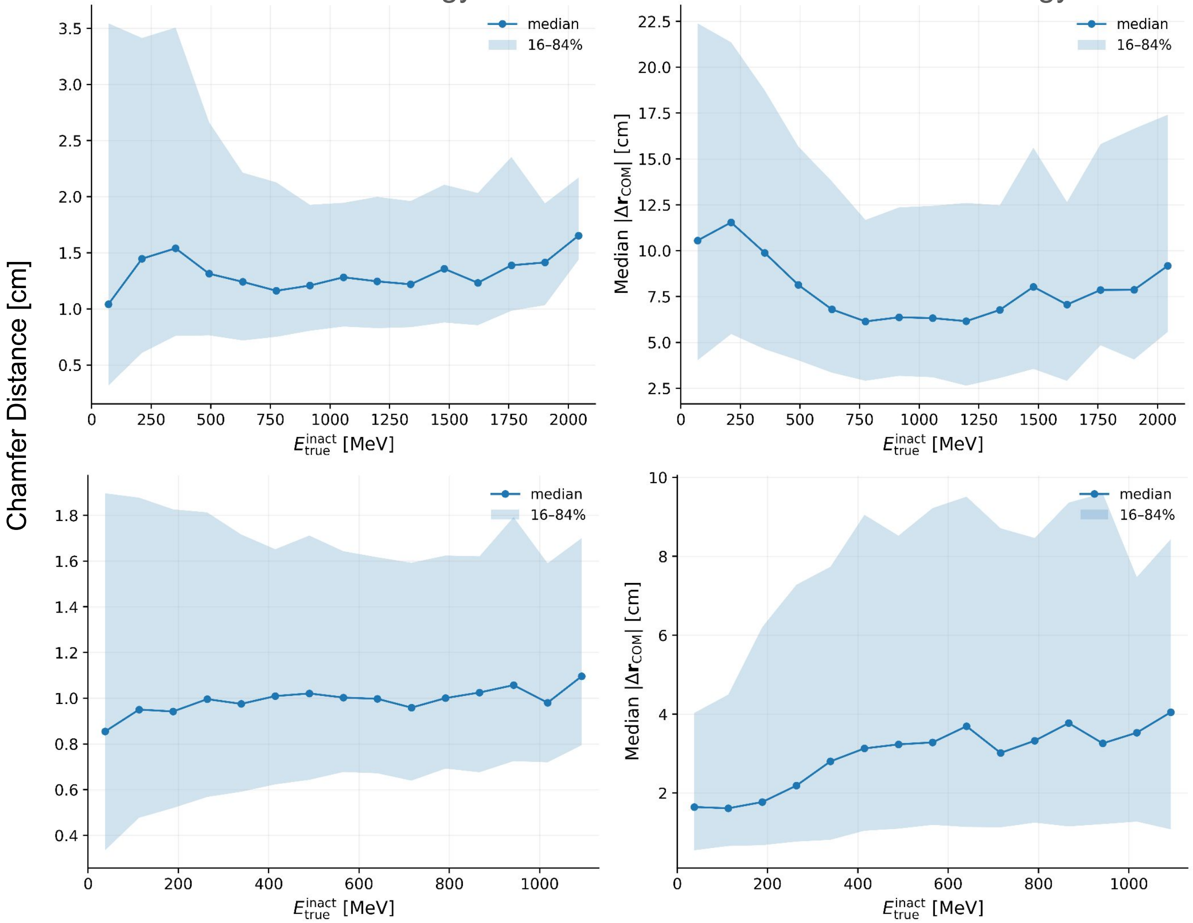
Plots

Chamfer Distance
vs True Inactive Energy

Change in Center of Mass
vs True Inactive Energy

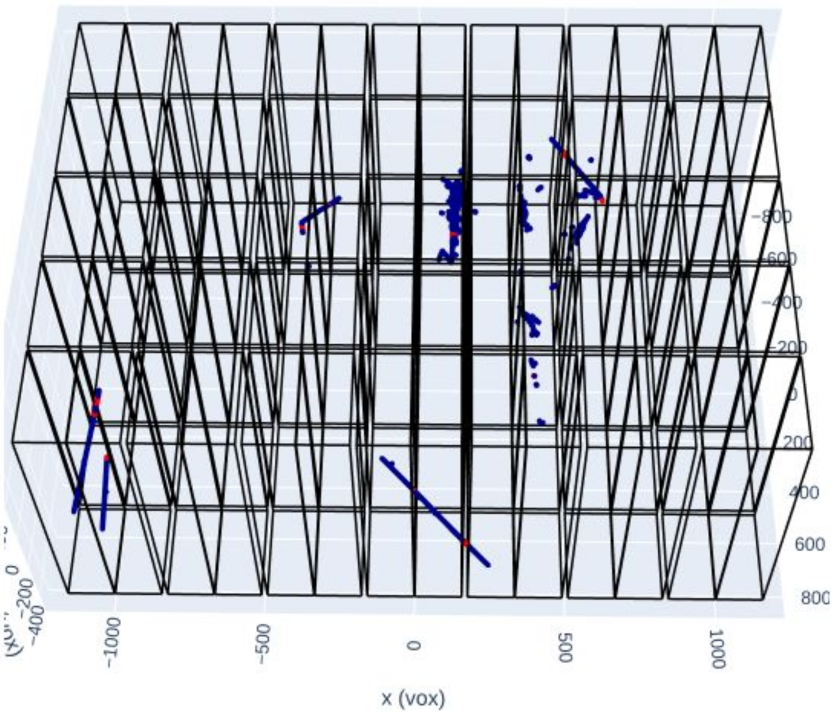
Spill

Electron Showers

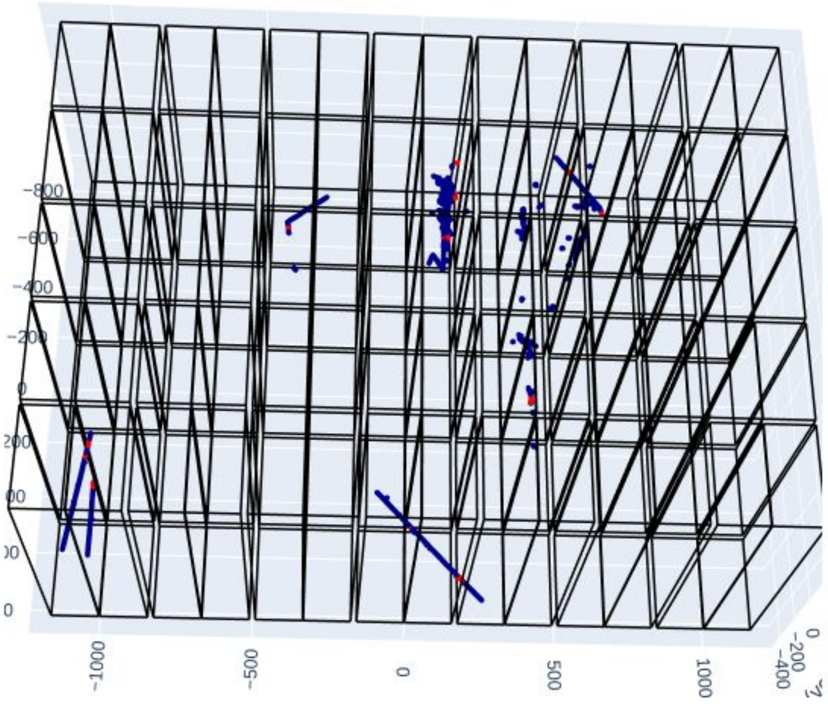


NDLAr

Prediction

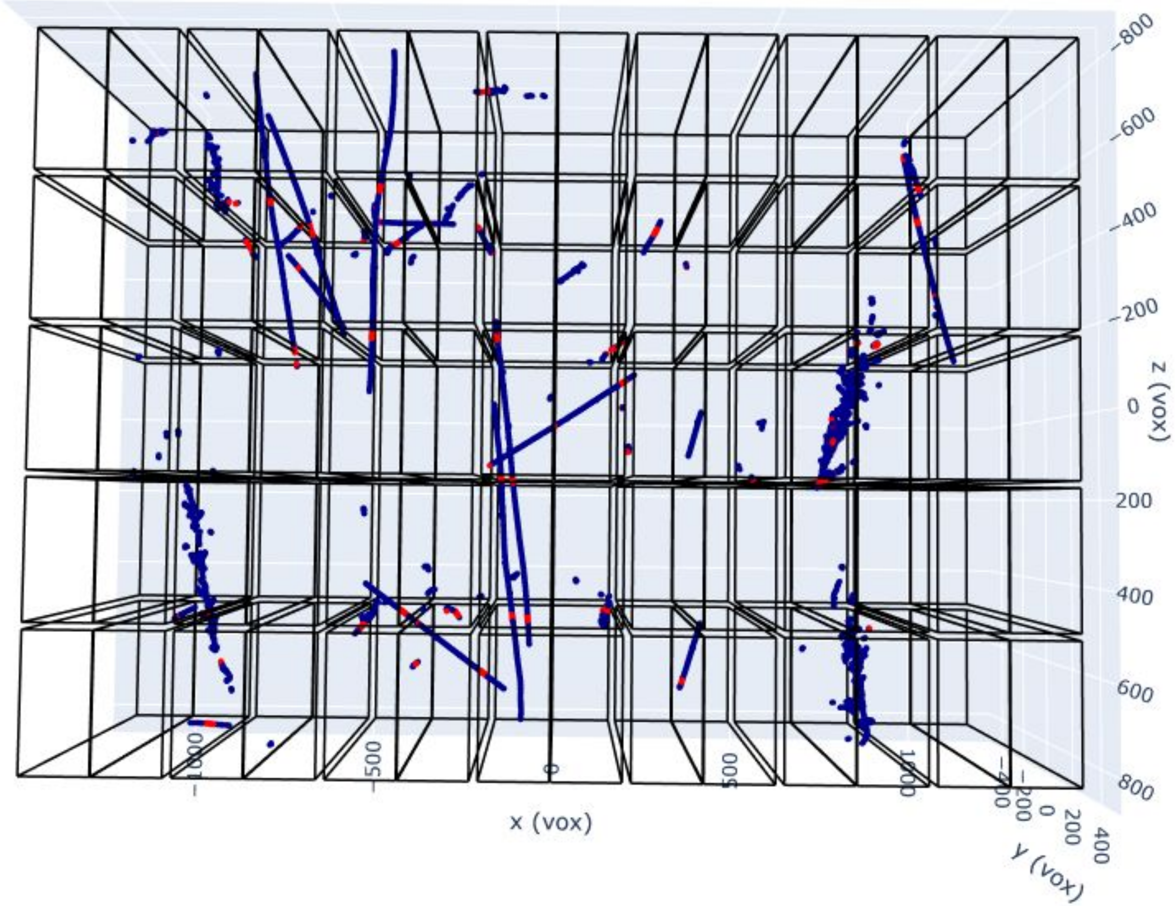


Truth



NDLAr

Prediction



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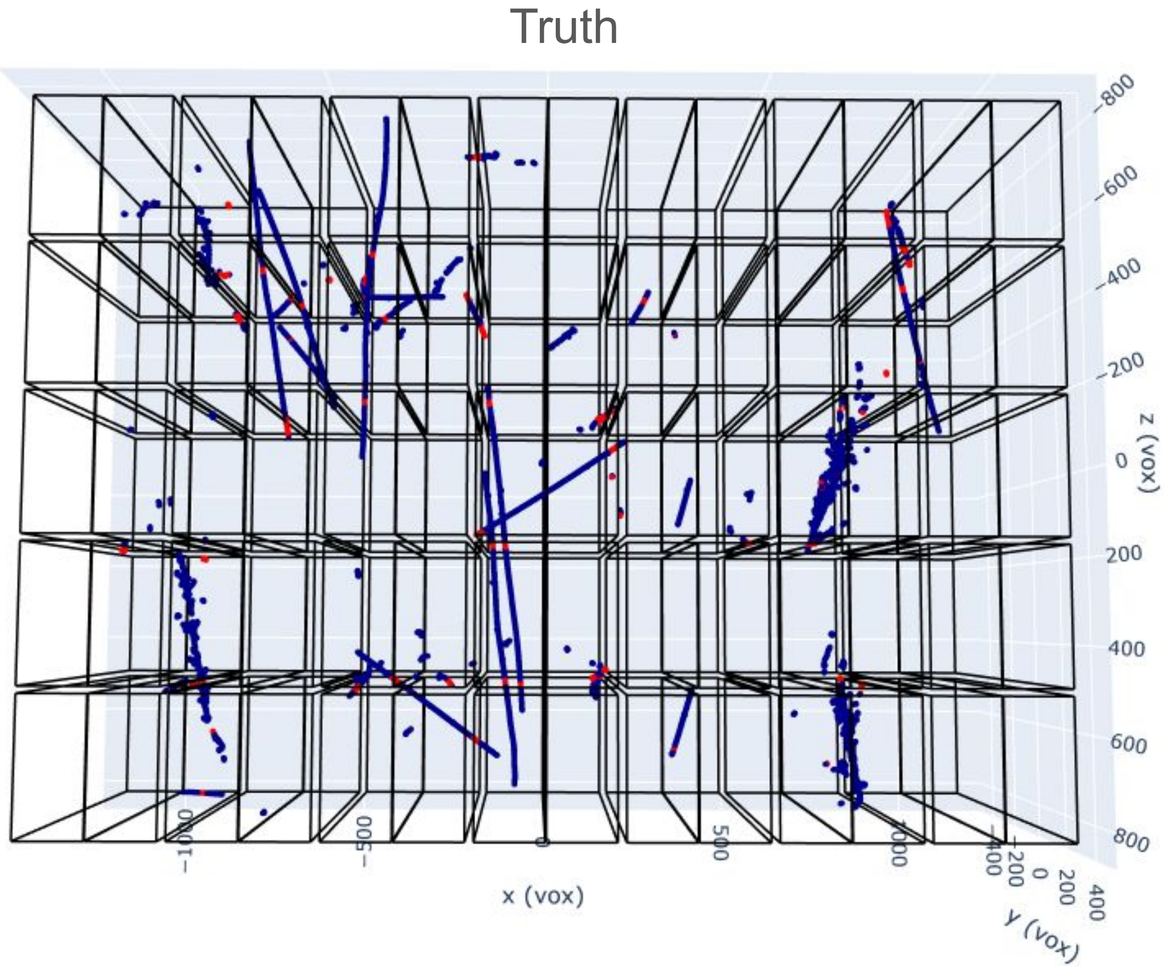
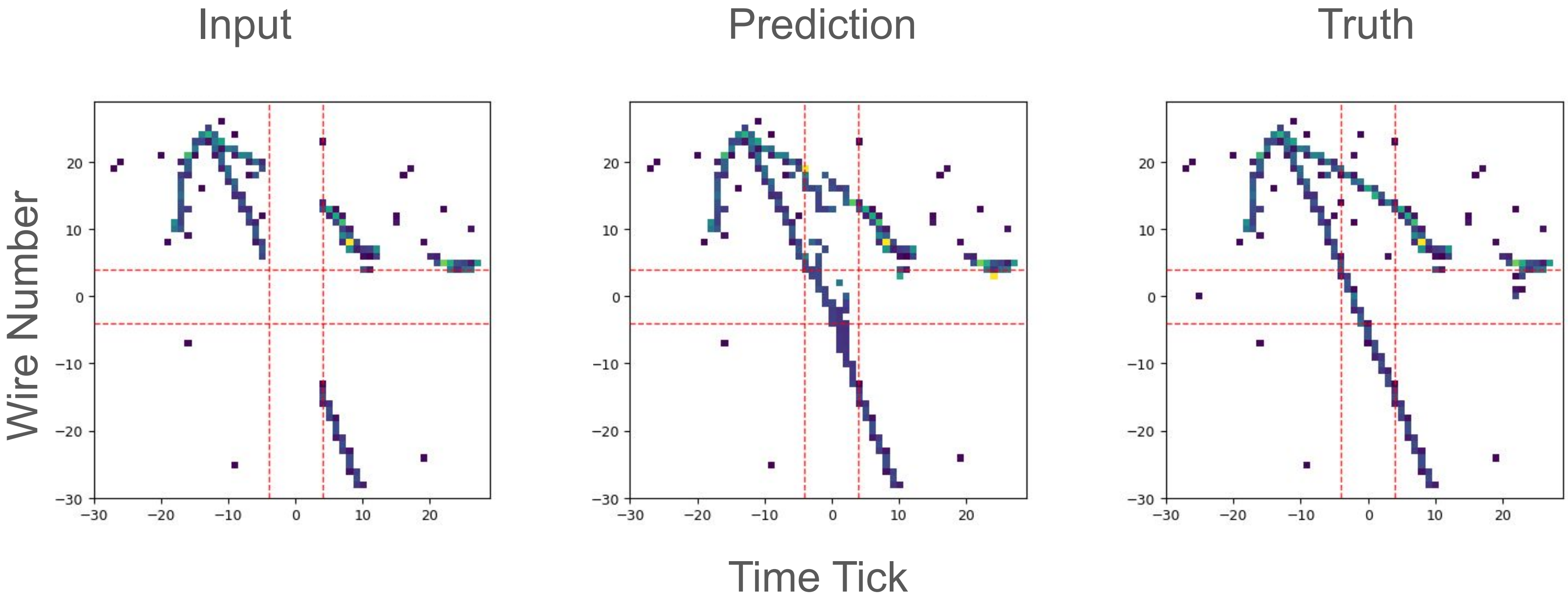
NDLAr



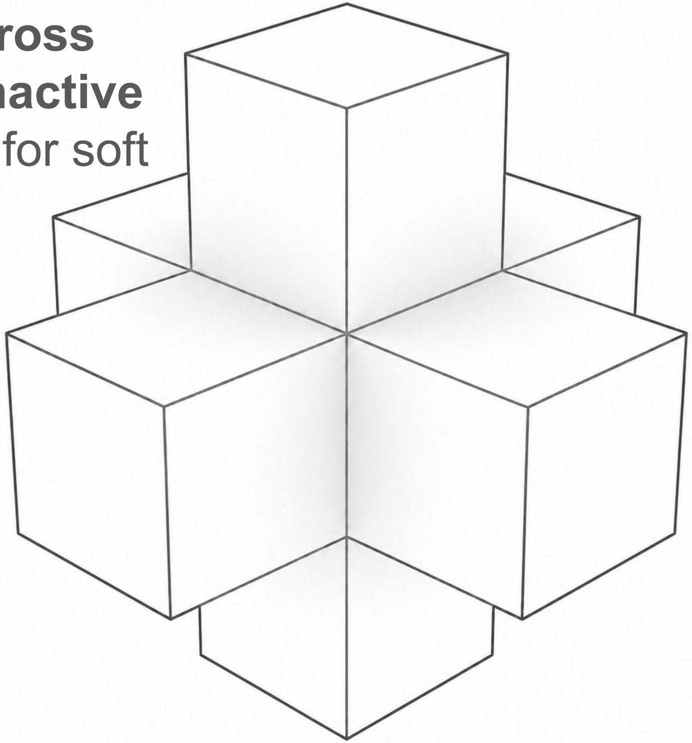
Truth
x (vox)
y (vox)
z (vox)
−1000
−500
0
500
1000
−400
−200
0
200
400
−800
−600
−400
−200
0
200
400
600
800

2D LArDRIP

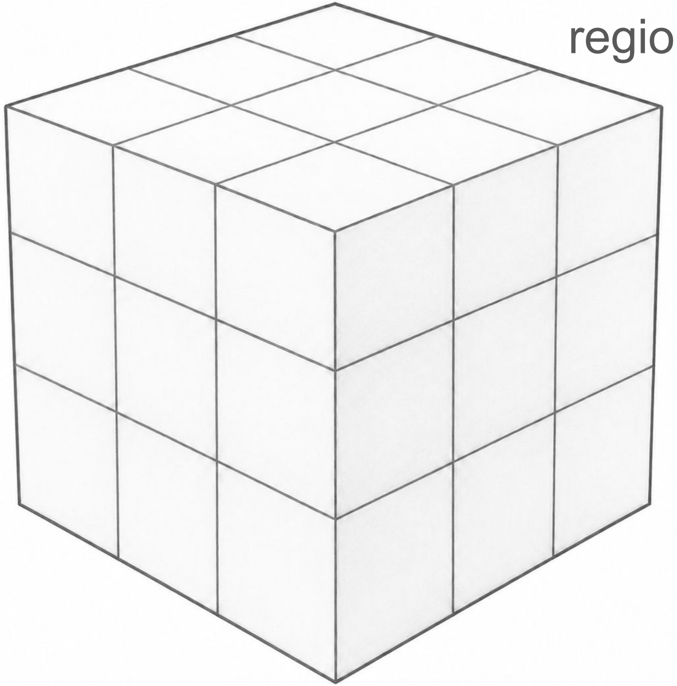


Future

Hypercross
In the **inactive** regions for soft labels



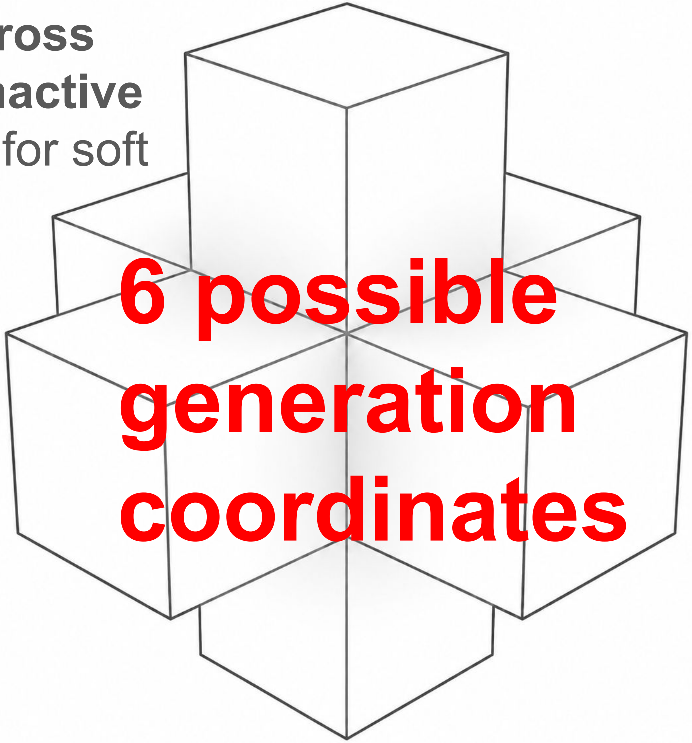
Hypercube
In the **active** regions



At the moment, LArDRIP uses a hyper cross for inactive region generation

Future

Hypercross
In the **inactive** regions for soft labels



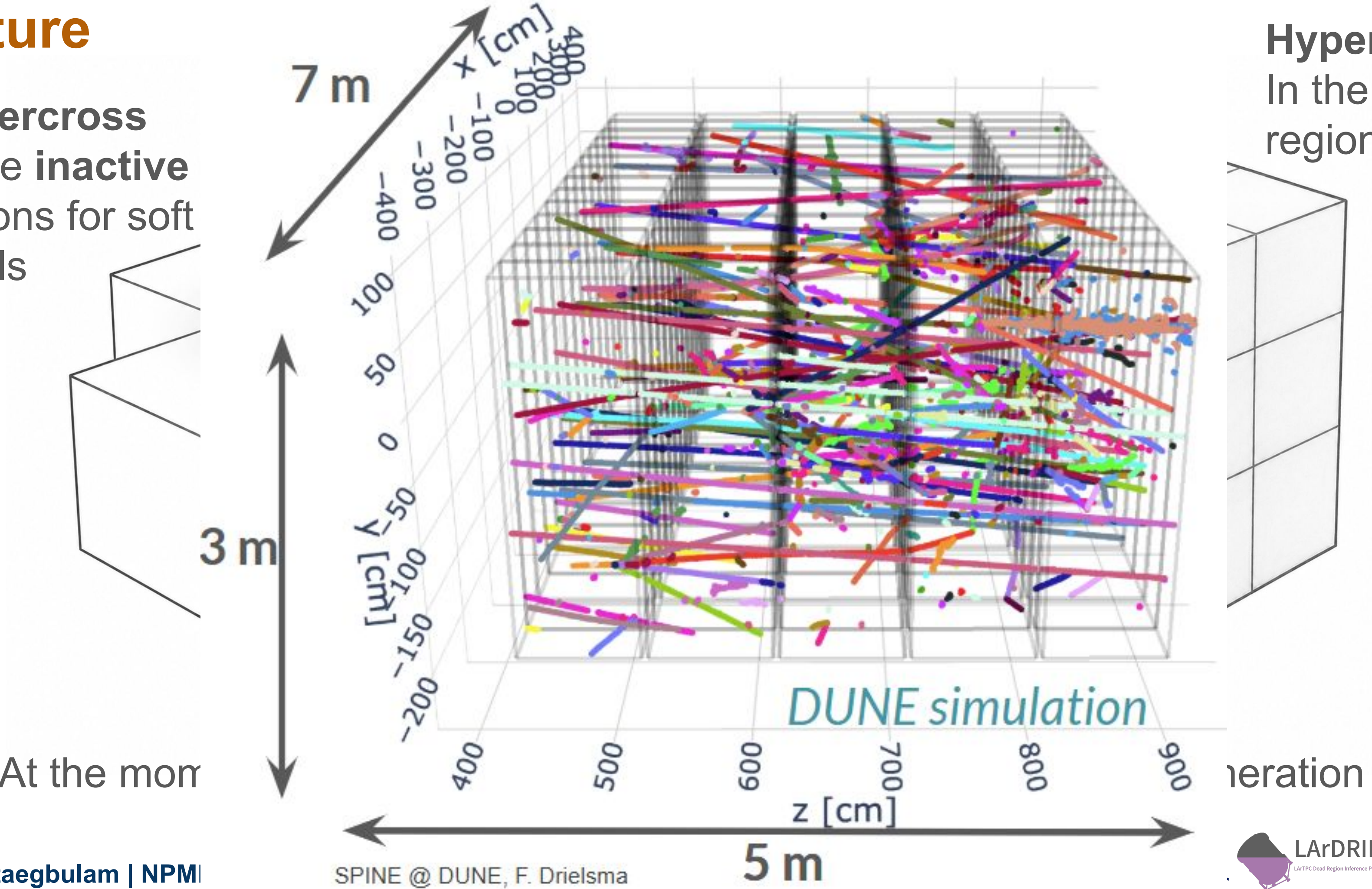
Hypercube
In the **active** regions

At the moment, LArDRIP uses a hyper cross for inactive region generation

Future

Hypercross
In the **inactive** regions for soft labels

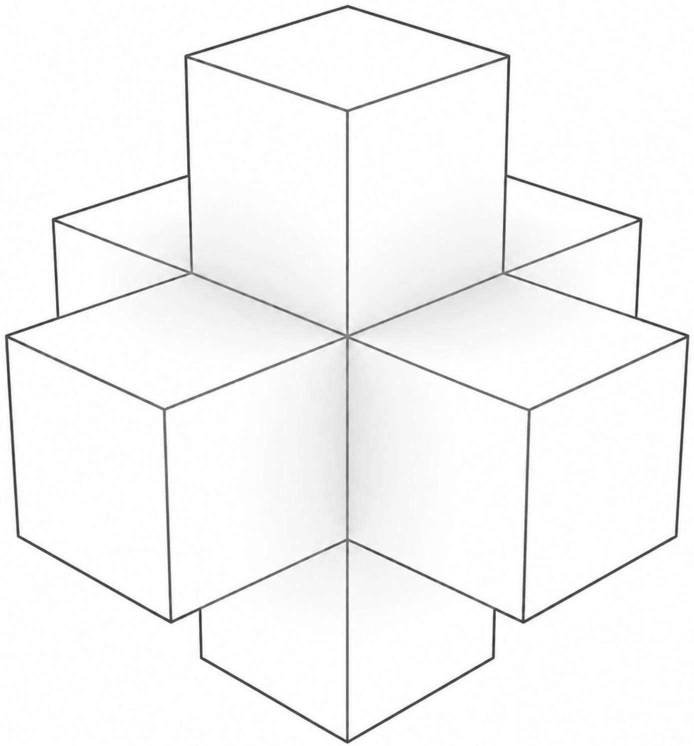
Hypercube
In the **active** regions



At the mom ... neration

Future

- At the moment, LArDRIP uses a hyper cross for BCE soft label inactive region generation
- Hypercross reduces the number of potential voxels for generation, thereby reducing memory and compute
- Currently transitioning to **hypercross everywhere**
- Improve voxel inference by using a smoothly decreasing function and hypercross; no more soft labels!
- Should allow for much more messiness -> NDLAr



Future

- Connect with SPINE
- Standalone, for other detector geometries
- Improve inference loss function/voxel generation

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